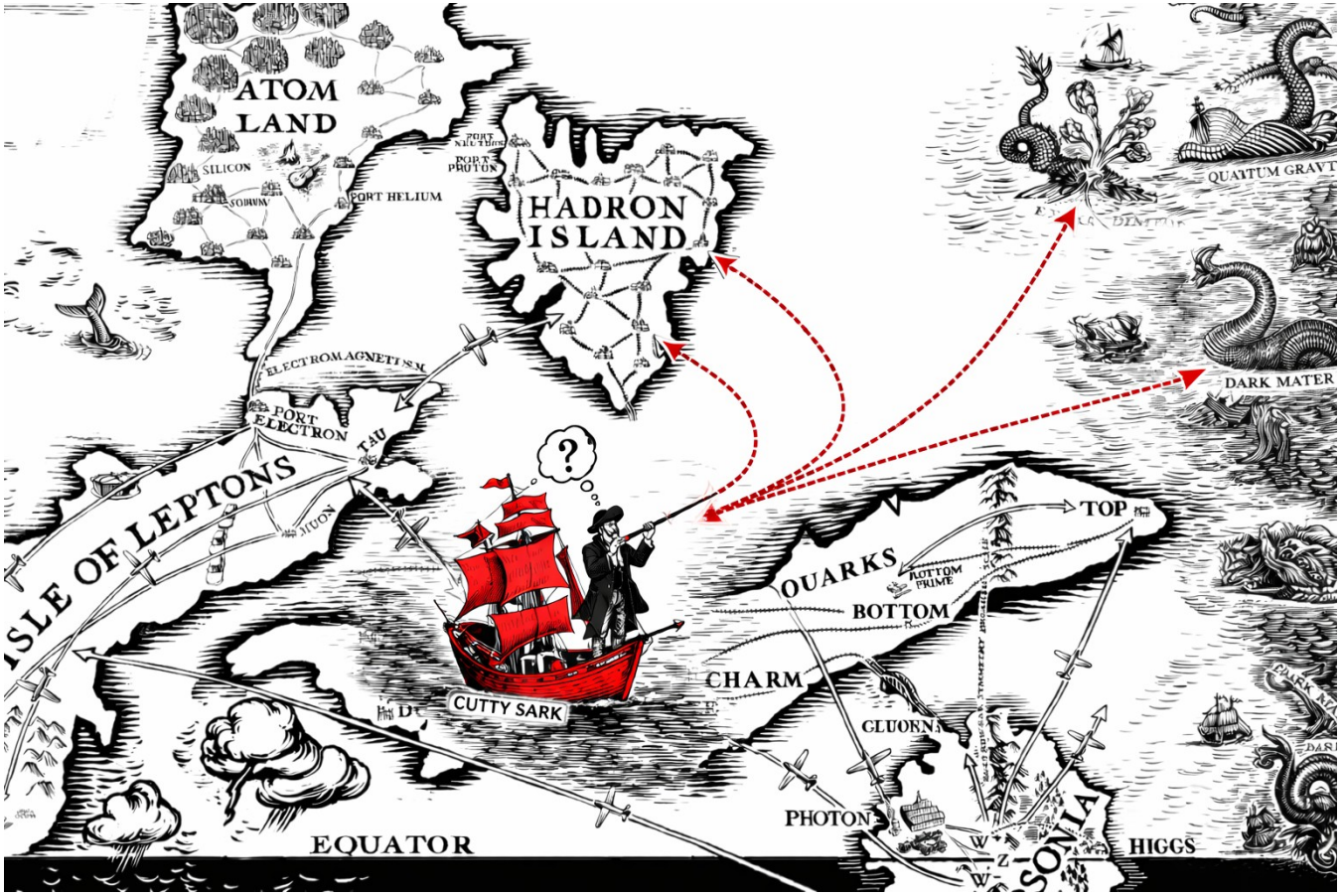
LHC Searches and the Next Standard Model: A Meta-Statistical View

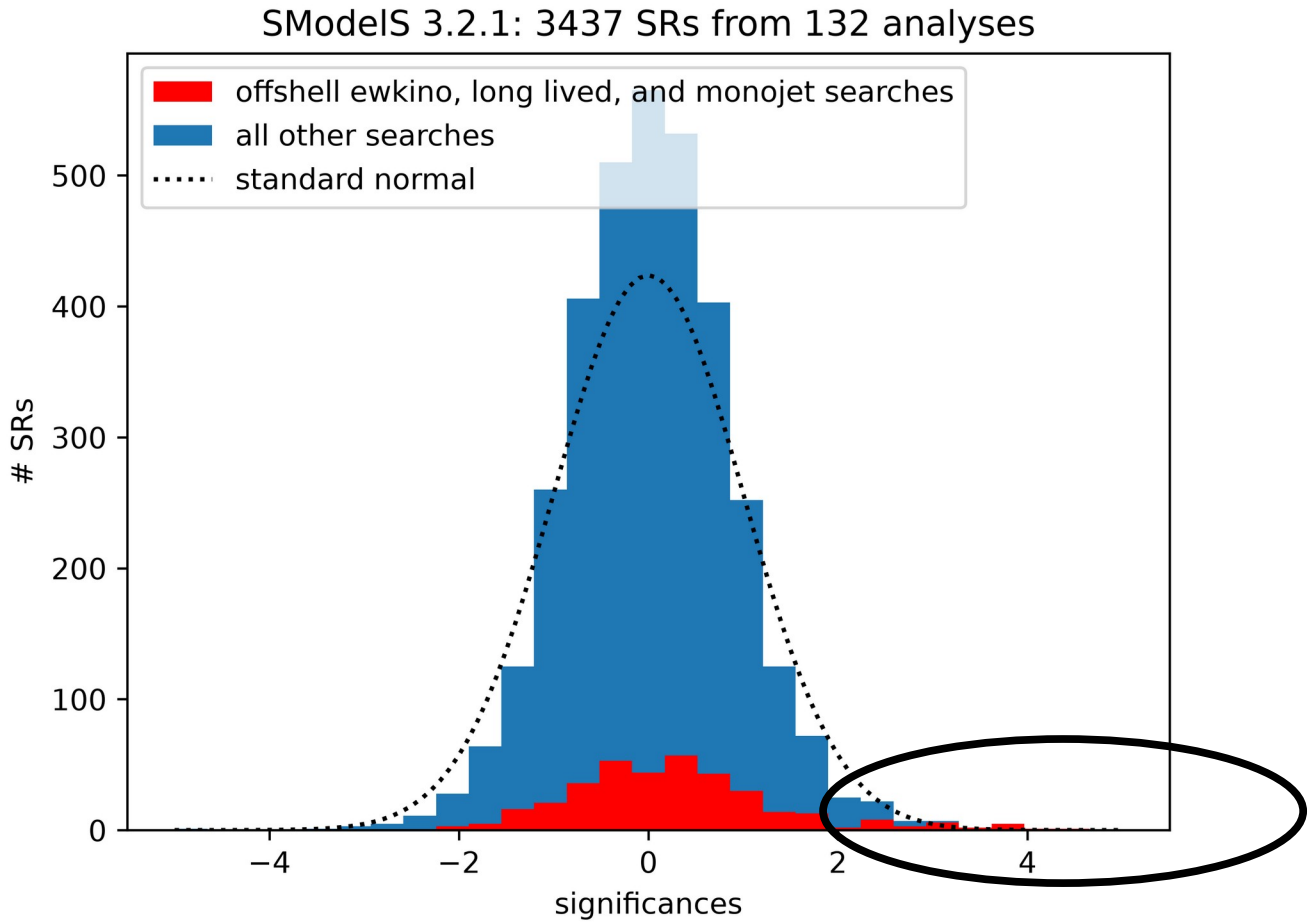


Wolfgang Waltenberger (MBI and Uni Wien)
For the SModelS / protomodels collaborations

BSM Workshop @ CERN
October 2026

image courtesy of Jon Butterworth, Chris Wormell and chatgpt

We, the SModelS collaboration have been collecting the published search results of CMS and ATLAS since the dawn of the LHC. Here is v3.2.1 of our database:



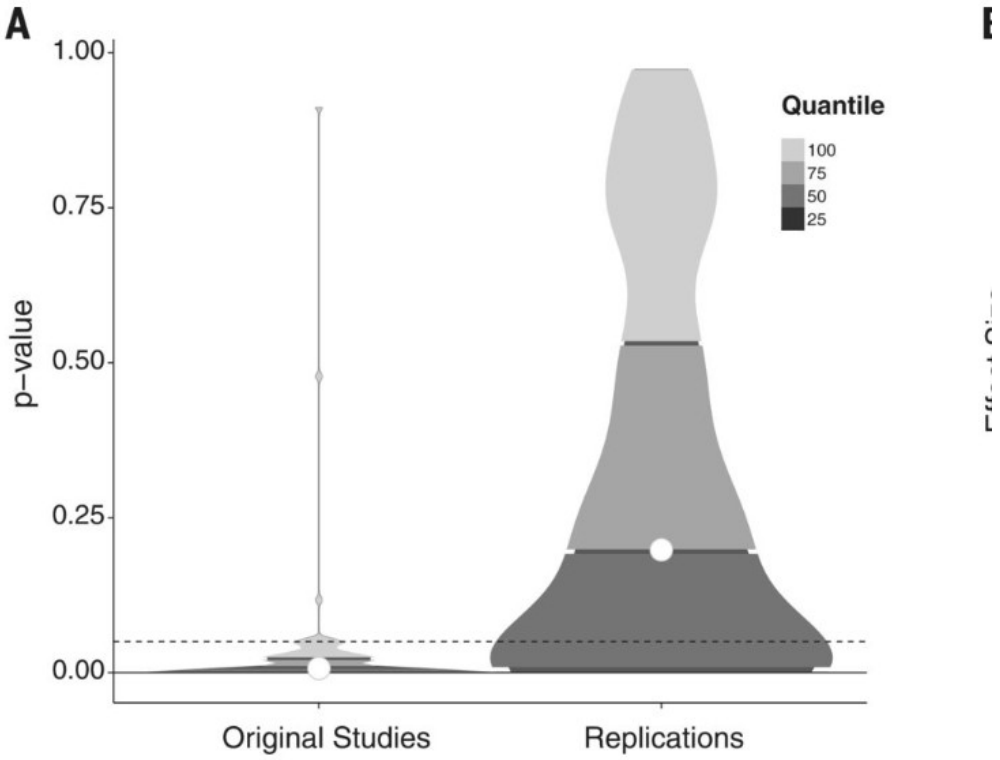
Distribution of significances of all the signal regions we have in our database, Standard Model hypothesis.

If no new physics is in the data, it should follow a standard normal (dashed black).

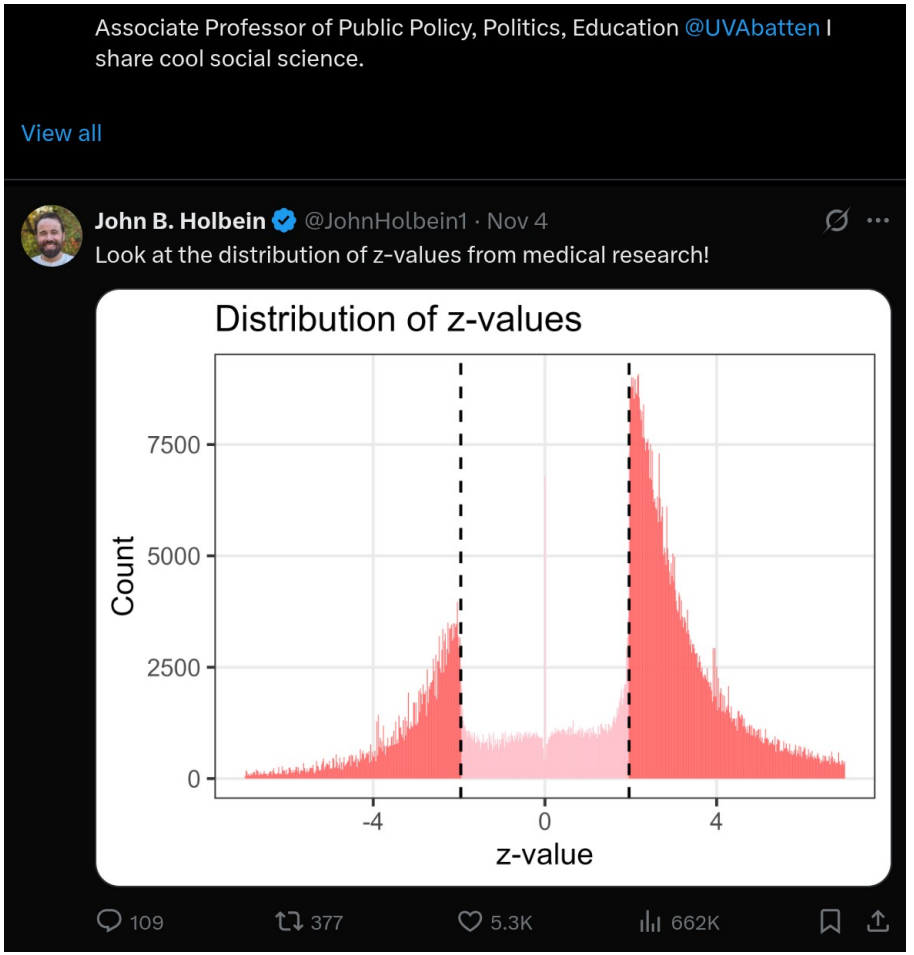
New pyhsics would be at $Z \rightarrow +\infty$, obs == exp are at $Z \sim 0$.



“Estimating the reproducibility of psychological science”



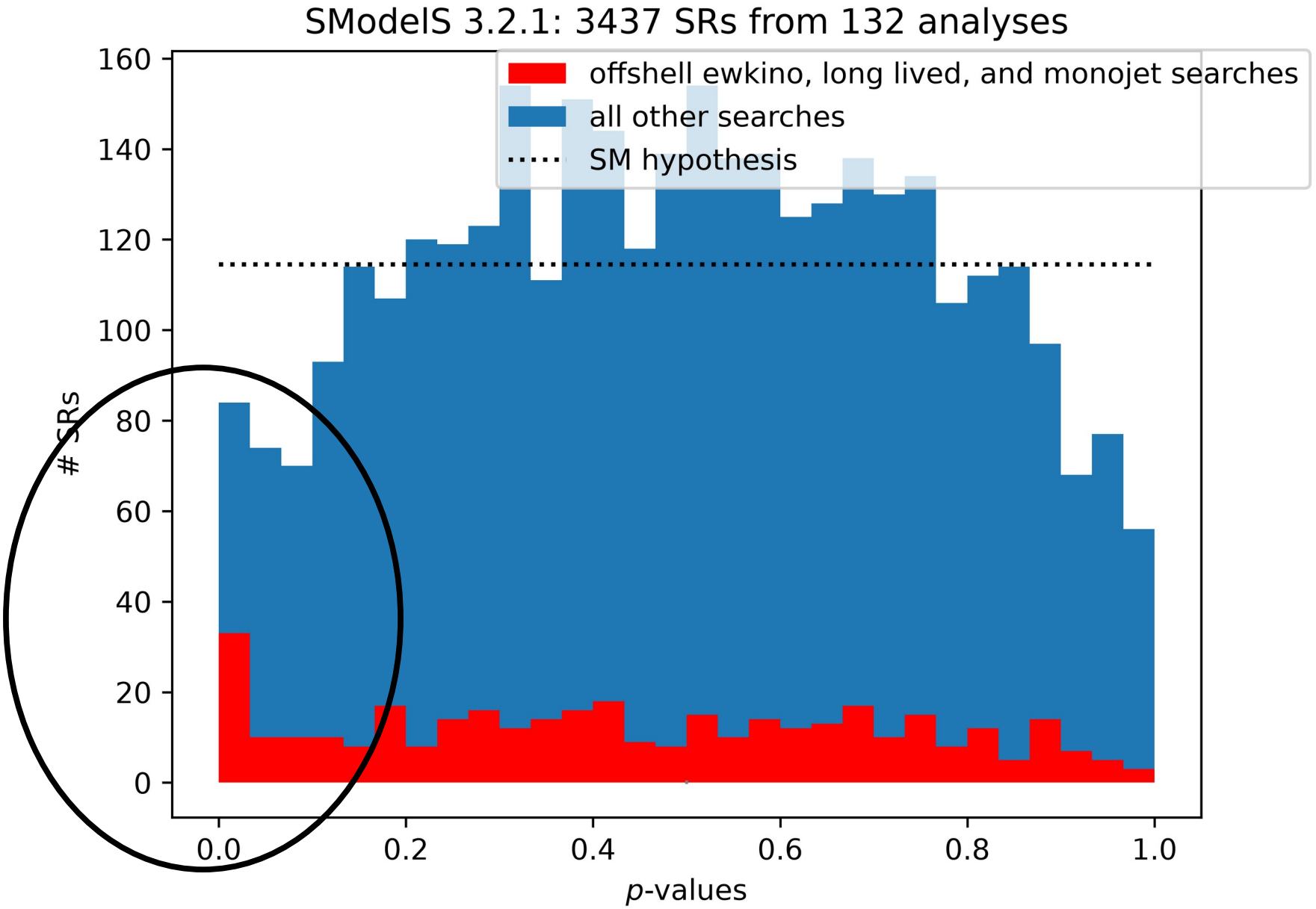
https://www.science.org/doi/10.1126/science.aac4716



https://medianwatch.netlify.app/post/z_values/

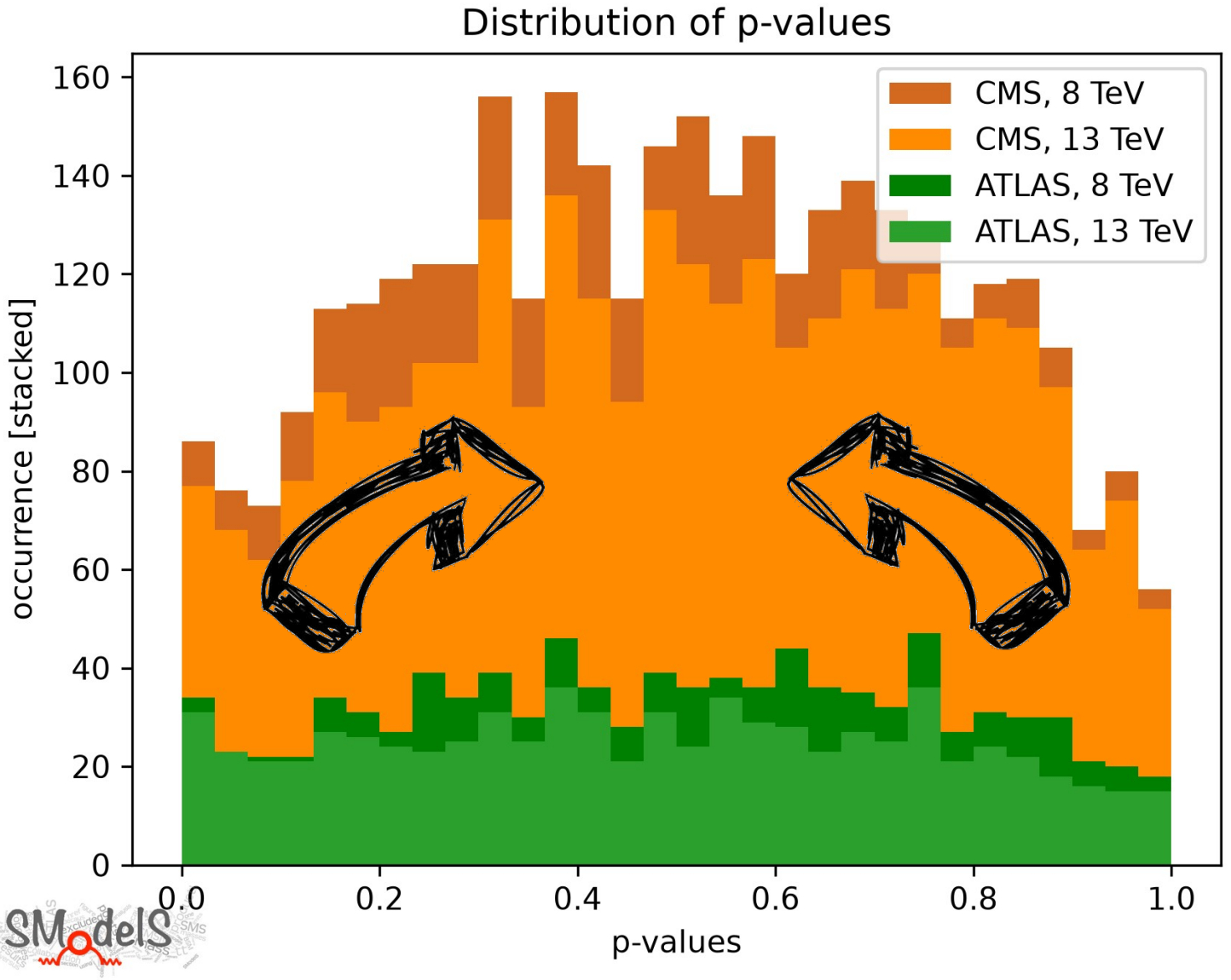
No offense meant, medical researchers, we are very happy you are doing all the hard research)

We have our own problems.
We do however not have these problems.



Similar information but p-values instead of significances. New physics would manifest at p ~ 0, but only in some classes of results. If observation == expectation, then p ~ 0.5, underfluctuations are at p ~ 1. Red: offshell electroweakino, long-lived, dark matter searches only.

Conservativeness effect



Systematic errors are systematically overestimated (as they should!): p-values tend towards 0.5

Conservativeness: CMS versus ATLAS

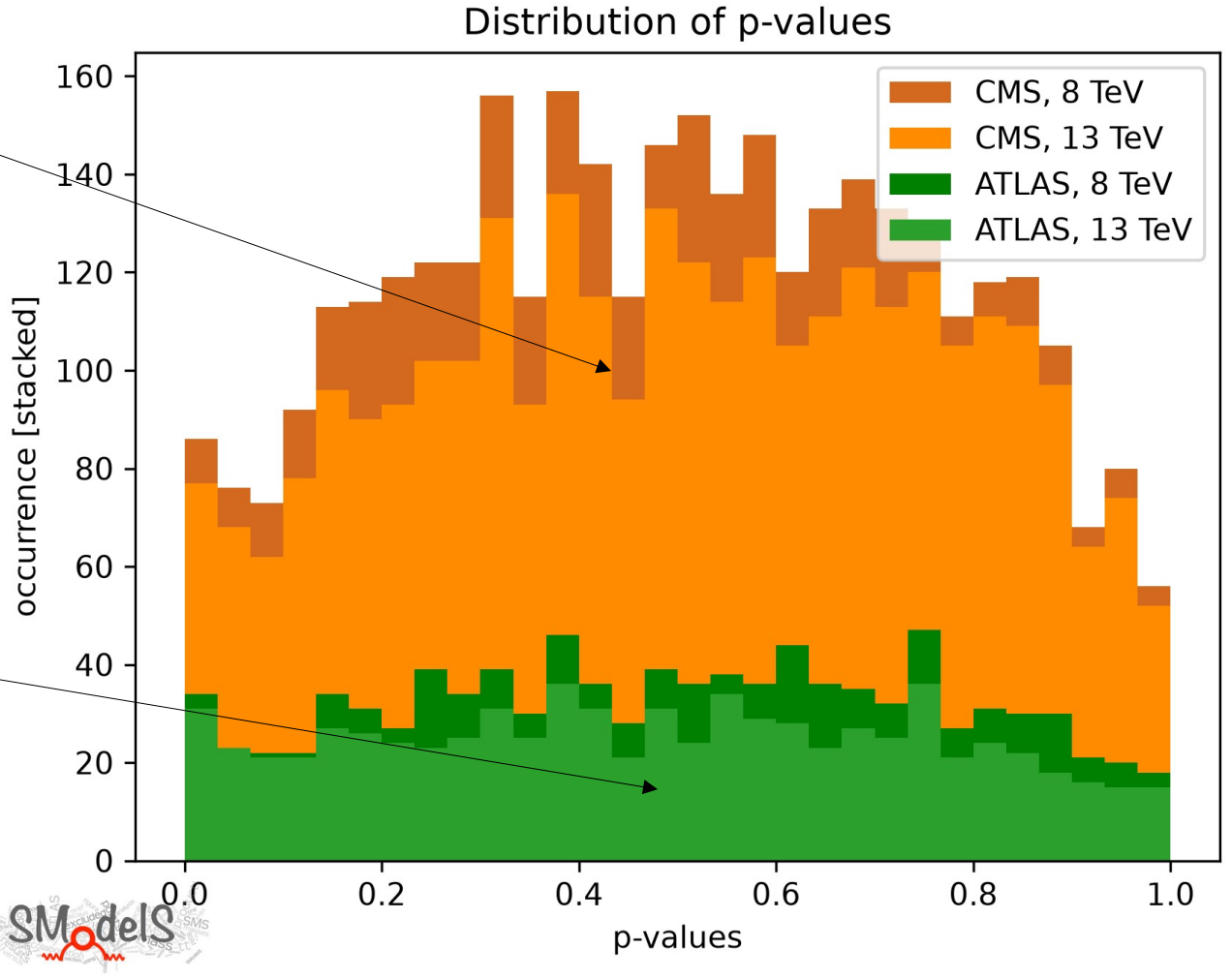
Systematic errors are conservatively estimated

CMS:

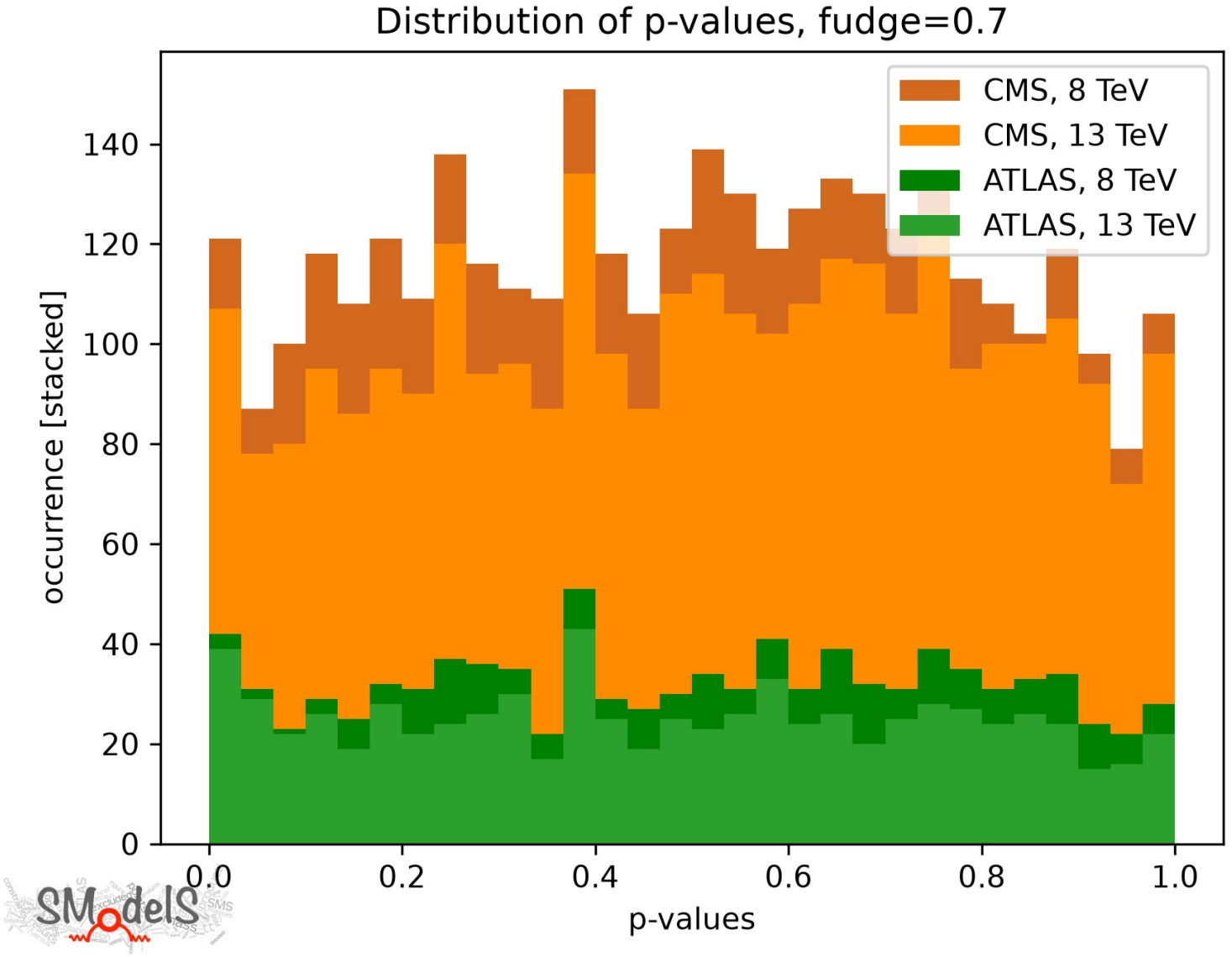
more fine grained SRs
→ more regions
→ more systematics dominated
→ more sensitive to conservativeness effect

ATLAS:

More inclusive SRs
→ fewer regions
→ more statistics dominated
→ less affected by conservativness effect

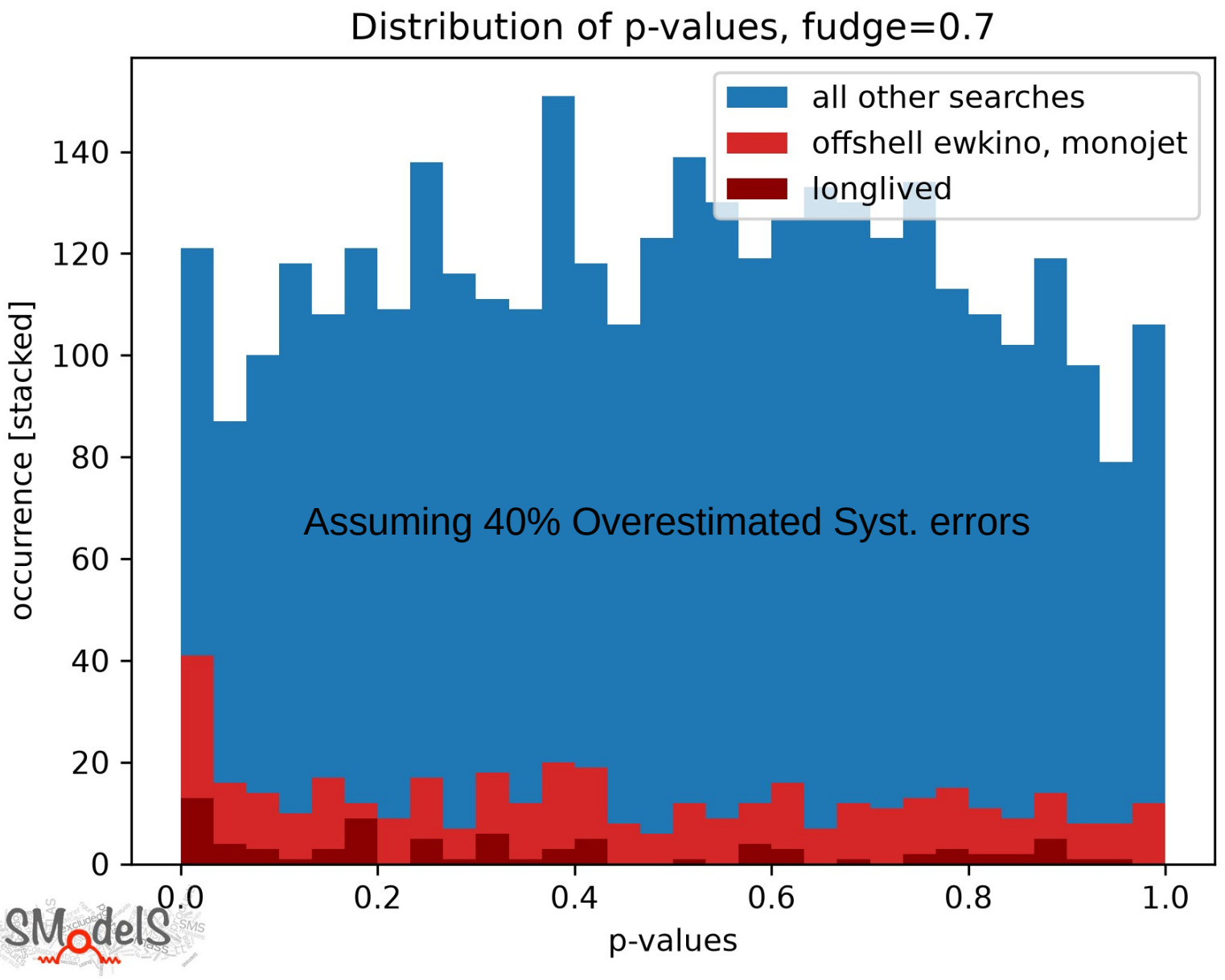


Correcting for conservativeness



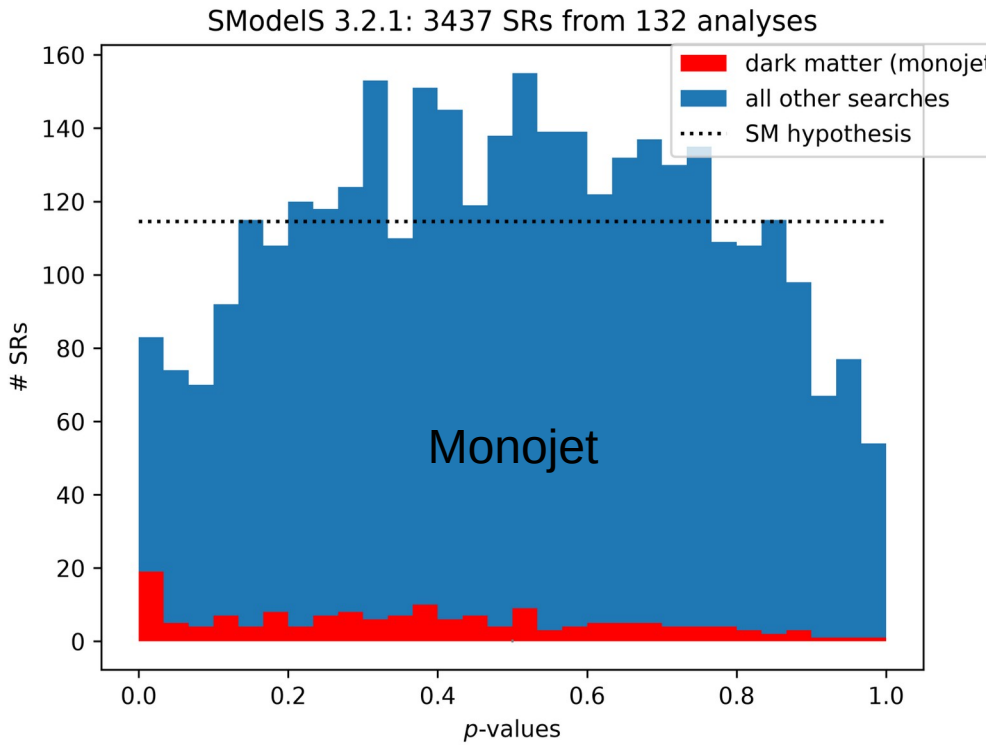
Assuming that all systematic errors are roughly 40% overestimated – and “globally” correcting for it -- results in an approximately flat distribution of p-values.

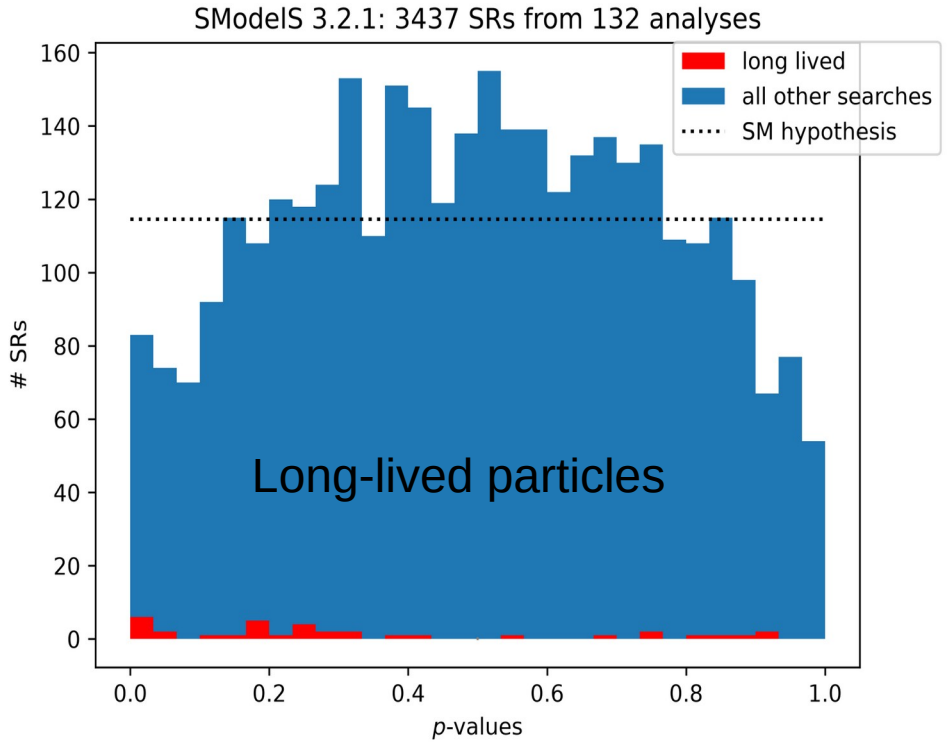
Looking great, all explained?

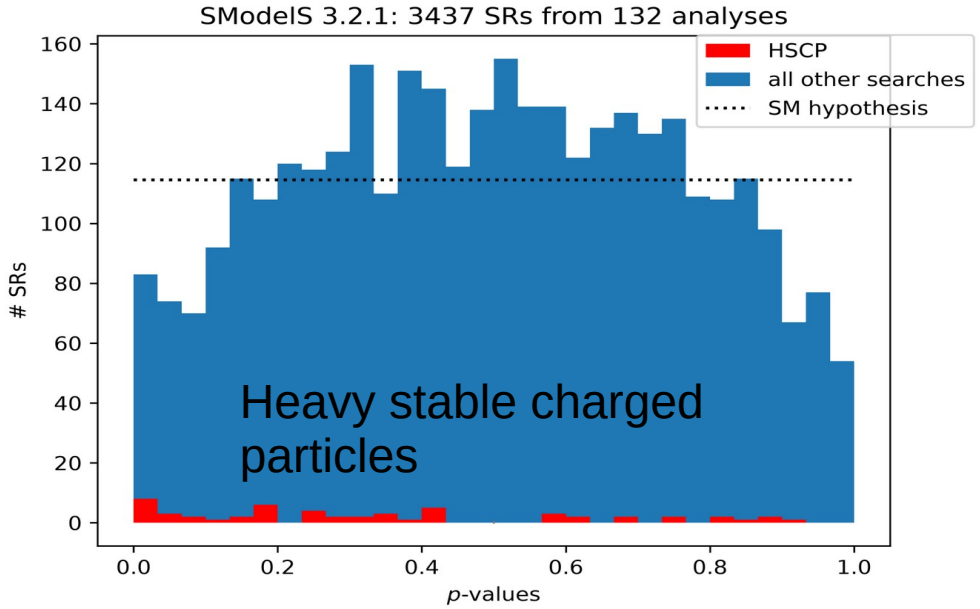


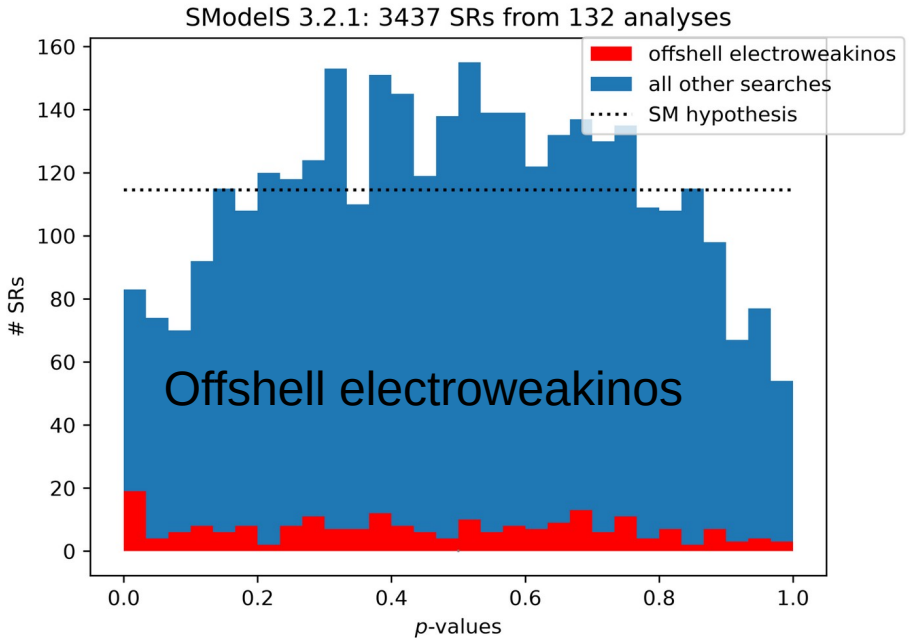
Red: “searches for dark matter and compressed scenarios”

More detailed decomposition (and this time NOT correcting for conservatism)

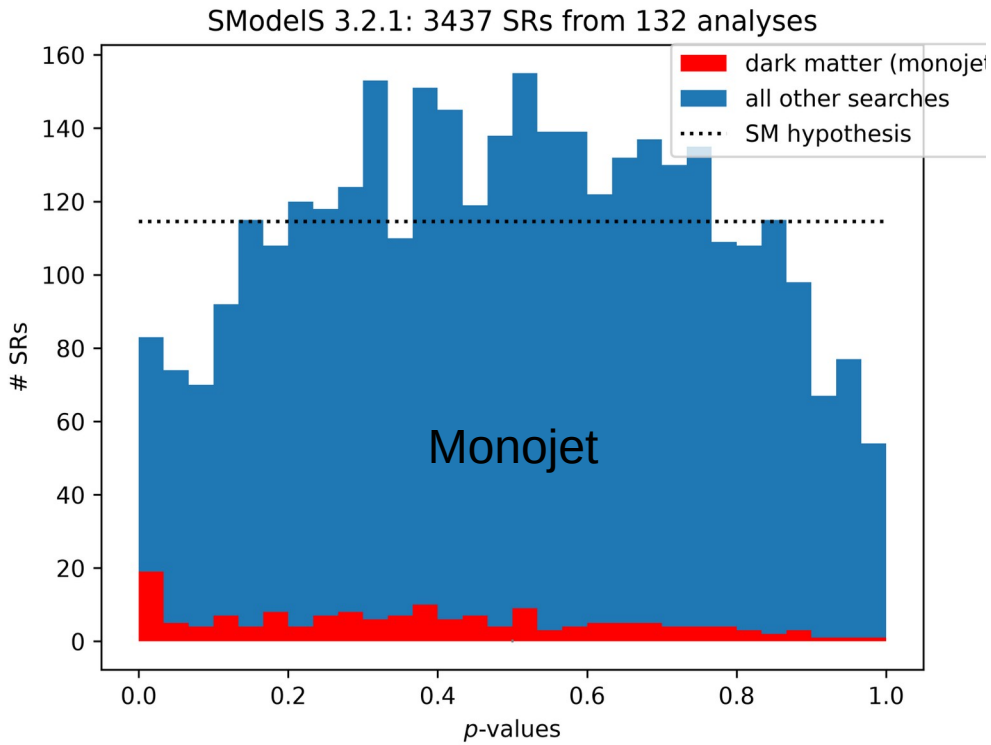


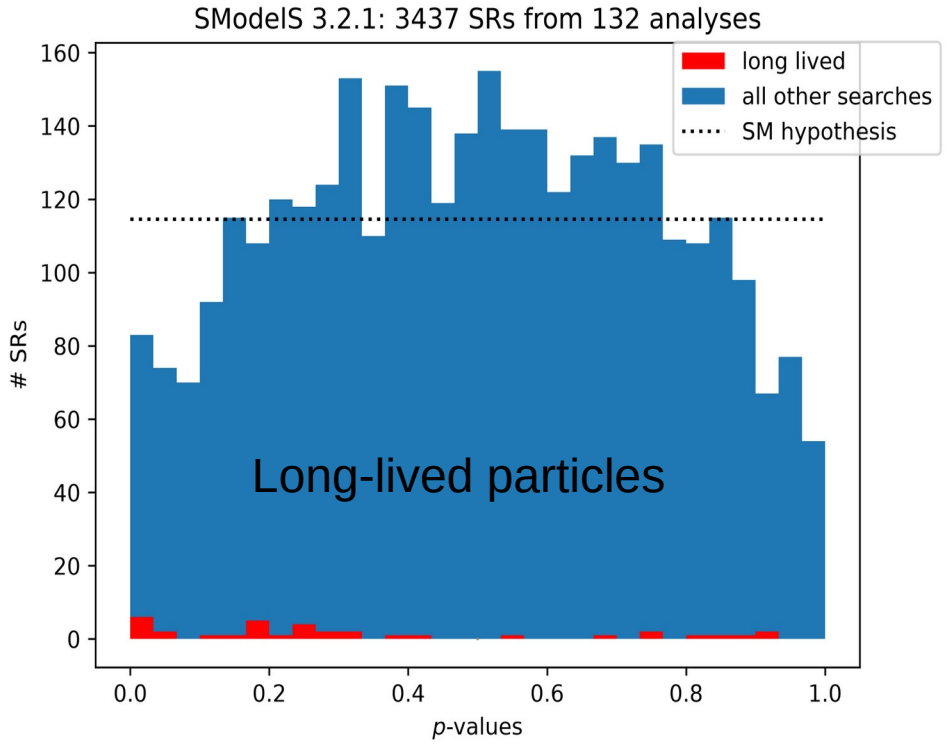


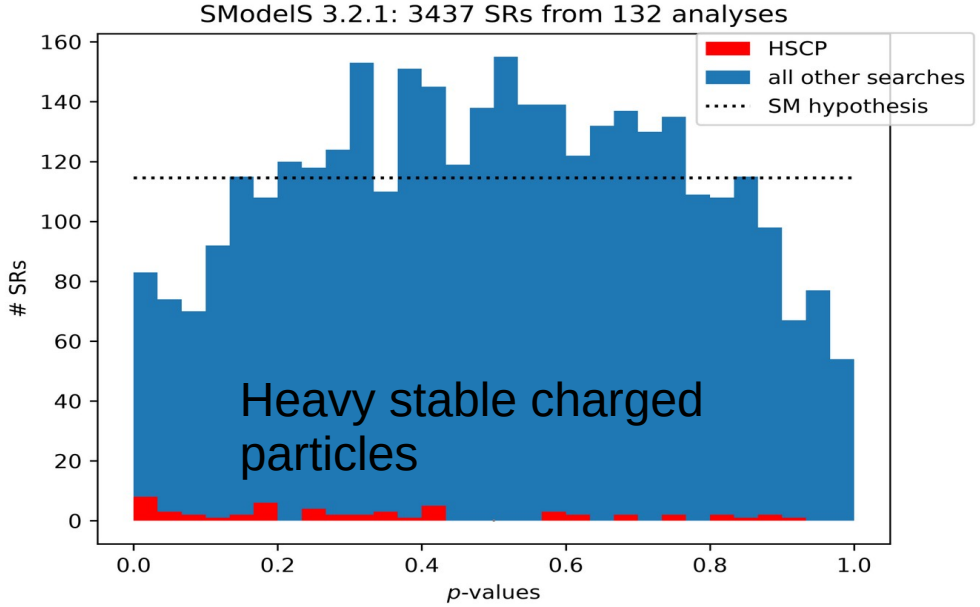




More detailed decomposition (and this time NOT correcting for conservatism)







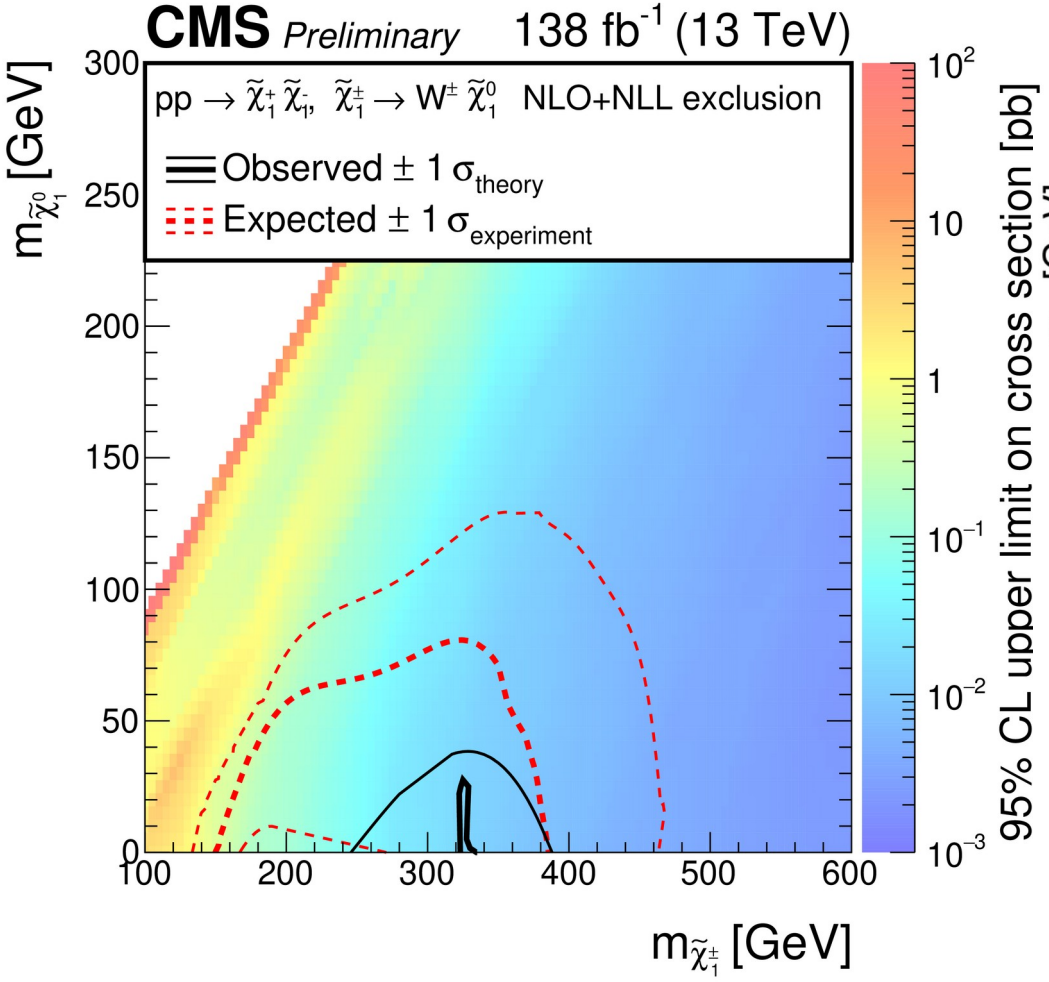
These three categories nicely fit
A minimal compressed scenario

So the meta-statistics of almost all classes of results seem to look as expected (congratulations, LHC community!) except for dark matter searches and things with small mass gap from dark matter.

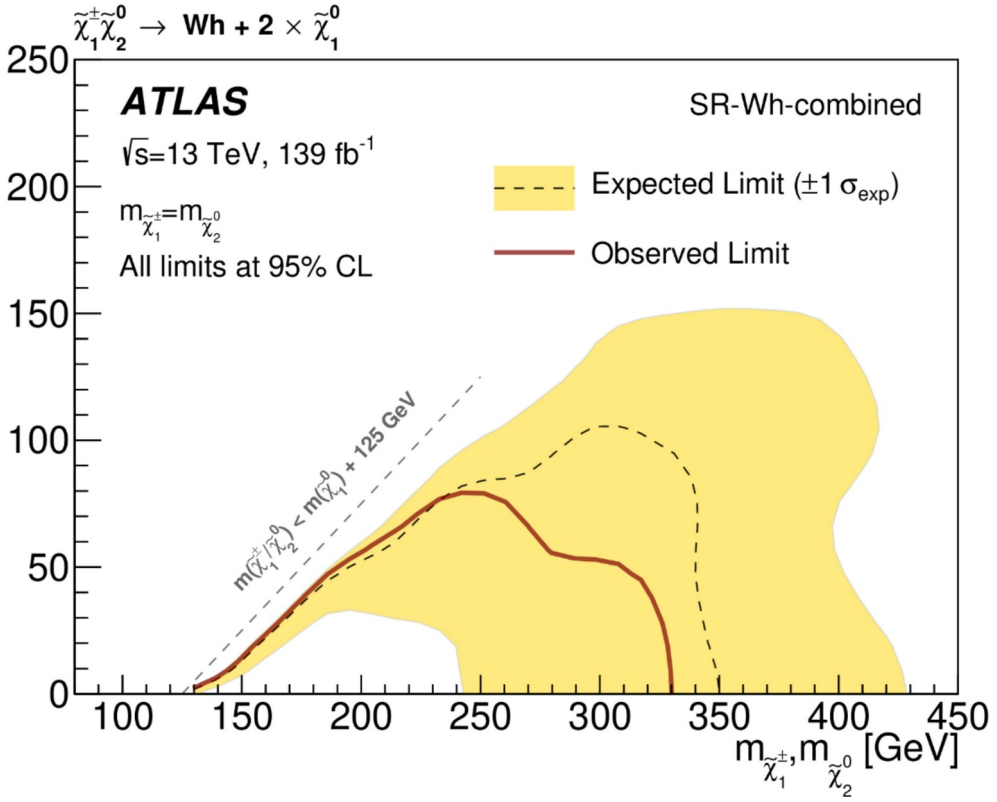
Results of about 50% of electroweakino/dark matter searches are not In the database (either not yet, or technical obstacles).

So what about missing, or more recent mono-X and electroweakino searches?

Results Not (yet) in SModelS

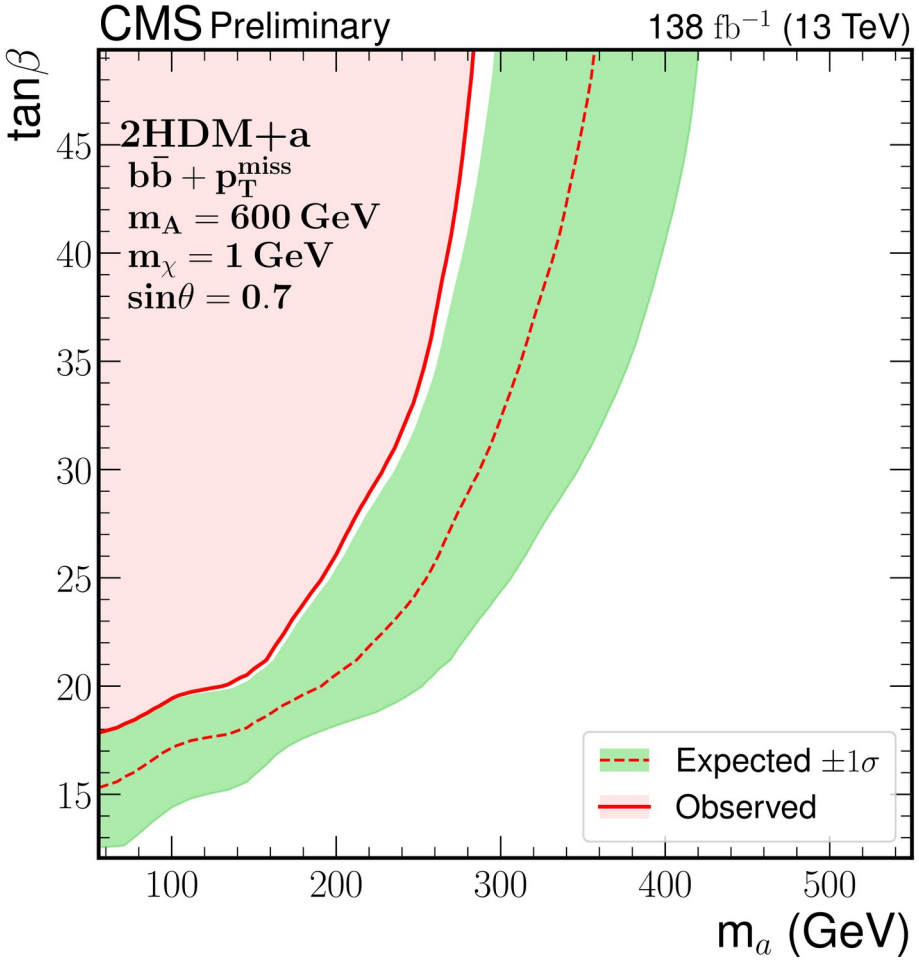


CMS-SUS-23-002, chargino pair production

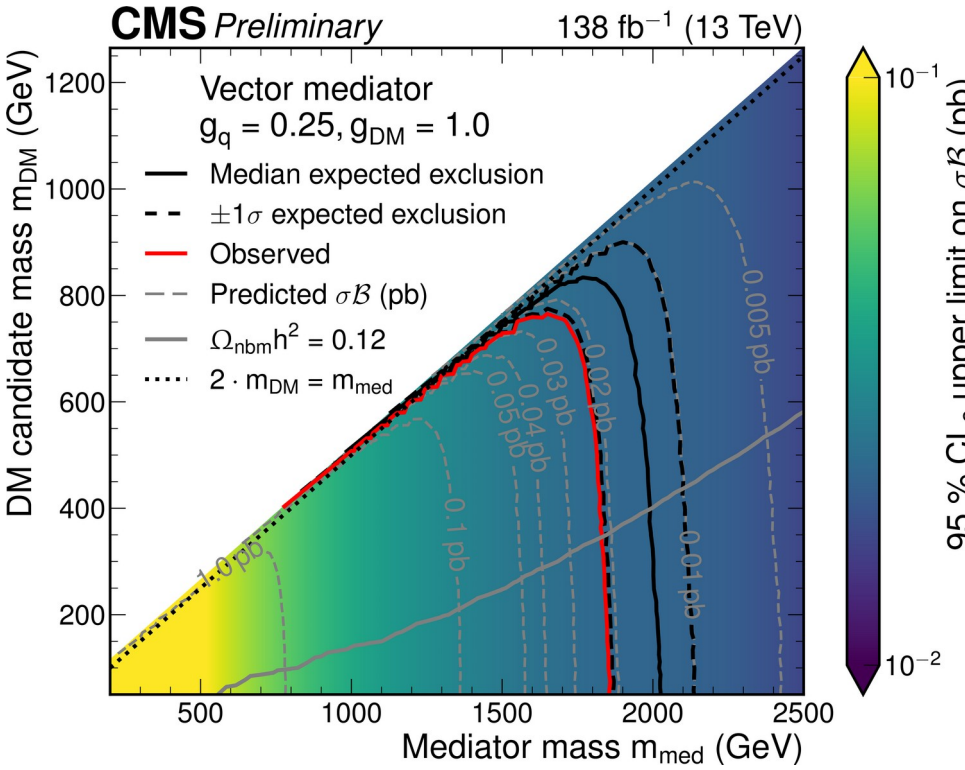


ATLAS-SUSY-2023-03,
Charginos + neutralinos

Results Not (yet) in SModelS

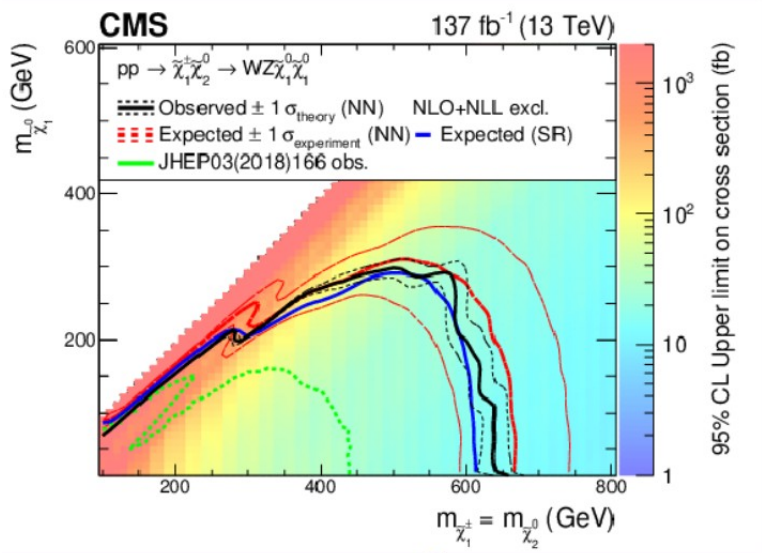


CMS-SUS-23-008, MET+bb

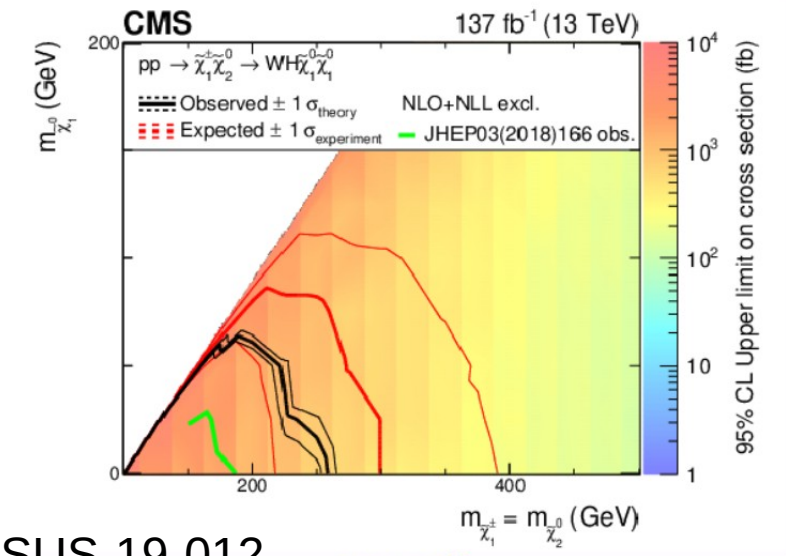


CMS-SUS-23-004, mono-top

Results Not (yet) in SModelS

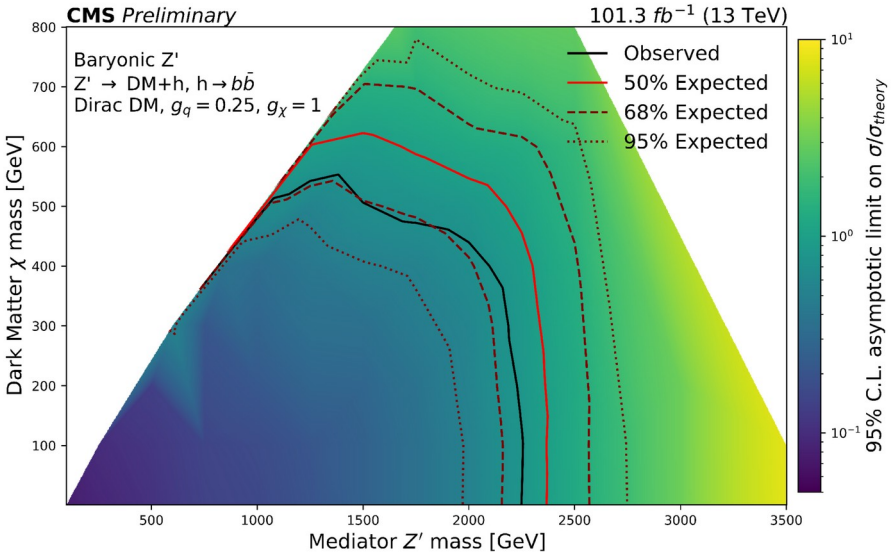


png pdf



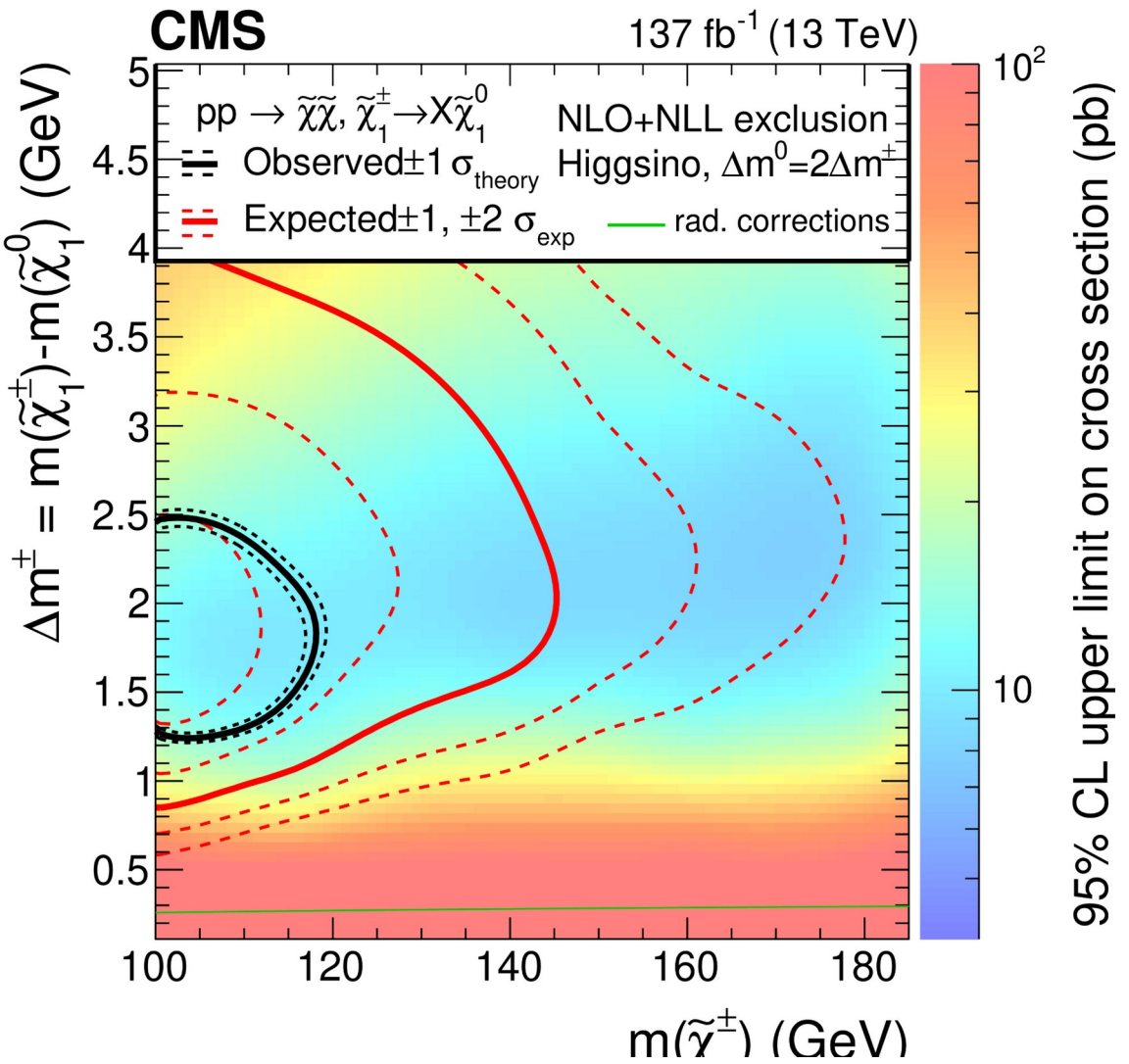
CMS-SUS-19-012, ewkinos

png pdf



CMS-SUS-24-007 MET + higgs

Results Not (yet) in SModelS



CMS-SUS-24-003

So overall:

too many small excesses for these classes of searches, while perfect matches for all other classes

Whither now?

Where do we take things from here?

We find it imperative for experimentalists to be radically conservative:

*) how can we argue this away?

e.g.: Can it be a problem with our MET? (But then it is odd that things are fine as soon as we add colored production)

e.g.: Can it be a problem with our ISR jets? (But then why do the electroweakino excesses nicely fit the dark matter excecsses)

e.g.: Can it be a problem with parton distribution functions?

*) can we really draw a consistent picture or are there inconsistencies between all these excesses?

*) and in case we cannot “argue this away”, let’s make sure all tools, all infrastructure that is needed for evidence/discovery of new physics, is in fact there

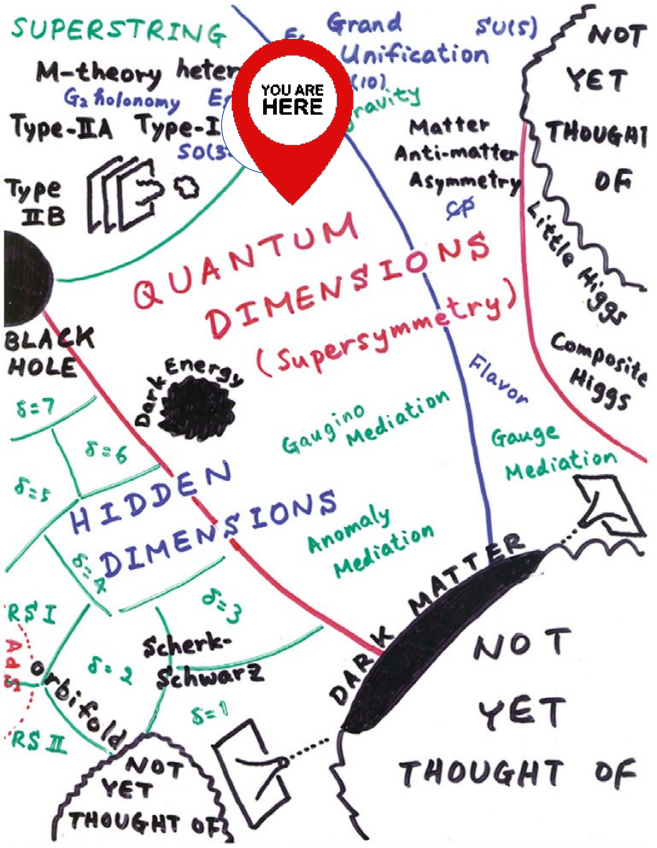
It is crucial for theorists to be conservatively radical:

*) can we really draw a consistent picture or are there inconsistencies between all these excesses?

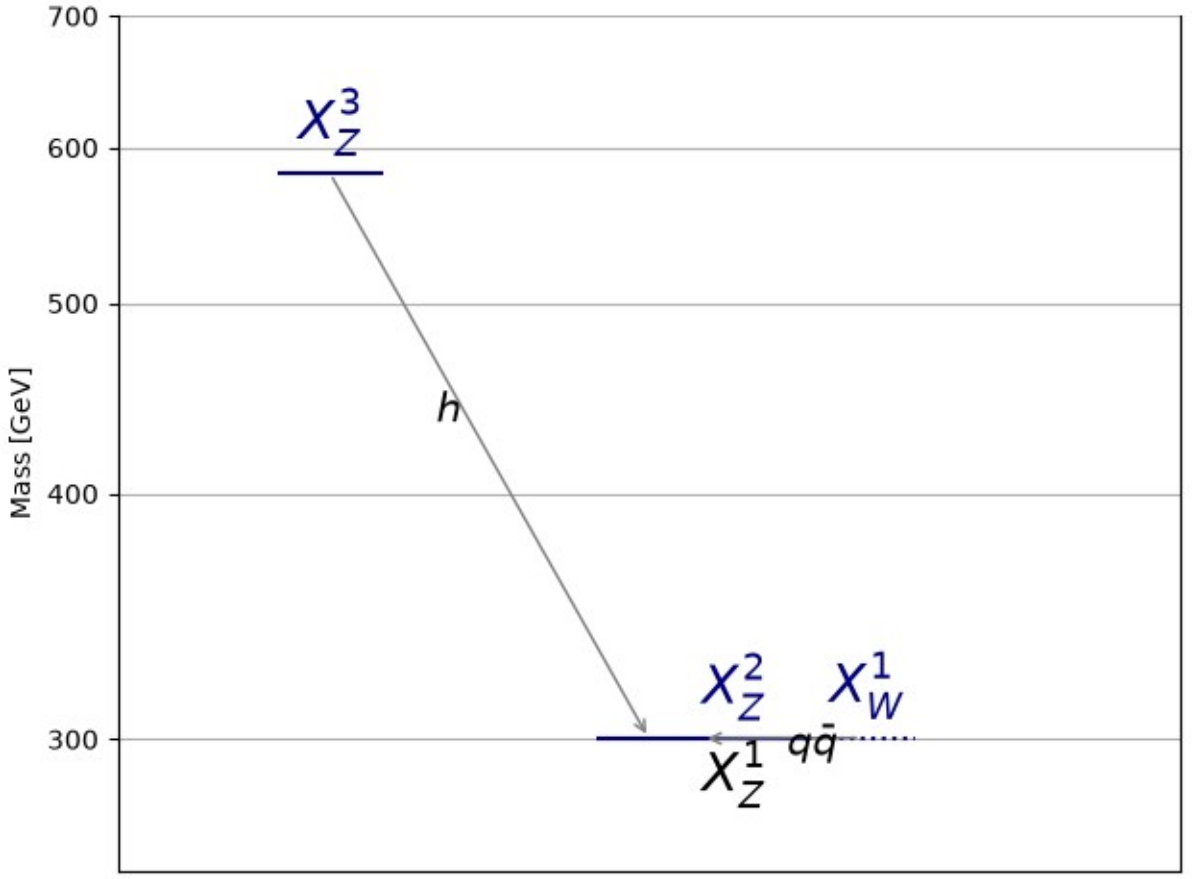
*) if it really is due to new physics,
what's the big picture,
what do we infer globally?

*) how would we construct a
Next Standard Model from
such experimental evidence?

A situation very much unlike that of the Higgs discovery!



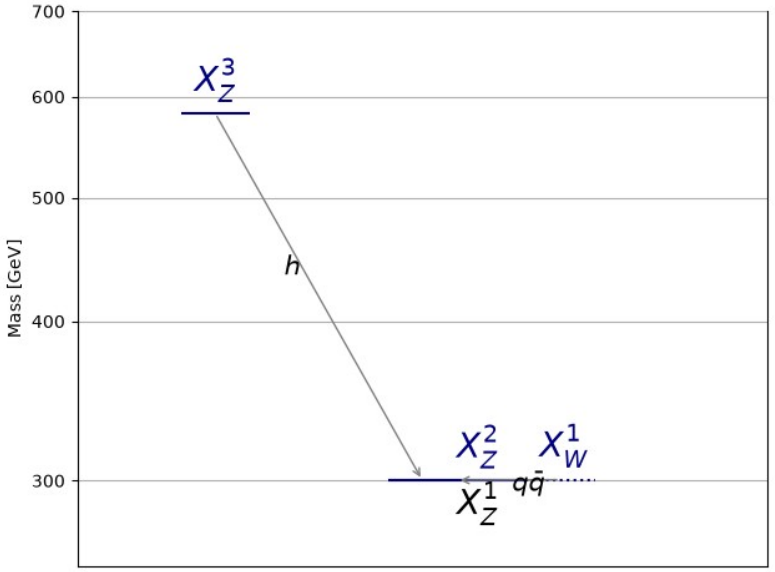
Here is a simple prototype model, a “protomodel”



X1Z: neutralino-like particle
X1W: chargino-like particle

Why do we consider this model to be intersting?

Why is this “protomodel” interesting?



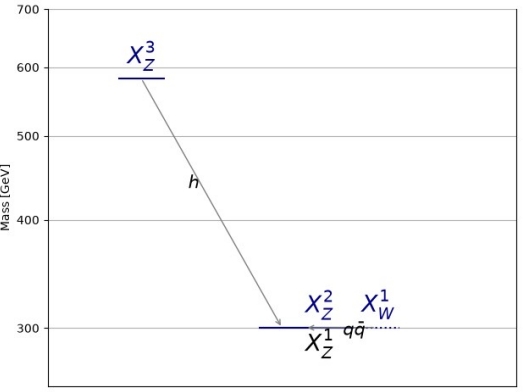
*) X1Z, X1W, X2Z: compressed scenario → inelastic DM? (compatible with LZ?)

*) Compatible with our negative results of 130+ publications

*) Contextualizes many of our biggest excesses

Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
ATLAS-EXOT-2018-06	em	EM10	TChiISR	413	359.00 +/- 10.00	2.4 σ	X^2_Z, X^1_Z
CMS-EXO-20-004	comb	-	TChiISR	4e+01 fb	3e+01 fb	2.2 σ	X^2_Z, X^1_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.2 σ	X^3_Z, X^2_Z
CMS-EXO-19-010	em	SR_nlay5	TDTM2F	6	3.70 +/- 1.24	0.8 σ	X^1_W, X^2_Z, X^1_Z
ATLAS-SUSY-2018-42	em	SR300_High_long	TDTM2F	7	5.10 +/- 0.50	0.7 σ	X^1_W, X^2_Z, X^1_Z
ATLAS-SUSY-2016-06	em	SR_EW	TDTM2F	9	11.80 +/- 3.10	-0.7 σ	X^1_W, X^2_Z, X^1_Z

Not in Best Combo	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
ATLAS-SUSY-2018-19	em	SR_EW	TDTM2F	3	3.00 +/- 0.70	-0.2 σ	X^1_W, X^2_Z, X^1_Z



Why is this "protomodel" interesting?

*) X1Z, X1W, X2Z: compressed scenario → inelastic DM? (compatible with LZ?)

*) Compatible with our negative results of 130+ publications

*) Contextualizes many of our biggest excesses

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CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.2 σ	X^3_Z, X^2_Z
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ATLAS-SUSY-2018-42	em	SR300_High_long	TDTM2F	7	5.10 +/- 0.50	0.7 σ	X^1_W, X^2_Z, X^1_Z
ATLAS-SUSY-2016-06	em	SR_EW	TDTM2F	9	11.80 +/- 3.10	-0.7 σ	X^1_W, X^2_Z, X^1_Z

Not in Best Combo	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
ATLAS-SUSY-2018-19	em	SR_EW	TDTM2F	3	3.00 +/- 0.70	-0.2 σ	X^1_W, X^2_Z, X^1_Z

monojet

N3 → h N1

disappearing track

Charged LLP

We consider the analyses to be approximately uncorrelated, their likelihoods seem mutually compatible. Combining them yields a model with (naively) a > 4.3*sigma deviation from the Standard Model (though with a HUGE look-elsewhere effect)

Recap: higgsinos!

In SUSY we have two higgsino doublets (b/c we need two higgs doublets b/c holomorphism of superpotential b/c anomaly cancellation)

$$\tilde{H}_u = \begin{pmatrix} \tilde{H}_u^+ \\ \tilde{H}_u^0 \end{pmatrix} \qquad \tilde{H}_d = \begin{pmatrix} \tilde{H}_d^0 \\ \tilde{H}_d^- \end{pmatrix}$$

Generally, the higgsinos mix with the Winos and Binos to form the neutralinos and charginos

- The **neutral components** $\tilde{H}_u^0$ and $\tilde{H}_d^0$ mix with the **bino** ($\tilde{B}$) and **wino** ($\tilde{W}^0$) to form four **neutralino** mass eigenstates $\tilde{\chi}_i^0$ ($i = 1, \ldots, 4$).
- The **charged components** $\tilde{H}_u^+$ and $\tilde{H}_d^-$ mix with the **charged winos** ($\tilde{W}^\pm$) to form two **chargino** mass eigenstates $\tilde{\chi}_j^\pm$ ($j = 1, 2$).
- They are coupled in the superpotential via the μ**-term**: $W \supset \mu\, H_u \cdot H_d$.

Let's now assume that the LSP ("N1") is an (almost) pure higgsino. In this case, N1, N2, C1 are approximately mass degenerate.

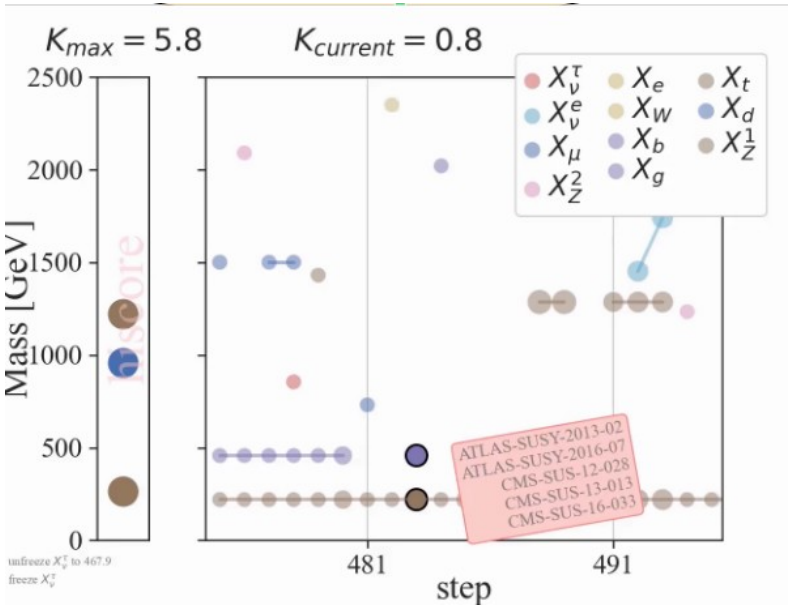
For very large $M_{1,2}$, **one-loop electroweak corrections** can dominate:

$$\Delta m_{\text{1-loop}} \sim \tfrac{\alpha_2}{4\pi}\, m_W \sim \mathcal{O}(\text{few hundred MeV})$$

- $\tilde{\chi}_1^\pm - \tilde{\chi}_1^0$ **splitting**: For very large $M_{1,2}$, the tree-level $\sim m_W^2/M$ vanishes, but **one-loop EW corrections** ($\sim \tfrac{\alpha}{4\pi} m_W \sim 300$ MeV) take over, because electromagnetism breaks the $SU(2)$ symmetry relating charged and neutral Higgsinos.
- $\tilde{\chi}_2^0 - \tilde{\chi}_1^0$ **splitting**: The tree-level $\sim m_Z^2/M$ contribution **always dominates**, because in the pure Higgsino limit the two neutralinos form a **Dirac pair** with a protecting symmetry. This symmetry is only broken by gaugino mixing — not by gauge interactions — so one-loop EW corrections **cannot** generate an M-independent splitting.

How did we obtain this model?

Protomodels



It's like an MCMC walk as used for e.g. Bayesian inference, except:

*) we allow the machine to add / remove particles, decay channels, production mechanisms (variable dimensionality)

*) the machine combines analysis results depending on a notion of "combinability" and "completeness"

*) we look at likelihood ratios – L(BSM)/L(SM) – not likelihoods

*) we thus look for that legal combination that **maximally violates The SM hypothesis**

*) we added the notion of "critics" that ensure we remain consisent with all negative results in our database

*) we think of production cross sections as free parameters

*) we published a prototype v1 in 2020:

https://arxiv.org/pdf/2012.12246

Artificial Proto-Modelling: Building Precursors of a Next Standard Model from Simplified Model Results

Wolfgang Waltenberger,[a,b,1] André Lessa,[c] Sabine Kraml[d]

[a] *Institut für Hochenergiephysik, Österreichische Akademie der Wissenschaften, Nikolsdorfer Gasse 18, A-1050 Wien, Austria*

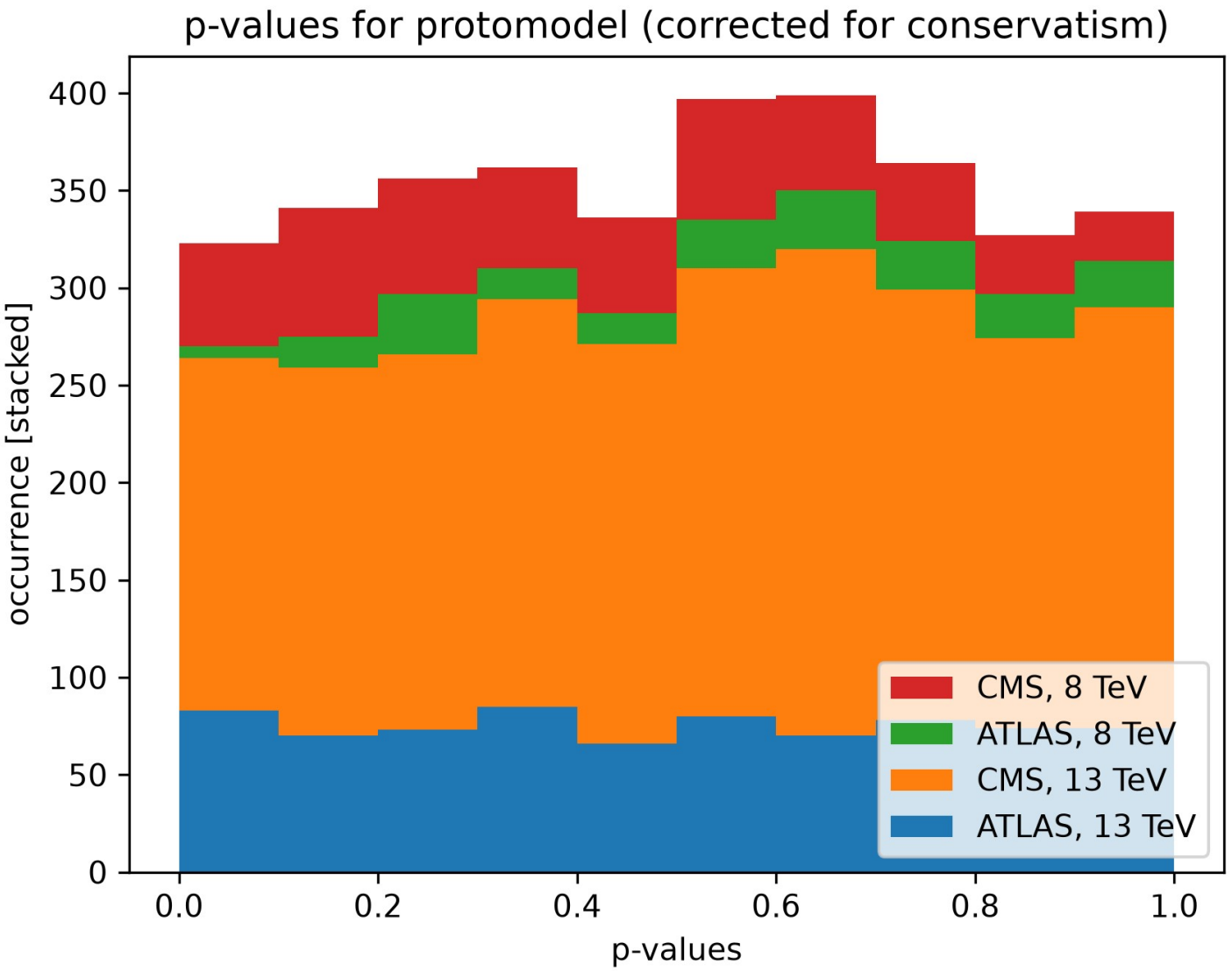
[b] *University of Vienna, Faculty of Physics, Boltzmanngasse 5, A-1090 Wien, Austria*

[c] *Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, Santo André, SP - Brasil*

[d] *Laboratoire de Physique Subatomique et de Cosmologie, Université Grenoble-Alpes, CNRS/IN2P3, 53 Avenue des Martyrs, F-38026 Grenoble, France*

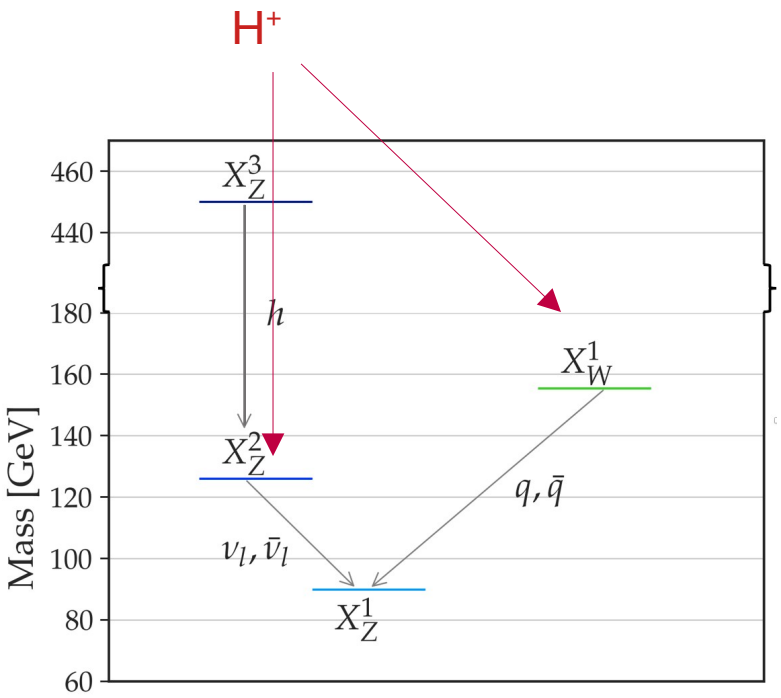
E-mail: wolfgang.waltenberger@oeaw.ac.at, andre.lessa@ufabc.edu.br, sabine.kraml@lpsc.in2p3.fr

p-values (BSM hypothesis, corrected for conservatism)



What do we with these protomodels?

Informing the next Searches

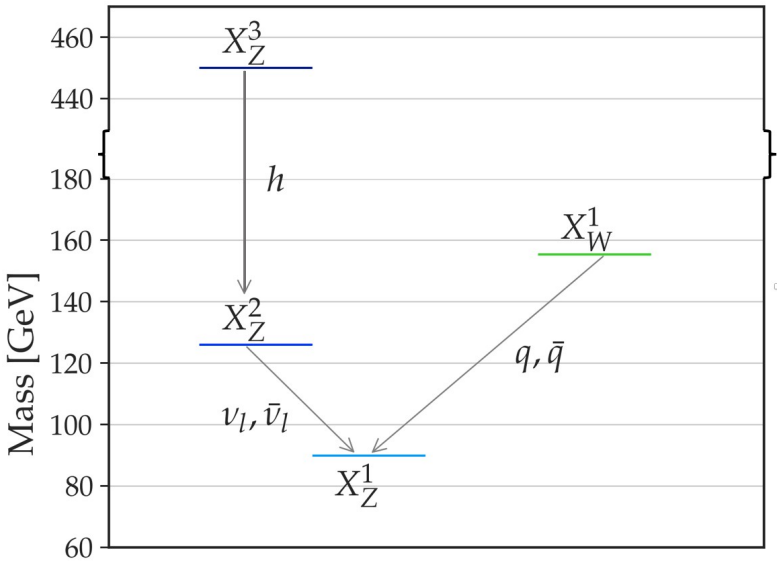


Example:
Say you are taking the protomodel to the left seriously. But also the excess In the charges Higgs search at $H^+ \sim 600$ GeV. An obvious question would then immediately be:

What about $H^+ \rightarrow X^1_W X^2_Z$?

(I havent yet thought carefully about such matters though. @Jonas?)

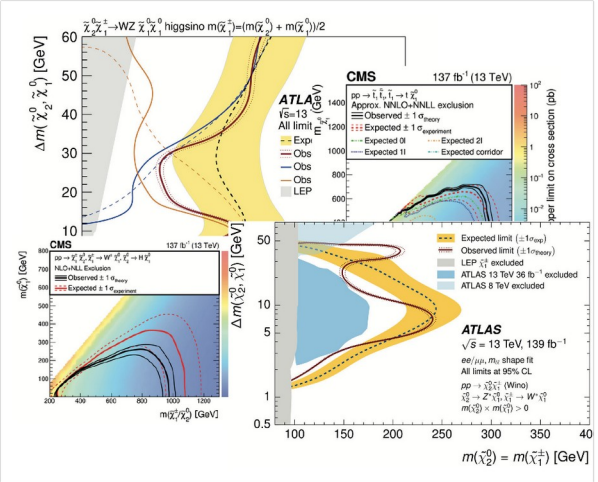
Informing the next Searches

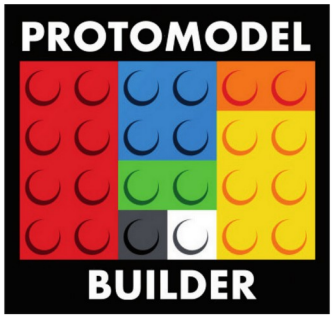


Differential diagnoses:
If e.g. a hadronic channel is open, search for the leptonic equivalent channel

What about $X^1_W \rightarrow l\ nu\ X^1_Z$?

LHC Results → Protomodels → UV Lagrangians





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$$\mathcal{L} = \mathcal{L}_{\rm SM} + \mathcal{L}_{\rm BSM}$$

$$\mathcal{L}' = \mathcal{L}_{\rm SM} + \mathcal{L}'_{\rm BSM}$$

$$\mathcal{L}'' = \mathcal{L}_{\rm SM} + \mathcal{L}''_{\rm BSM}$$

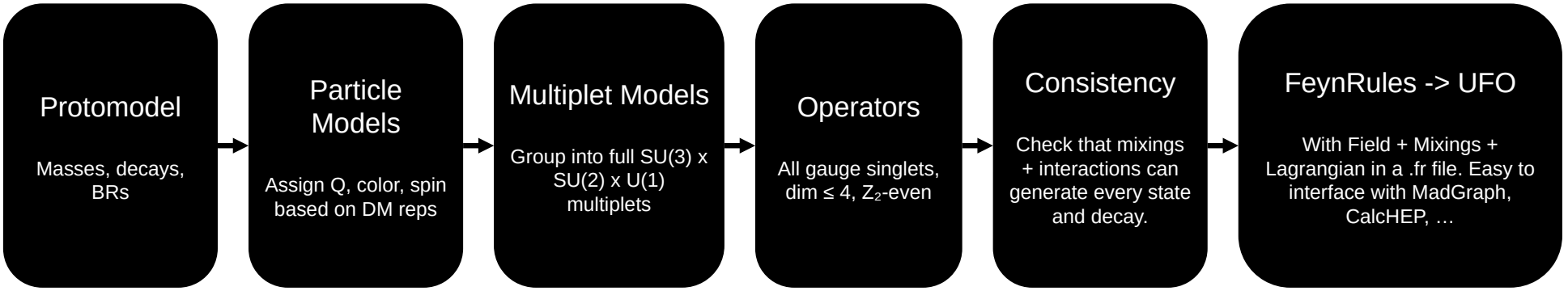
$$\vdots$$

- Goal: Automatically build minimal, UV-consistent, gauge-invariant Lagrangians from a given Protomodel

(Meant as a path in addition to more well-defined UV-completions, not instead)

Lucas Heck, Andre Lessa

Procedure / Closure



EXAMPLE

Particle	Mass (GeV)
Z1 (DM)	109
Z2	290
Z3	431
W1	200

With W, Z/h decays

With Majorana DM (input)

3 Majorana + 1 Dirac = 5 Weyl fields

3 possibilities:

$\{\mathbf{1}_0, \mathbf{1}_0, \mathbf{1}_0, \mathbf{1}_{+1}, \mathbf{1}_{-1}\}$,

$\{\mathbf{2}_{+1/2}, \mathbf{2}_{-1/2}, \mathbf{1}_0\}$,

$\{\mathbf{3}_0, \mathbf{1}_0, \mathbf{1}_0\}$

For the doublet:

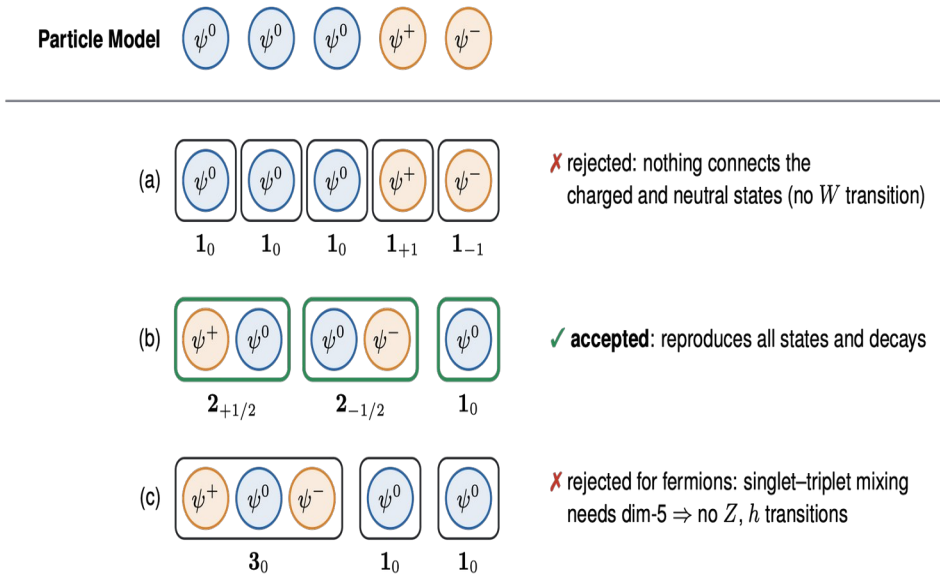
$SS, \quad D_d \cdot D_u,$

$S(D_d \cdot H), \quad S(H^\dagger D_u)$

Only $\{\mathbf{2}_{+1/2}, \mathbf{2}_{-1/2}, \mathbf{1}_0\}$ Survives

.fr -> UFO

With hand-fitted parameters, reproduces the input masses and BRs



Derived Lagrangian (4 parameters)

$$\mathcal{L} \supset -\tfrac{1}{2} M_S \, SS \; - \; M_{ud} \, D_d \cdot D_u \; - \; y_d \, S(D_d \cdot H) \; + \; y_u \, S(H^\dagger D_u) \; + \; \text{h.c.}$$



Majorana DM (user choice)

$\{\mathbf{2}_{+1/2}, \mathbf{2}_{-1/2}, \mathbf{1}_0\}$

Only the input model survives; with fitted parameters the UFO reproduces the (fake) protomodel

Lucas Heck, Andre Lessa

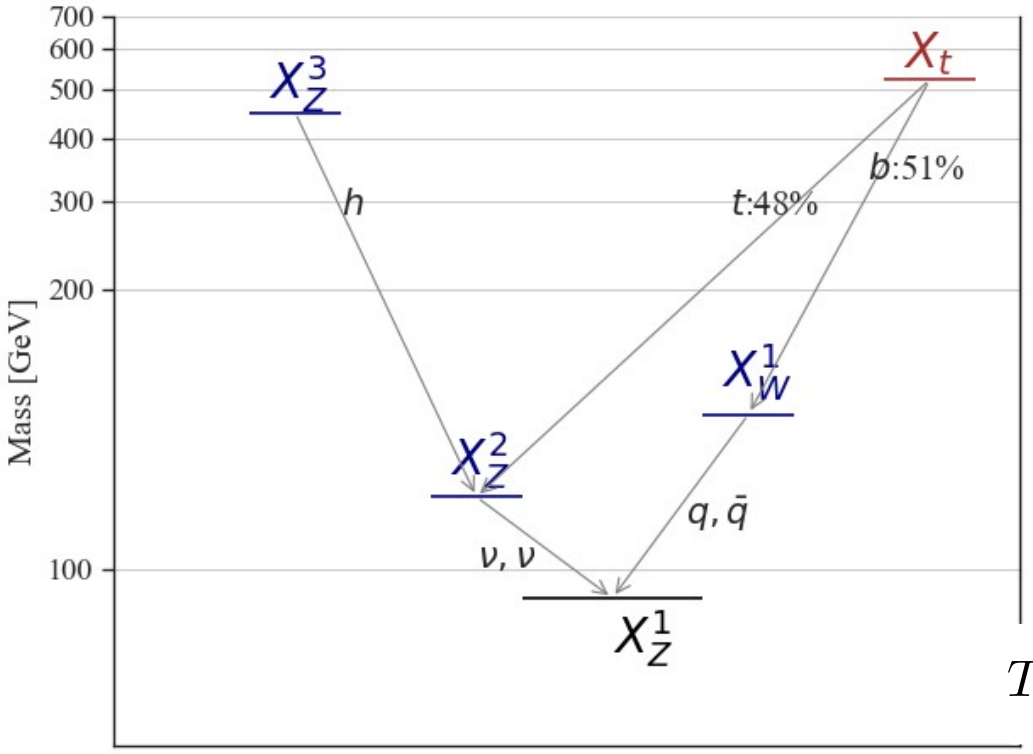
Want to collaborate?

TODO

- Implement more analyses in our database
- machine learn more surrogate statistical models to speed up evaluation
- work on hs3/combine support in our software
- work on code that can simplify statistical models, to speed things up
- combinations within the MSSM, NMSSM, etc
- introduce gradients in our inference chain
- extend the space of theories we can cover
- Use protomodels to proposed targeted, refined LHC searches
- Recast older analyses, recast for new simplified models
- Compare more with e.g. dark matter relic density, LZ, XENONnT, flavor physics
- Connect with LHC measurements, EFTS
- Protomodels but full-scale combinations (possibly within the experimental collaborations?)
- Run interesting models with full recasts

- Identification of compatible UV complete theories

Thank you



Here is another such protomodel

$$T_L := 2 \ln \left(\frac{L^c_{\mathrm{BSM}}(\hat{\mu}) \cdot \pi_{\mathrm{BSM}}}{L^c_{\mathrm{SM}} \cdot \pi_{\mathrm{SM}}} \right) = 39.06 = 6.25^2$$

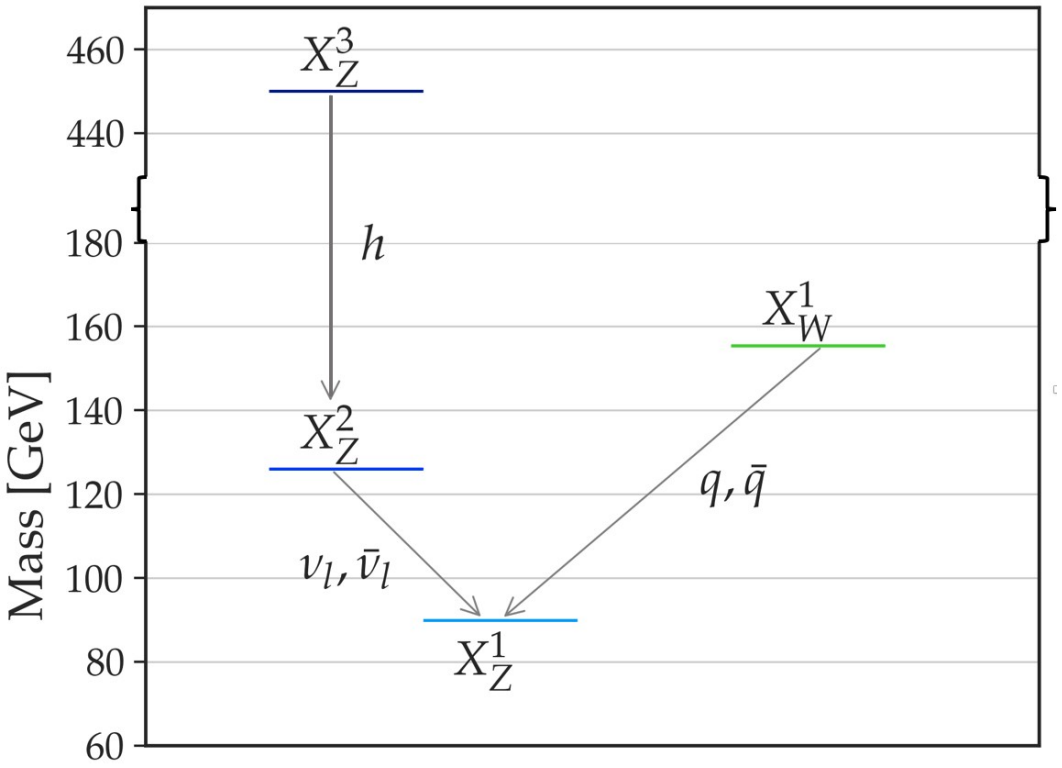
Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-SUSY-2018-12	em	SRBTT	T2tt	67	46.00 +/- 7.00	2.1 σ	X_t, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.0 σ	X^3_Z, X^2_Z
ATLAS-SUSY-2018-08	em	SRSF140	T2tt	9	5.10 +/- 0.90	1.5 σ	X_t, X^2_Z
ATLAS-SUSY-2016-16	em	tN_med	T2tt	50	36.30 +/- 6.60	1.5 σ	X_t, X^2_Z
ATLAS-SUSY-2013-16	em	SRA4	T2tt	4	2.40 +/- 0.70	0.9 σ	X_t, X^2_Z
ATLAS-SUSY-2013-15	em	tNboost	T2tt	5	3.30 +/- 0.70	0.9 σ	X_t, X^2_Z
CMS-PAS-SUS-13-015	em	pTmiss350_Nb2	T2tt	15	8.60 +/- 7.10	0.8 σ	X_t, X^2_Z
CMS-SUS-16-050-agg	comb	-	T2tt	2 fb	2 fb	-0.2 σ	X_t, X^2_Z
CMS-SUS-13-011	em	LM150	T2tt	227	251.00 +/- 50.00	-0.5 σ	X_t, X^2_Z

Let’s try and find a simple prototype BSM theory that explains the excessive number of excesses!

Apart from it being interesting in its own right, such an attempt may help us also answering the radically conservatives’ questions of where the problem might lie (if there is a problem)

Here is such a prototype model, a “protomodel”

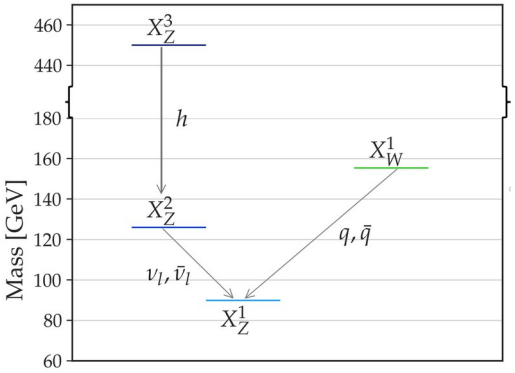
(We introduced this notion of a “protomodel”. It’s an incomplete BSM model, driven by data, not by theory considerations. LHC production cross sections are free parameters, BSM particle spins are left unspecified)



X^i_Z = “neutralino-like”

X^i_W = “chargino-like”

Why do we consider this to be an interesting protomodel?



Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

*) It contextualizes three of the biggest excesses we are seeing in the databases, that are considered mutually independent.

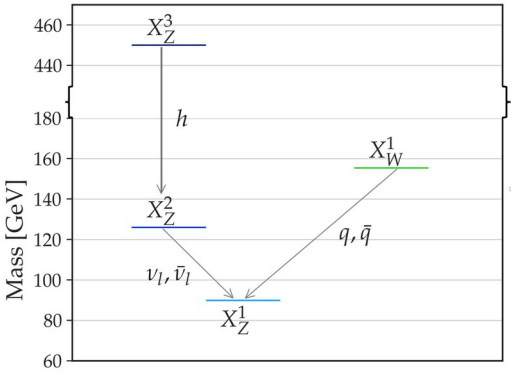
*) we can thus approximately combine the likelihoods of these three analyses – by multiplying them.

*) Another result exists, but is correlated, so we cannot add it in that fashion (but it would also exhibit an excess)

$$T_L := 2 \ln \left(\frac{L^c_{\mathrm{BSM}}(\hat{\mu}) \cdot \pi_{\mathrm{BSM}}}{L^c_{\mathrm{SM}} \cdot \pi_{\mathrm{SM}}} \right) = 33.69 = 5.8^2$$

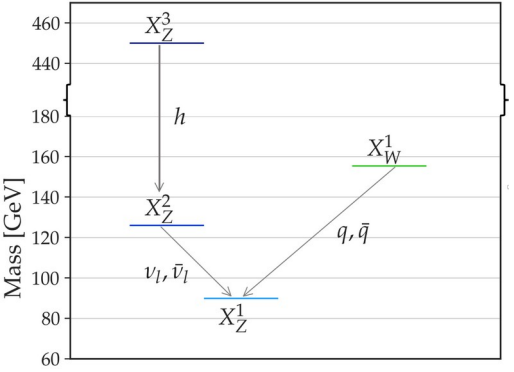
"local 5.8 sigma" (if ndf=1) – with a GIGANTIC look elsewhere effect!

But let's focus on the simpler protomodel



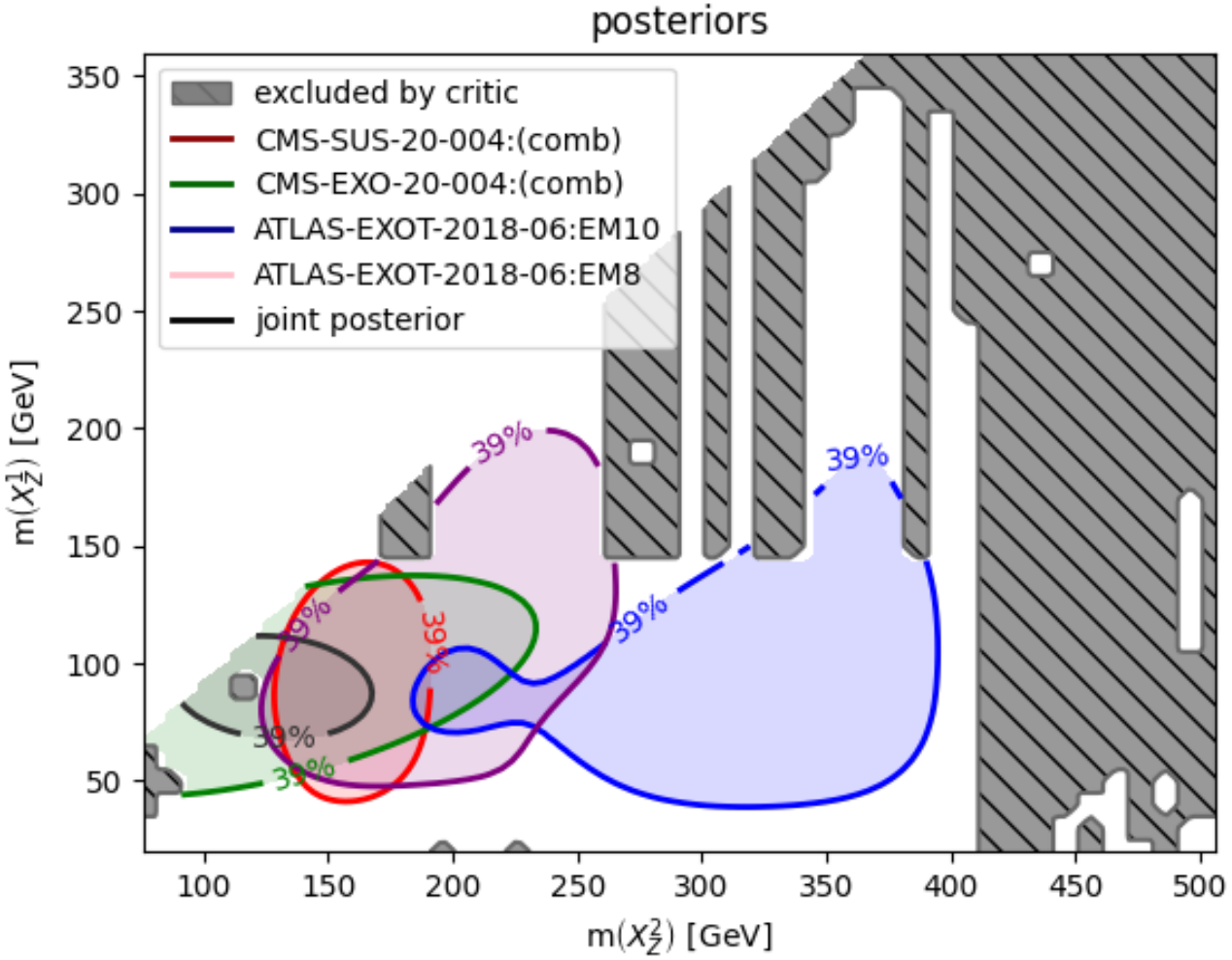
Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Do we consider these excesses to be mutually compatible?

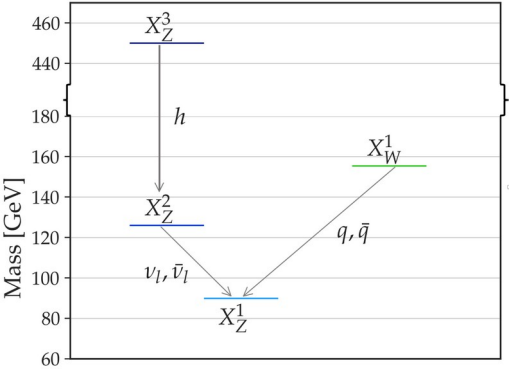


Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Combining Likelihoods

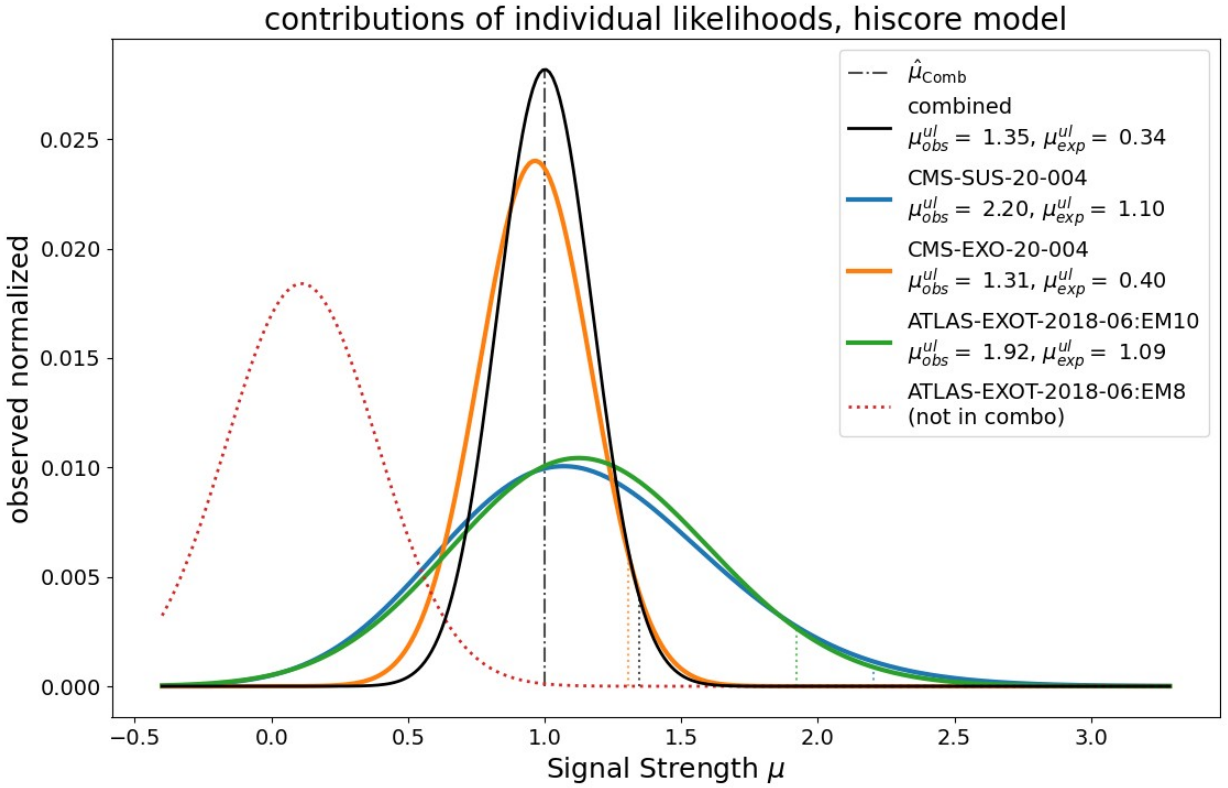


Likelihoods as functions of particle masses



Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Combining Likelihoods



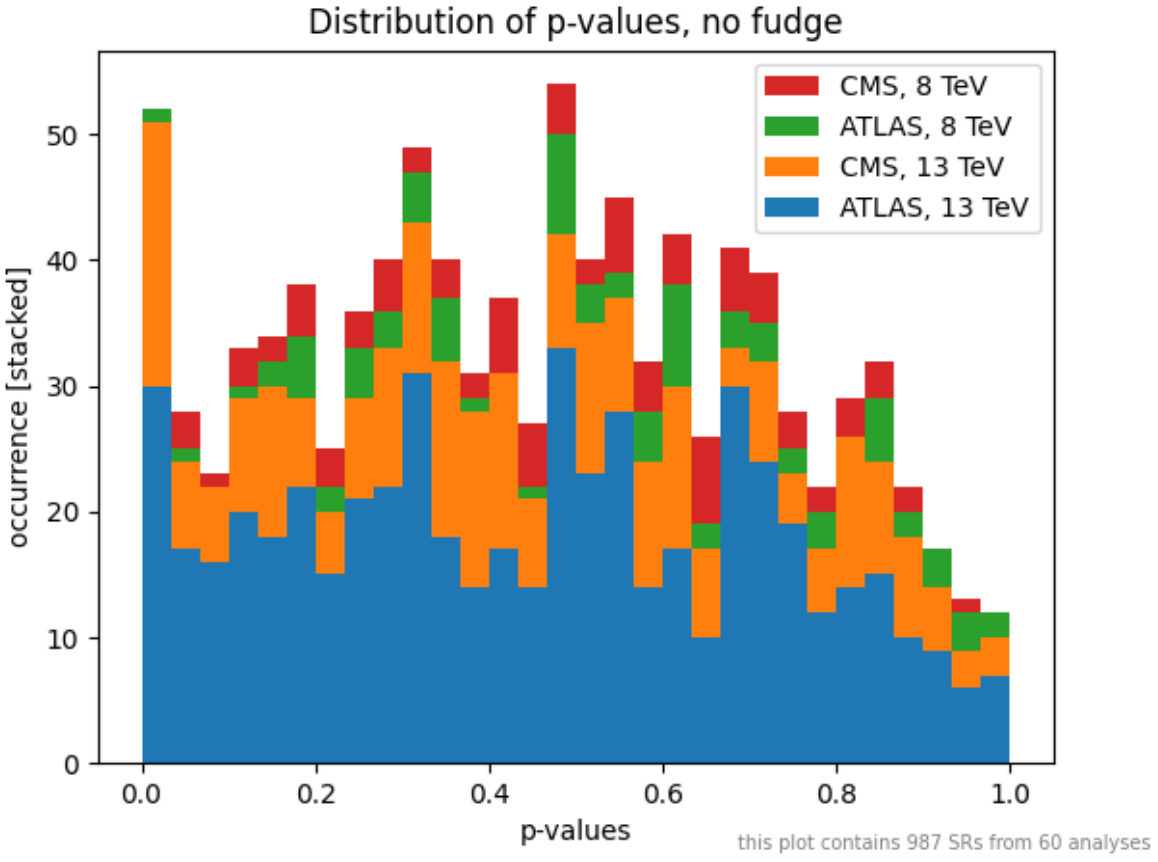
Likelihoods as functions of an overall signal strength multiplier

Taking this model with mu as the only free parameter, we could formulate a likelihood ratio test:

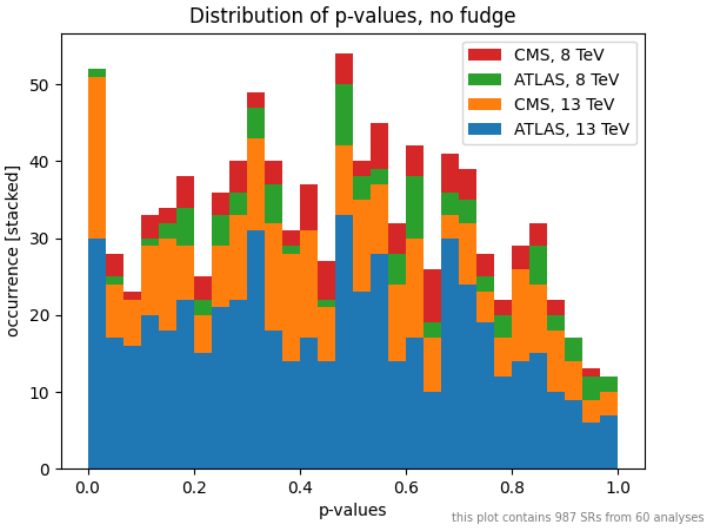
$$T_L := 2 \ln \left(\frac{L^c_{\text{BSM}}(\hat{\mu}) \cdot \pi_{\text{BSM}}}{L^c_{\text{SM}} \cdot \pi_{\text{SM}}} \right) = 33.69 = 5.8^2$$

Now that we have a protomodel, what about the meta-statistics plots that I showed you earlier? We have a new hypothesis, so we can compute the p-values for that!

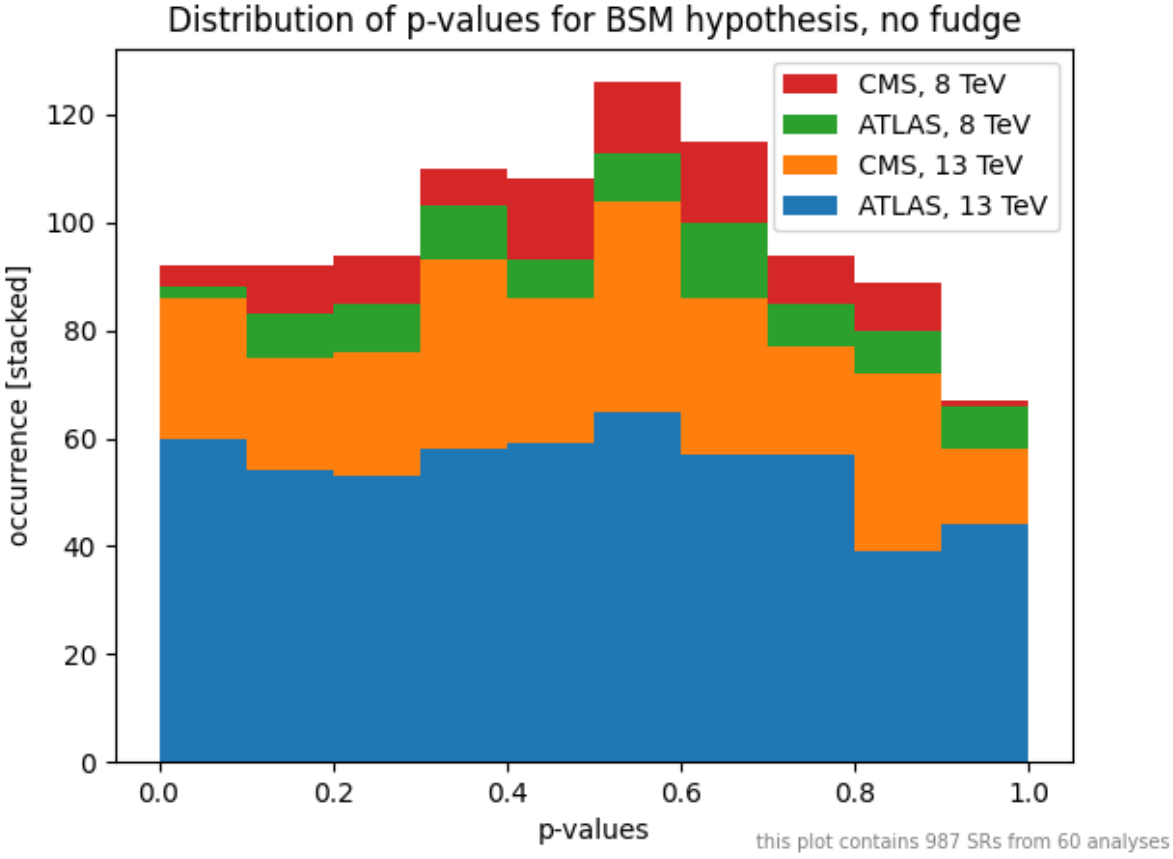
For the SM hypothesis:



p-values for
The SM hypothesis

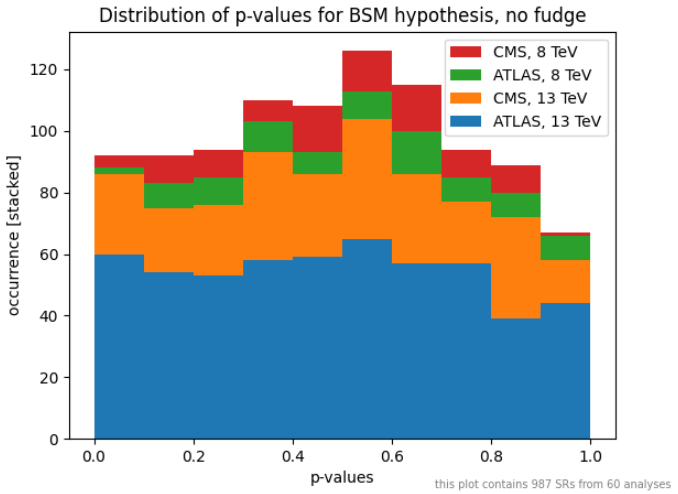


Instead of the SM, let's compute the p-values for the BSM hypothesis:

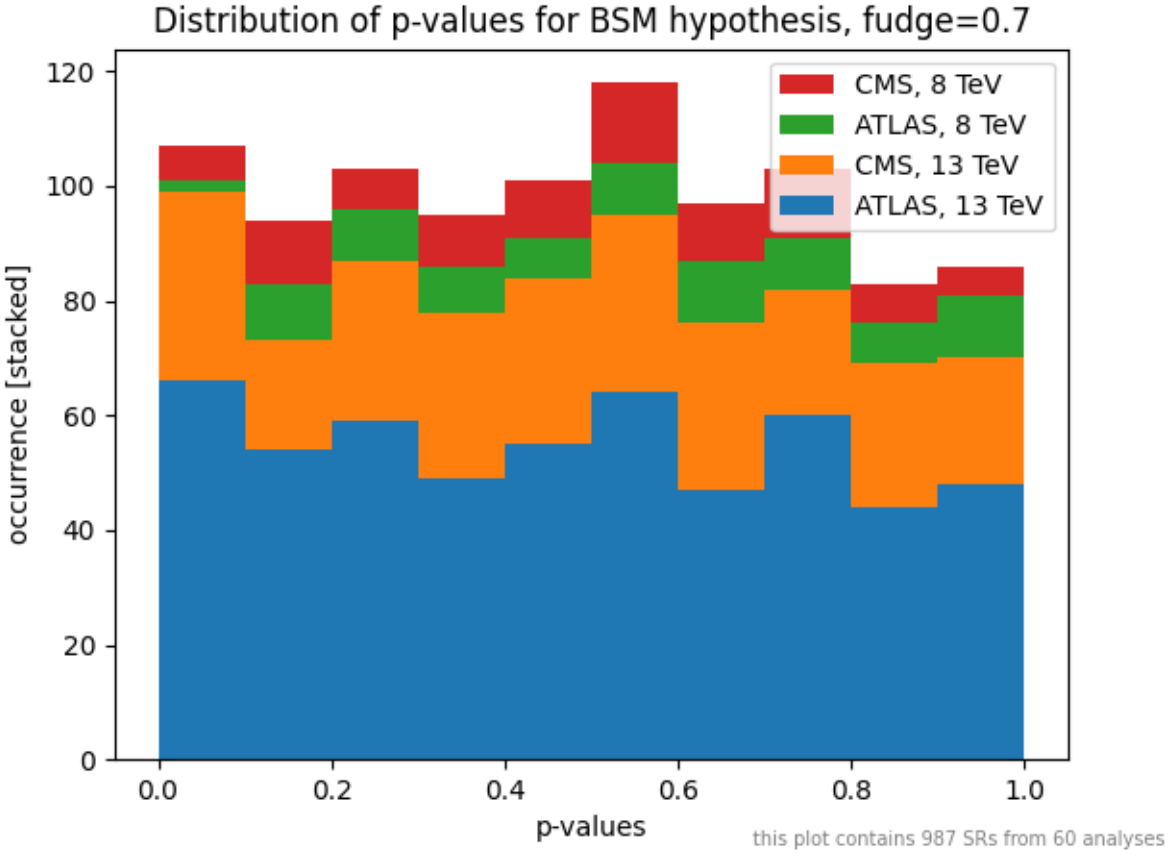


Instead of the SM, let's compute the p-values for the BSM hypothesis:

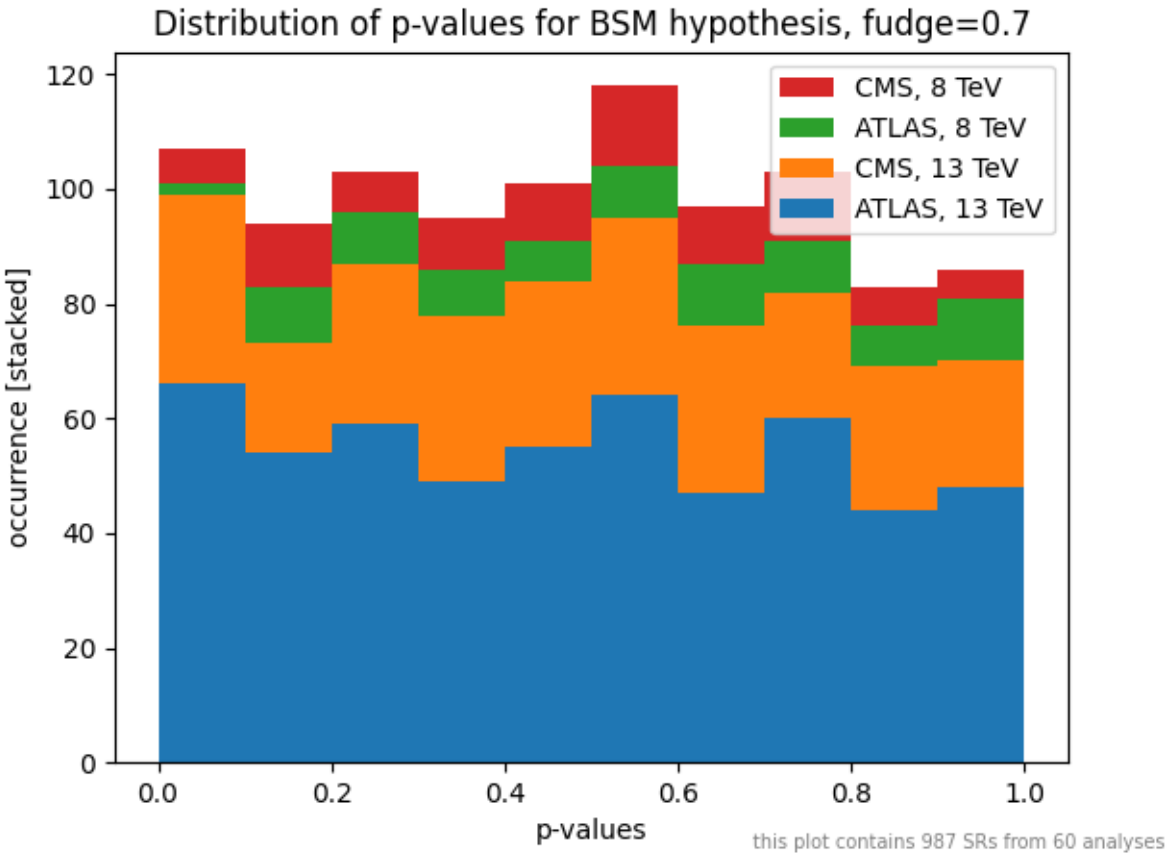
The convex shape makes sense if we assume the systematic errors to be conservatively estimated (as they should).



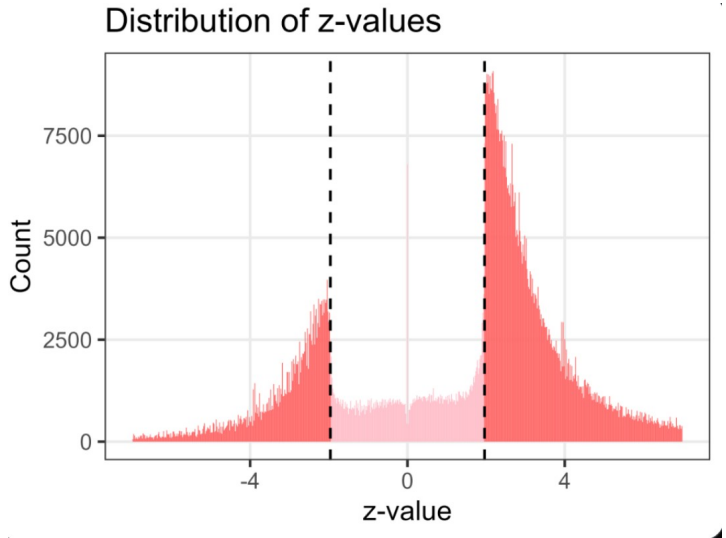
Here is the distribution of the P-values for the protomodel hypothesis, assuming all **systematic** errors to be ~ 40 % overestimated



Here is the distribution of the p-values for the protomodel hypothesis, assuming all systematic errors to be ~ 40 % overestimated



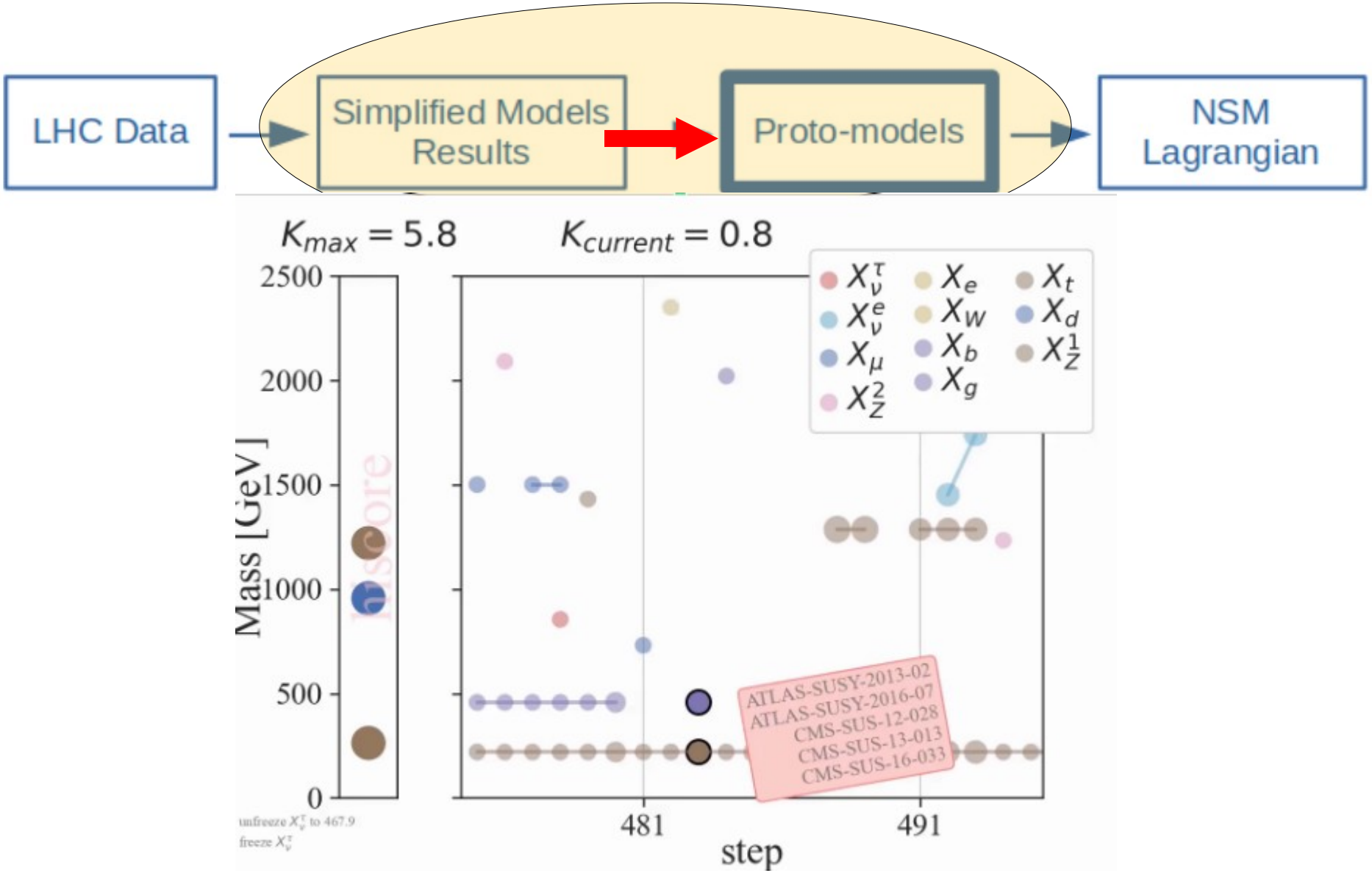
(This feels like nothing short of a miracle. Remember the medical research plot?)



Did anyone say LZ?

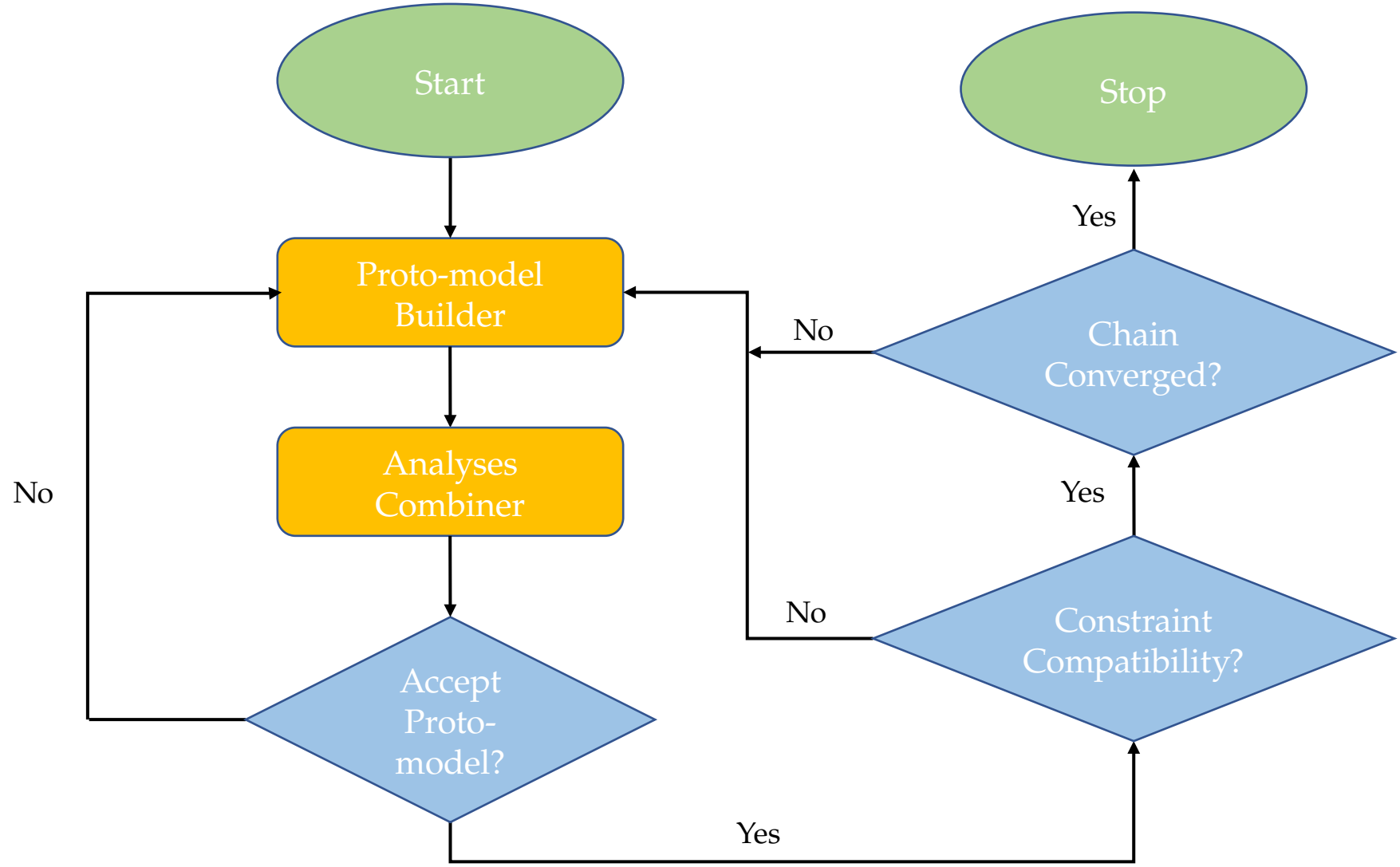
But how did we obtain that protomodel?

Protomodels

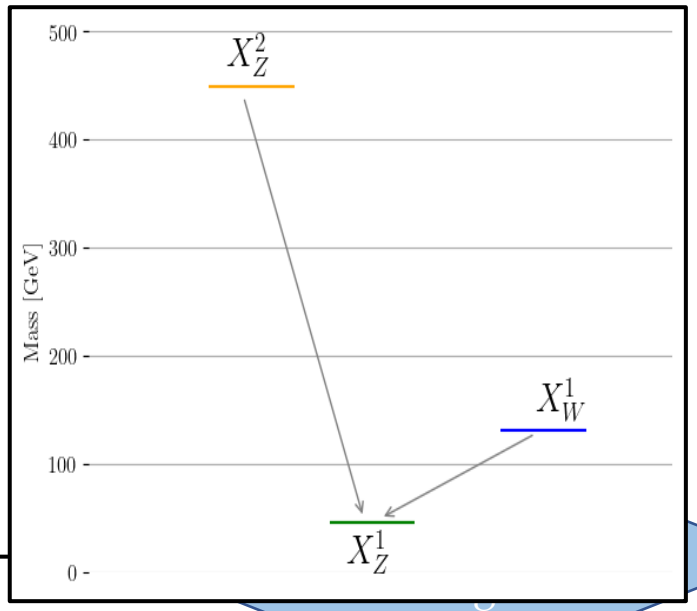


"Likelihood-based automated model building" (in a wider sense of the word 'model building')

Algorithm Flowchart



Algorithm Flowchart



Start

Proto-model Builder

Analyses Combiner

Accept proto-model?

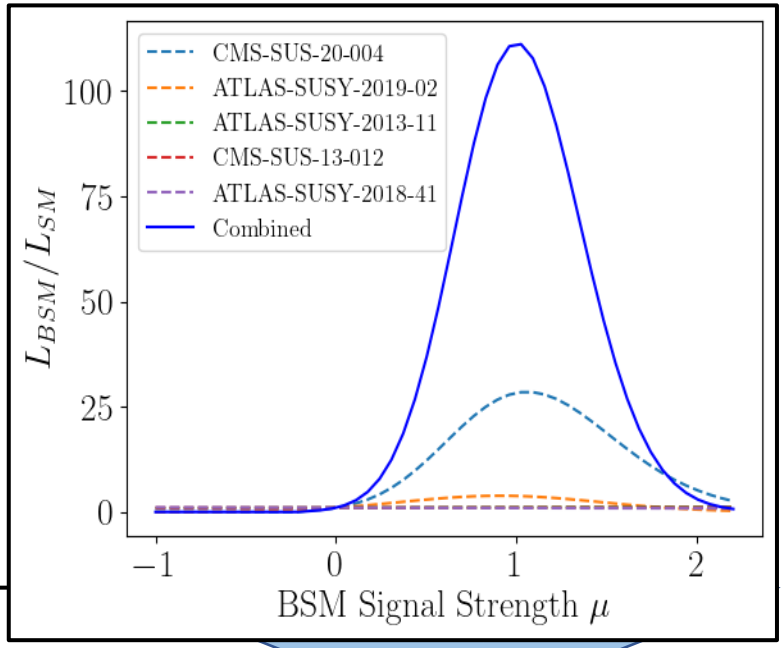
No

Yes

Proto-model Builder

Creates proto-models, randomly adding or removing new particles and changing any of the proto-model parameters.

Algorithm Flowchart



Start

Proto-model Builder

Analyses Combiner

Accept proto-model?

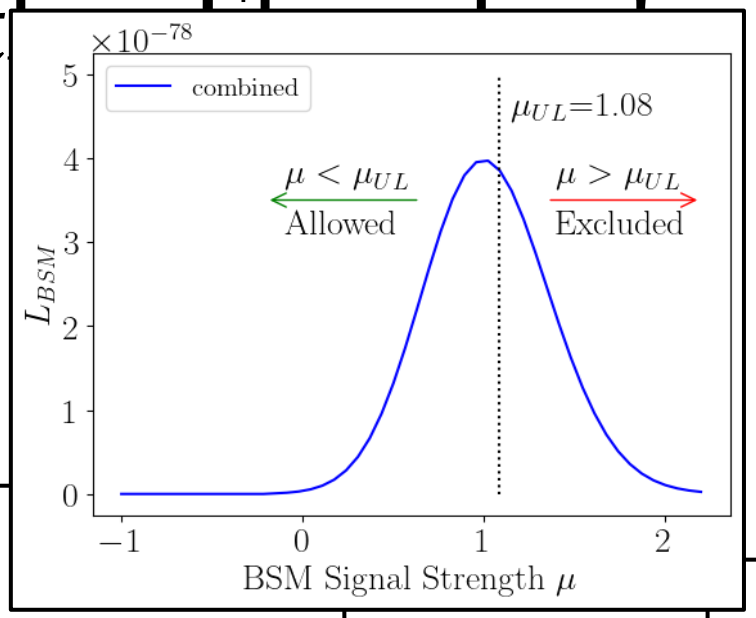
No

Yes

Analyses Combiner

Identifies the best possible combination of experimental results and constructs a combined likelihood for the proto-model

Algorit



Stop

Yes

Chain Converged?

No

Yes

Constraint Compatibility?

No

Yes

Analyses

No

"Critic"

Constraint Compatibility

Checks the compatibility of the proto-model against the extensive database of results

The Test Statistic

The test statistic K^c is a likelihood-ratio test that quantifies how much better the proto-model describes the data than the Standard-Model (plus a penalty for model complexity).

Quantifies violation of Standard Model hypothesis

Penalizes for model complexity (among other things)

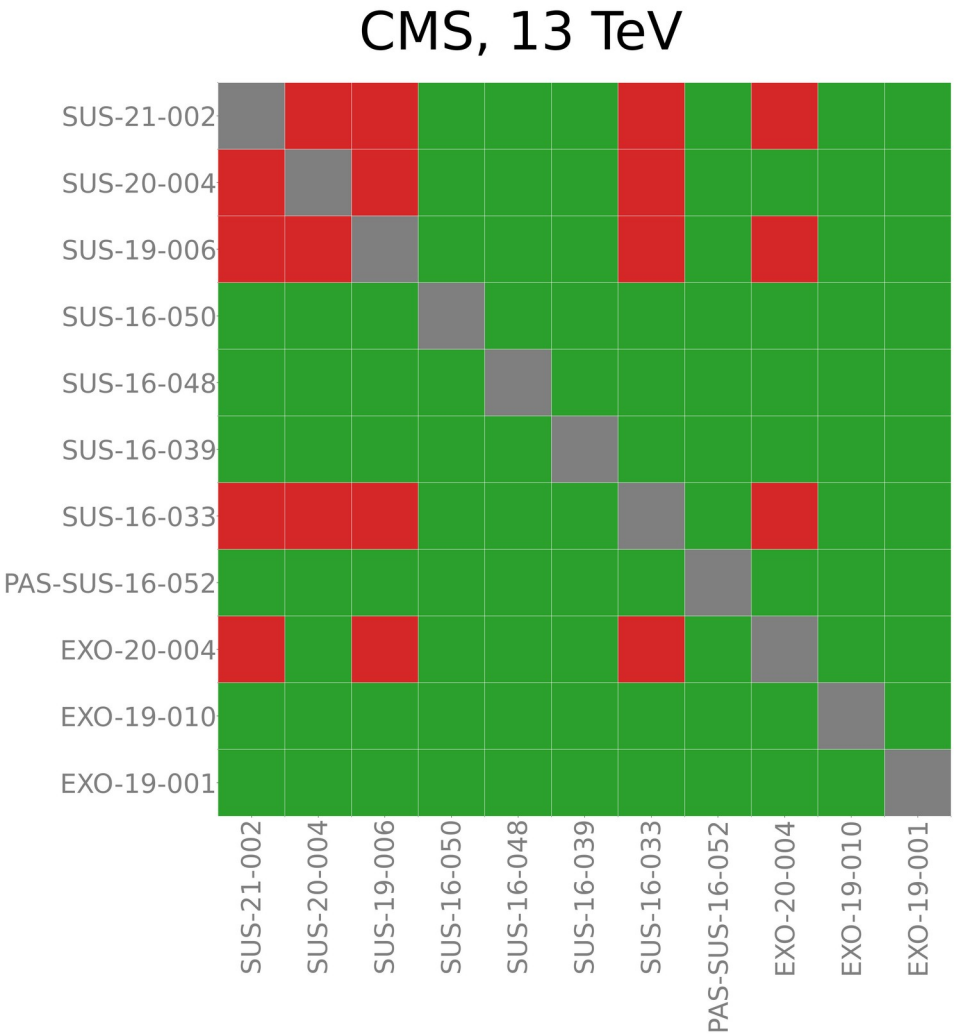
$$K^c := -2 \ln \frac{\mathrm{L}^c_{\mathbf{SM}}}{\mathrm{L}^c_{\mathbf{BSM}}(\hat{\mu})} \cdot \frac{\pi(\mathbf{SM})}{\pi(\mathbf{BSM})}$$

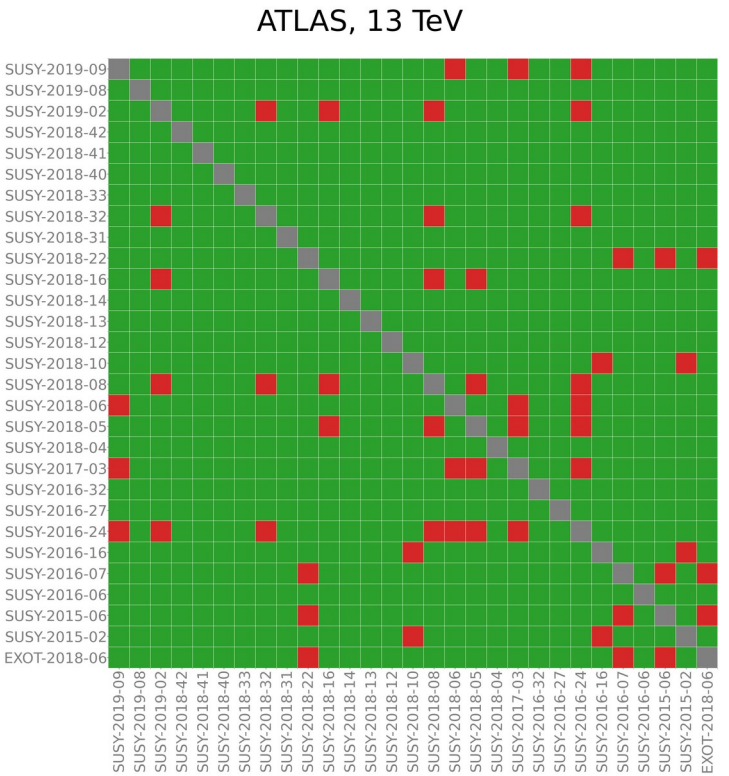
Main penalty is similar to Akaike Information Criterion:

$$\mathrm{AIC} = 2k - 2\ln(\hat{L})$$

We search for proto-models and combinations of results / likelihoods that maximize K^c *while remaining compatible with all negative results (and the most sensitive combination thereof) in our database*.

Combinabilities





Green: can combine
Red: cannot combine

*) Only 11 (ATLAS: 29) Run-2 CMS analyses published with all material needed to be usable for protomodels, though situation has been greatly improving recently

*) Longer term vision: full stats model for all analyses, naming conventions for nuisances. Correct combinations, taking into account correlations.

ProtoModels v2

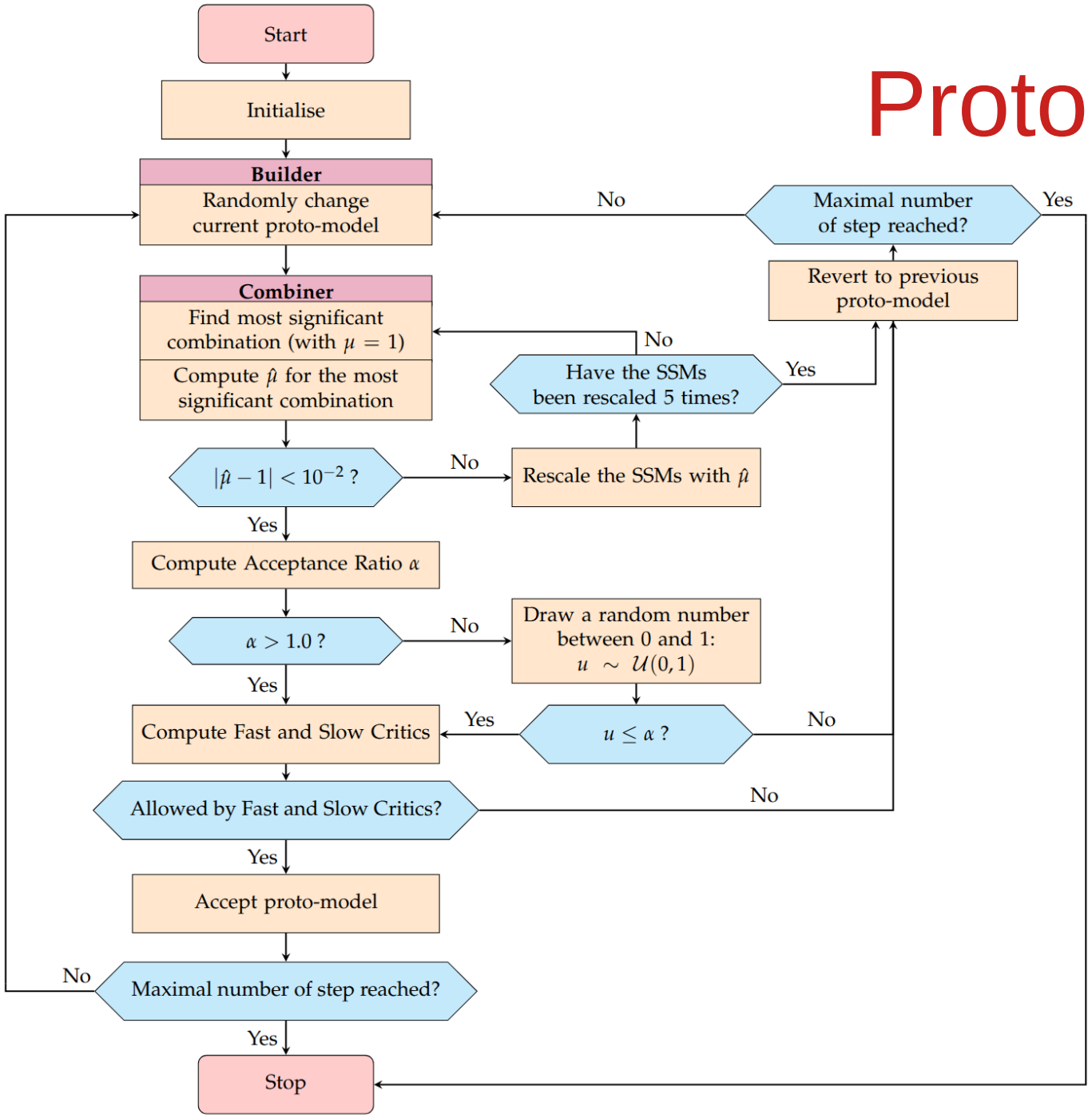
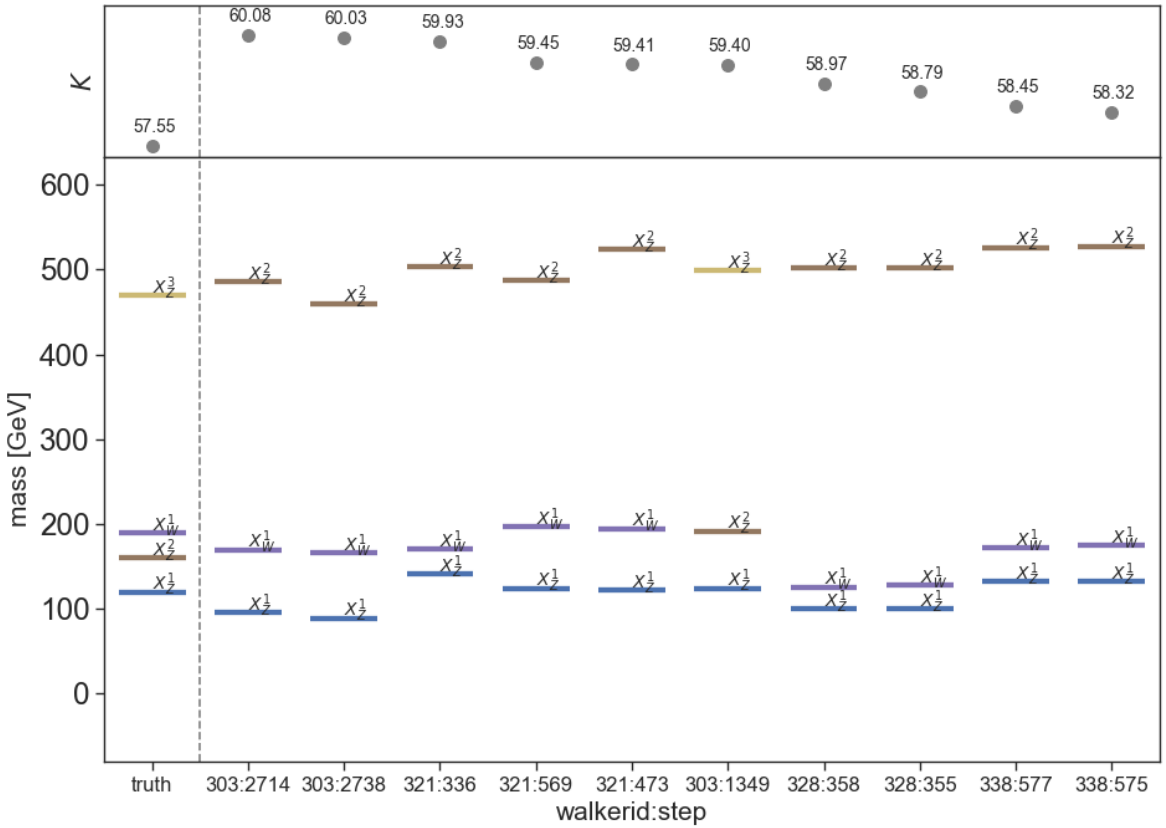


FIGURE 6.2: Simplified flowchart of the new code architecture.

Here is a flow chart of our much refined v2 of the machinery

Closure test

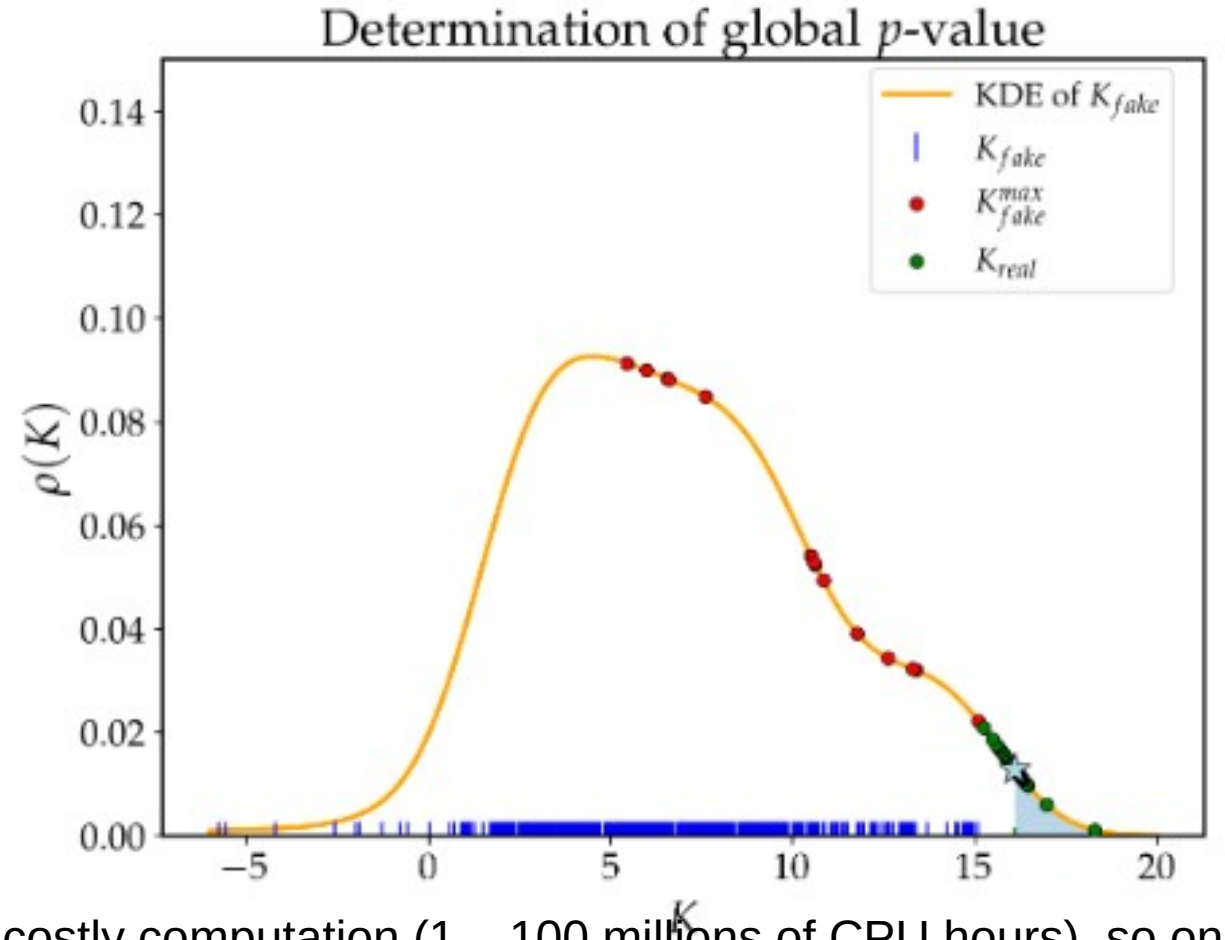
If we create fake data with an injected signal, then run our algorithm over it, do we get the truth back out?



For protomodels similar to our hiscore model, we do.

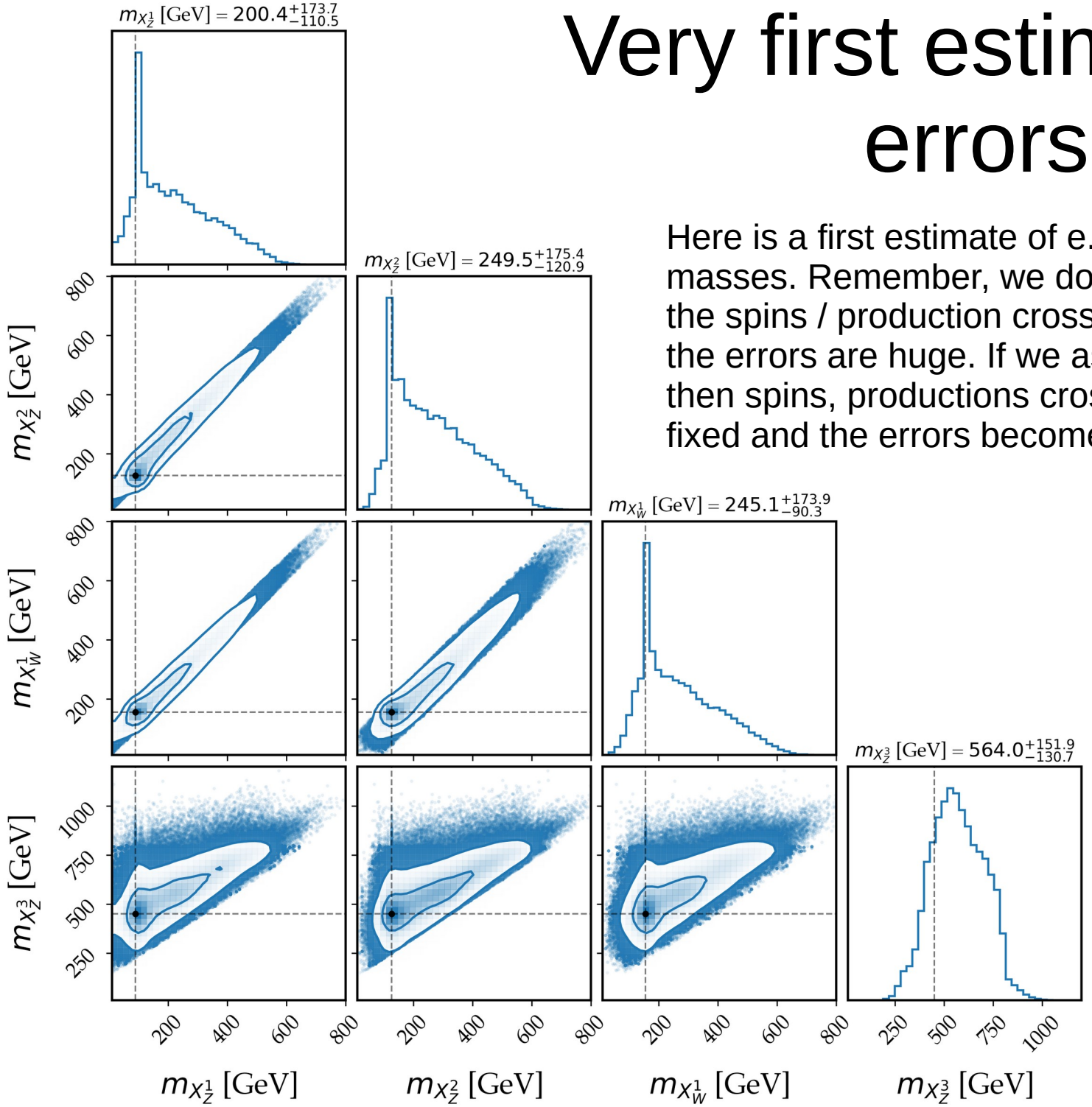
global p-value

If we synthesize SM-only versions of our database, and run our algorithm on these synthetic data, we can compute a p-values for the SM-hypothesis



Extremely costly computation (1 – 100 millions of CPU hours), so only very rough estimate of global p-value (though the estimate is likely conservative). We currently end up at p ~ 0.01 (very rough estimate!)

Very first estimates of errors



Here is a first estimate of e.g. errors on particle masses. Remember, we do not fixate the spins / production cross sections of our model, so the errors are huge. If we asssume e.g. the MSSM, then spins, productions cross sections etc are (more) fixed and the errors become much smaller

Other pheno groups

Have other pheno groups noticed / noted on these similar compatibilities between excesses?

A joint explanation for the soft lepton and monojet LHC excesses in the wino-bino model

Diyar Agin*[1], Benjamin Fuks†[1], Mark D. Goodsell‡[1], and Taylor Murphy§[2]

GUT-based NMSSM with Singlino Dark Matter: Describing the ATLAS/CMS Excesses

E. Bagnaschi,[a] M. Chakraborti,[b] S. Heinemeyer[c,*] and I. Saha[d]

partners still allowed with masses as low as a few hundred GeV. Over the last years, searches for the "golden channel", $pp \to \tilde{\chi}_2^0 \tilde{\chi}_1^\pm \to \tilde{\chi}_1^0 Z^{(*)}\, \tilde{\chi}_1^0 W^{\pm(*)}$ show consistent excesses between CMS and ATLAS in the 2 soft-lepton and 3 soft-lepton plus $\not{E}_T$ searches, favoring mass scales of $m_{\tilde{\chi}_2^0} \approx m_{\tilde{\chi}_1^\pm} \gtrsim 200$ GeV and $\Delta m_{21} := m_{\tilde{\chi}_2^0} - m_{\tilde{\chi}_1^0} \approx 20$ GeV. We analyze these excesses in the framework of the Next-to-Minimal Supersymmetric Standard Model (NMSSM). We assume

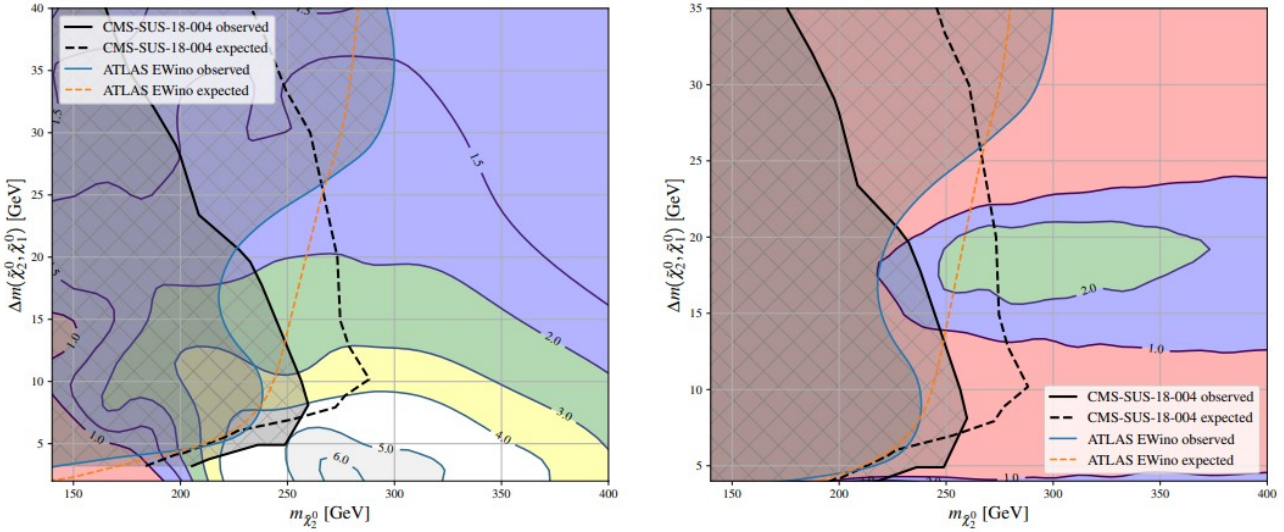


Figure 4: Results for CMS monojet searches (left) and ATLAS soft-lepton+$E_{\mathrm{T}}^{\mathrm{miss}}$ (right) in the wino-bino scenario, overlaid with soft-lepton excluded regions reported by ATLAS and CMS. Top: Expected and observed limits. Bottom: Contours of Bayes factor B_{10}.

Soft dilepton excess wants ~ 200, 250 GeV neutralinos with ~ 20 GeV mass gap. Compatible with our findings (and we didn't have dileptons result)

Other pheno groups

Mini summary of MSSM:

A) wino/bino DM with chargino co-annihilation ($|M_1| \sim M_2 \lesssim \mu$)

relic DM density 100% fulfilled

$\Rightarrow$ fits well the excesses!

Never: $M_1 \sim M_2/2$ $\Rightarrow$ not GUT-based

D) higgsino DM: $m_{\tilde{\chi}_1^0} \sim m_{\tilde{\chi}_2^0} \sim m_{\tilde{\chi}_1^\pm} \sim \mu$ ($\mu \lesssim |M_1|, M_2$)

relic DM density as upper limit (full relic density: $m_{\tilde{\chi}_1^0} \sim 1$ TeV)

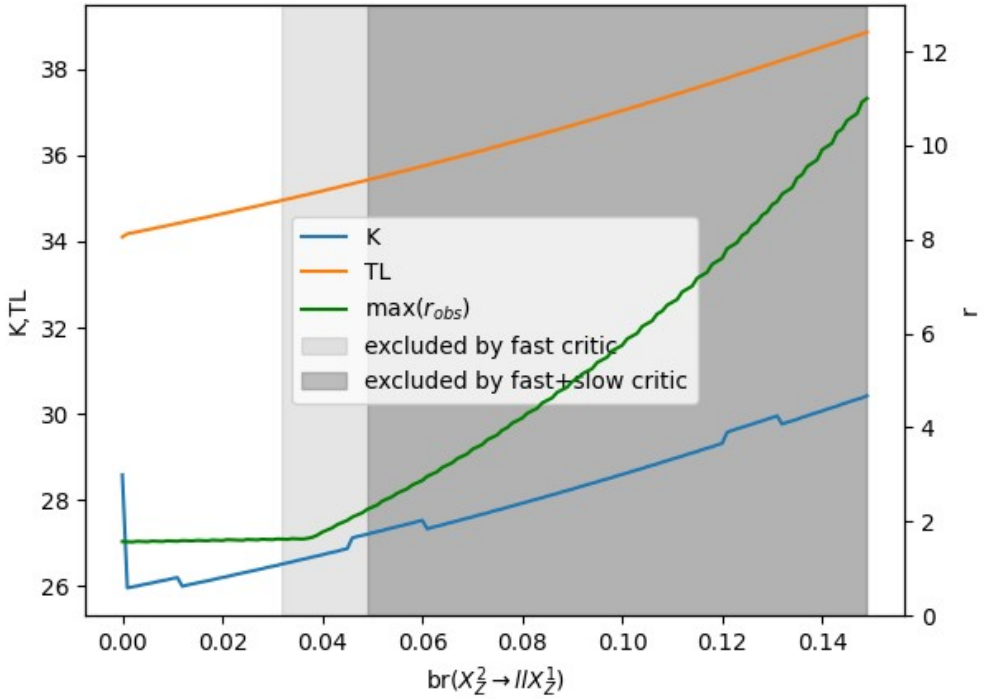
$\Rightarrow$ does not fit the excesses Possible: $M_1 \sim M_2/2$ $\Rightarrow$ can be GUT-based

No higgsinos within MSSM!!!

S. Heinemeyer, SUSY2026

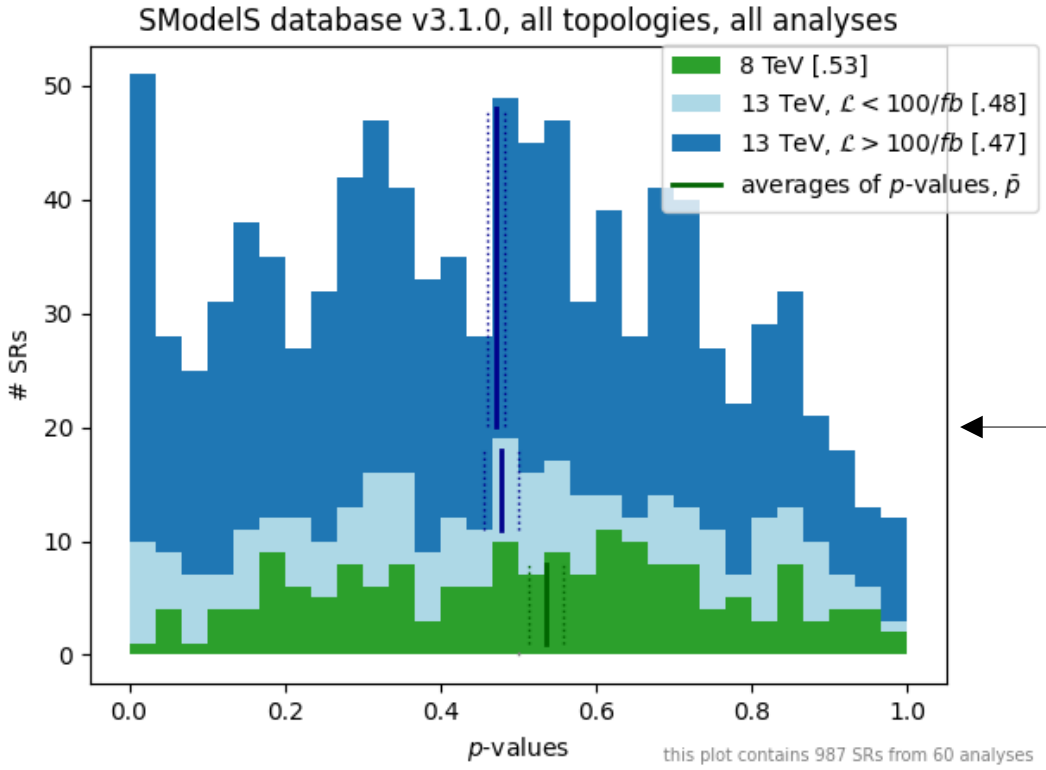
They added dileptons to the X2Z decays. We could, too.

The biggest excesses in soft dileptons + MET are not (yet) in the SModelS database. But even the older ones that we have, would actually fit the picture – the excesses are just not big enough yet.



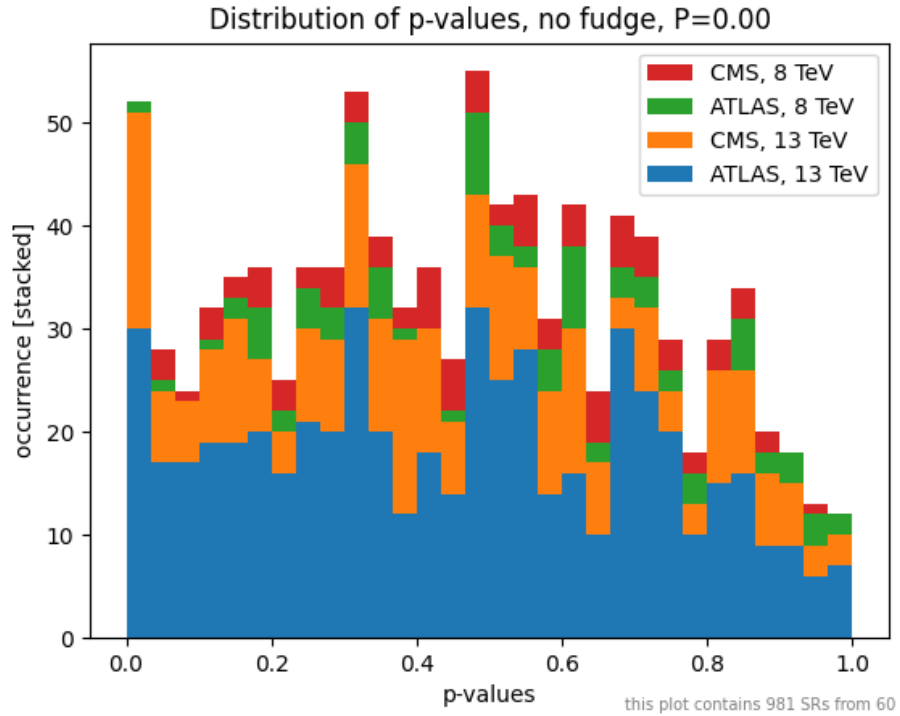
Protomodels – what will we do with them?

(apart from them serving as a high-level summary of what we are seeing, possibly informing the search programmes about where to look next)

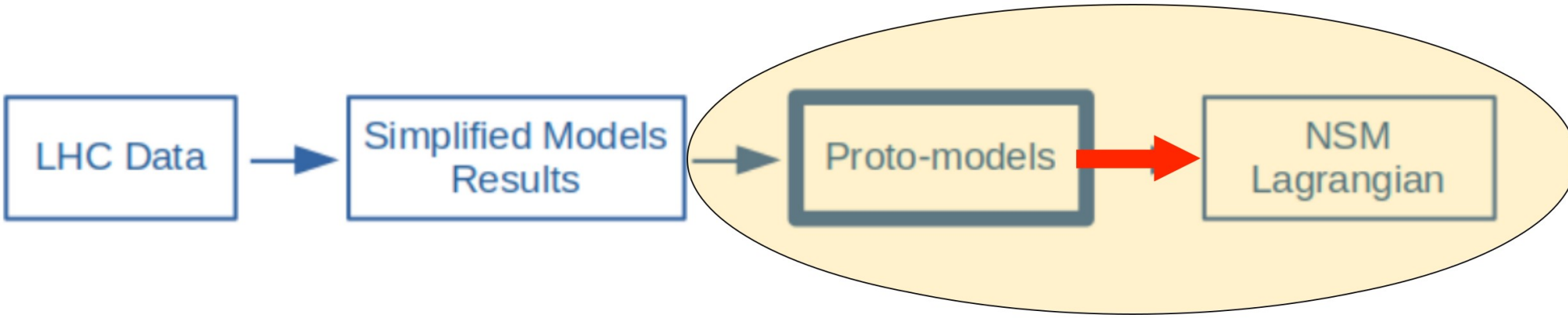


If this peak is due to new physics at the right energy scales, it's plausible that we didn't see anything at run-1 yet, we should starting seeing something at run-2 low-lumi, then things should get more pronounced at run-2 high-lumi.

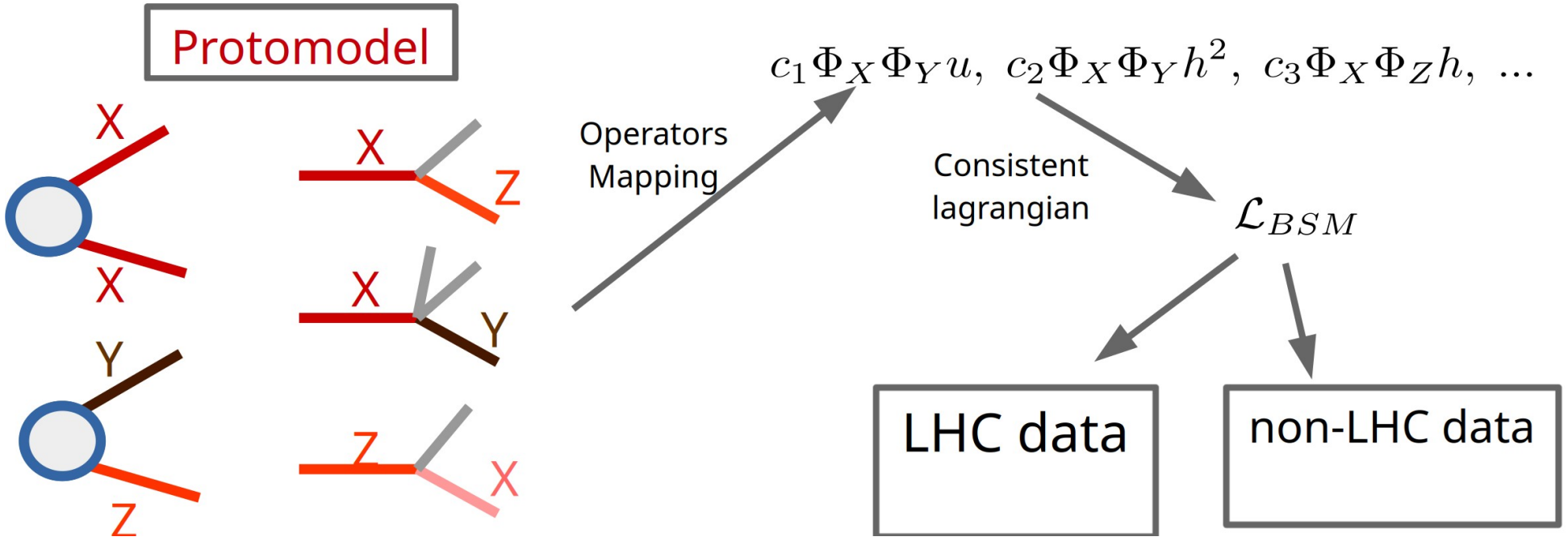
Given that the ATLAS and CMS physics programmes are very similar, we should be seeing similar behavior for both experiments.



NSM Lagrangian from the proto-models, work in progress



Work in progress. Lots of combinatorics:
we cannot differentiate between light quark flavors. Observed particles might be degenerate.
From particles to fields.
Singlets versus doublets versus multiplets. From broken to unbroken Lagrangians.



Wanna Collaborate?

v1 was a feasibility study

v2 will (soon) be the first version producing protomodels that are of physics relevance

v3 should give a comprehensive summary of much of the LHC programme (especially SUSY)

We have started to plan a v3 of the project

There is more than plenty to do, from curating the inputs to speeding things up with ML and gradients to connecting the output protomodels with UV complete models.

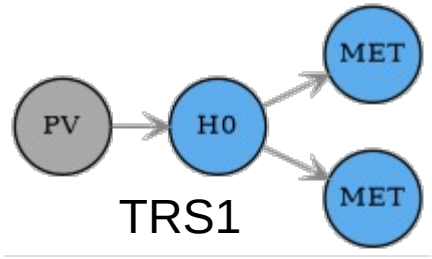
Make connection with dark matter physics, flavor physics.

Full scale combinations within the collaborations, etc etc.

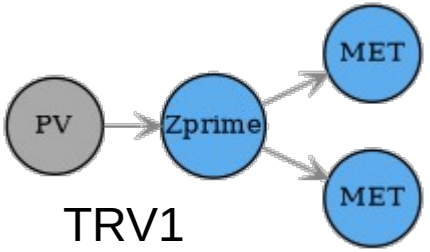
Thank you!

Naively: (*without* assuming conservatism) we would expect
<= 960 * Φ(2.40) ≈ 8 SRs.
We have 28 SRs with Z > 2.40, 4 SRs with Z < - 2.40

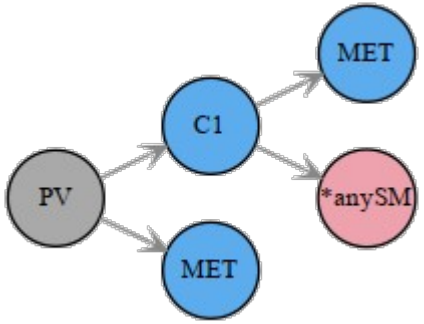
List Of SRs with Z > 2.40



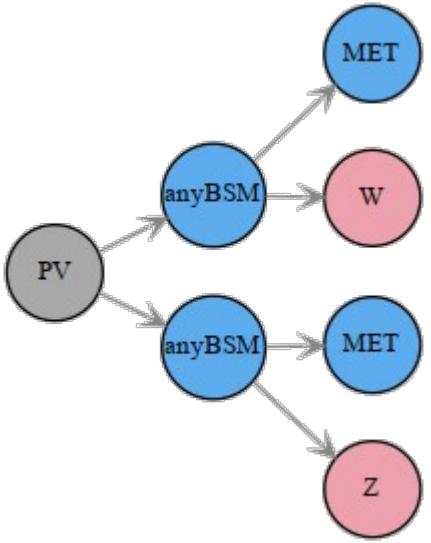
TRS1



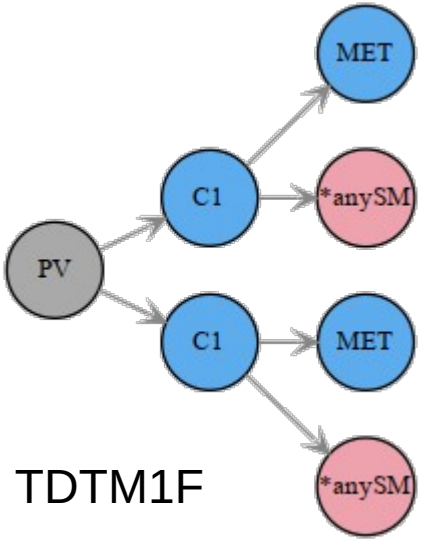
TRV1



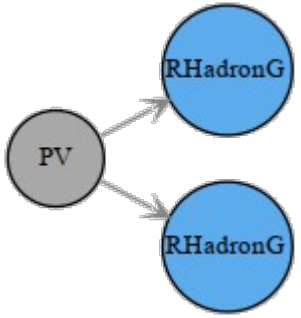
TDTM2F



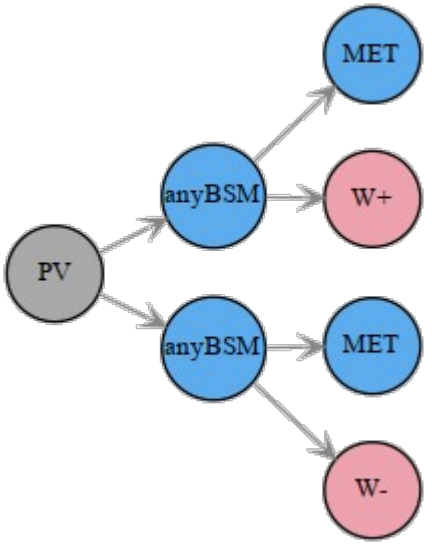
TChiWZ(off)



TDTM1F

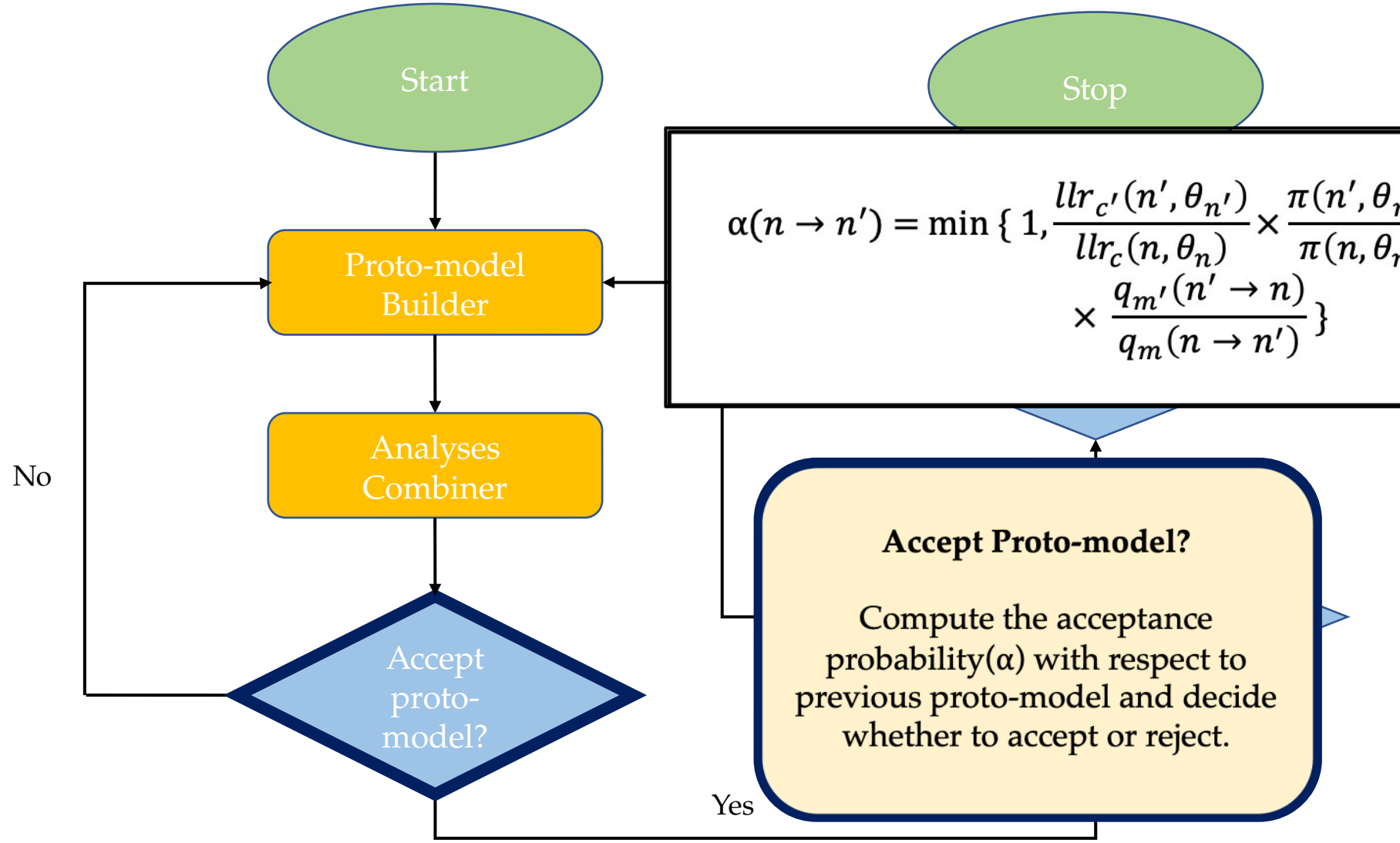


Z	analysis	SR	topo	obsN	expected
4.47	CMS-EXO-20-004	SR_2018_8	TChiISR ...	8529	7720.98±152.03
4.31	CMS-EXO-20-004	SR_2018_4	TChiISR ...	35833	33037.13±618.12
4.11	CMS-EXO-20-004	SR_2018_6	TChiISR ...	14266	13209.01±250.28
3.85	CMS-EXO-20-004	SR_2018_9	TChiISR ...	5406	4914.64±101.04
3.80	ATLAS-SUSY-2018-42	SR1400_High_long	T1DispT ...	7	0.70±0.40
3.78	CMS-EXO-20-004	SR_2018_1	TChiISR ...	191829	178470.12±3615.00
3.72	CMS-EXO-20-004	SR_2018_2	TChiISR ...	104522	97428.42±1896.81
3.63	CMS-EXO-20-004	SR_2018_3	TChiISR ...	59398	55592.19±1059.45
3.46	ATLAS-SUSY-2018-42	SR1500_High_long	TDTM1FT ...	6	0.60±0.40
3.35	CMS-EXO-20-004	SR_2018_11	TChiISR ...	2671	2428.39±55.08
3.30	ATLAS-SUSY-2018-16-hino	SR_ewk_2l_med	TChiWWo ...	170	112.40±12.60
3.21	ATLAS-SUSY-2018-42	SR1300_High_long	TDTM1FT ...	6	0.80±0.40
3.21	ATLAS-SUSY-2018-16	SR_ewk_2l_ee_high_d	TChiWZoff	16	4.80±1.60
3.17	CMS-EXO-20-004	SR_2018_12	TChiISR ...	1613	1441.83±35.64
3.17	ATLAS-SUSY-2018-42	SR1600_High_long	T1DispT ...	5	0.54±0.35
3.01	ATLAS-SUSY-2018-42	SR1200_High_long	T1DispT ...	6	0.90±0.50
3.01	CMS-EXO-20-004	SR_2018_7	TChiISR ...	11730	10991.22±213.74
2.93	ATLAS-SUSY-2017-03	SR3l_ISR	TChiWZ	12	3.90±1.00
2.83	ATLAS-SUSY-2018-42	SR1800_High_long	T1DispTRHadGM1	4	0.44±0.32
2.75	CMS-EXO-20-004	SR_2018_5	TChiISR ...	21854	20714.72±391.42
2.56	ATLAS-SUSY-2019-09	SRlow_0Jd	TChiWZoff	77	54.00±4.00
2.51	CMS-SUS-20-004	SR11	T5HHTChiHH	4	0.06±0.11
2.51	CMS-EXO-20-004	SR_2018_10	TChiISR ...	3285	3065.69±66.79
2.49	ATLAS-SUSY-2018-42	SR2400_High_long	T1DispTRHadGM1	3	0.29±0.28
2.47	ATLAS-EXOT-2018-06	EM10	TChiISR ...	413	359.00±10.00
2.45	ATLAS-SUSY-2018-42	SR2600_High_long	T1DispTRHadGM1	3	0.25±0.31
2.44	ATLAS-SUSY-2018-42	SR2000_High_long	T1DispTRHadGM1	3	0.36±0.25
2.42	ATLAS-SUSY-2018-42	SR2200_High_long	T1DispTRHadGM1	3	0.33±0.28



*) CMS-EXO-20-004 is yes overrepresented
(simplified likelihood fit possible culprit?)
*) “small mass gap + MET” topologies are dominant

Algorithm Flowchart



Our Inverse Problem

To connect our observations with contender theories for the Next Standard Model,

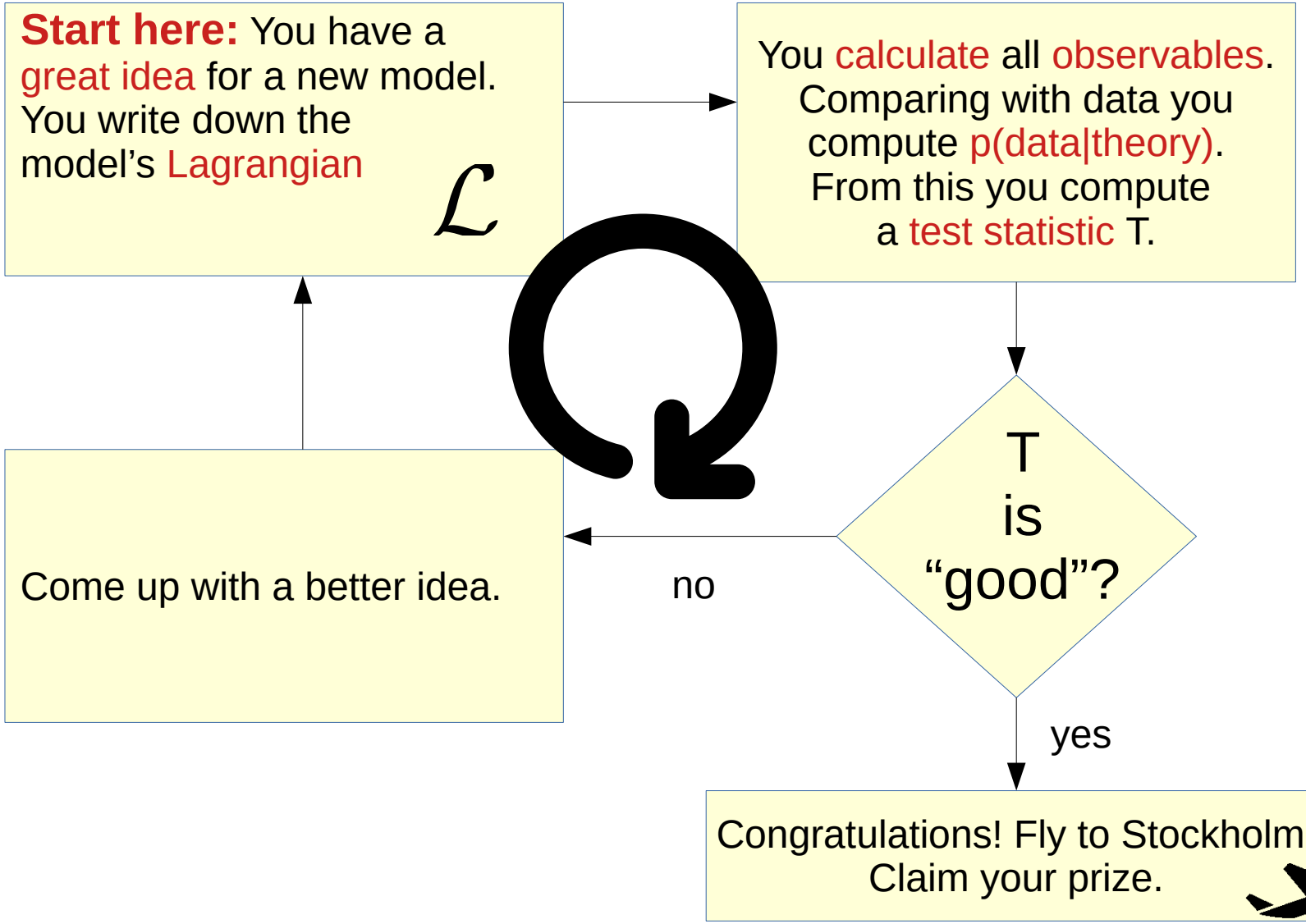
- In addition to lots of other information, we need statistical models (in one way or another)

- we can either start by:
  postulating a theory, then fit its parameters
  (“top-down”)

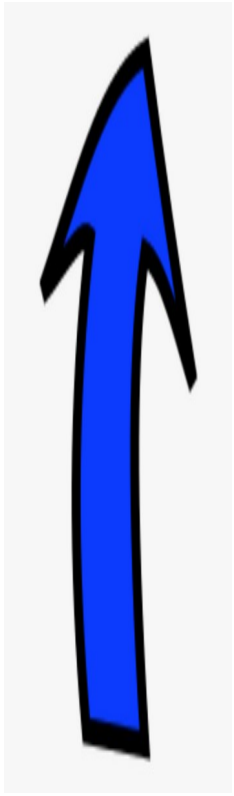
- or:
  try to build our model directly from data
  (“bottom-up”)

Top-down versus bottom-up

Top-Down:



Top-down versus bottom-up



Bottom-Up:

Only now do you think about symmetries, gauge groups, etc that may underlie all observations. Construct your Lagrangian.

$\mathcal{L}$

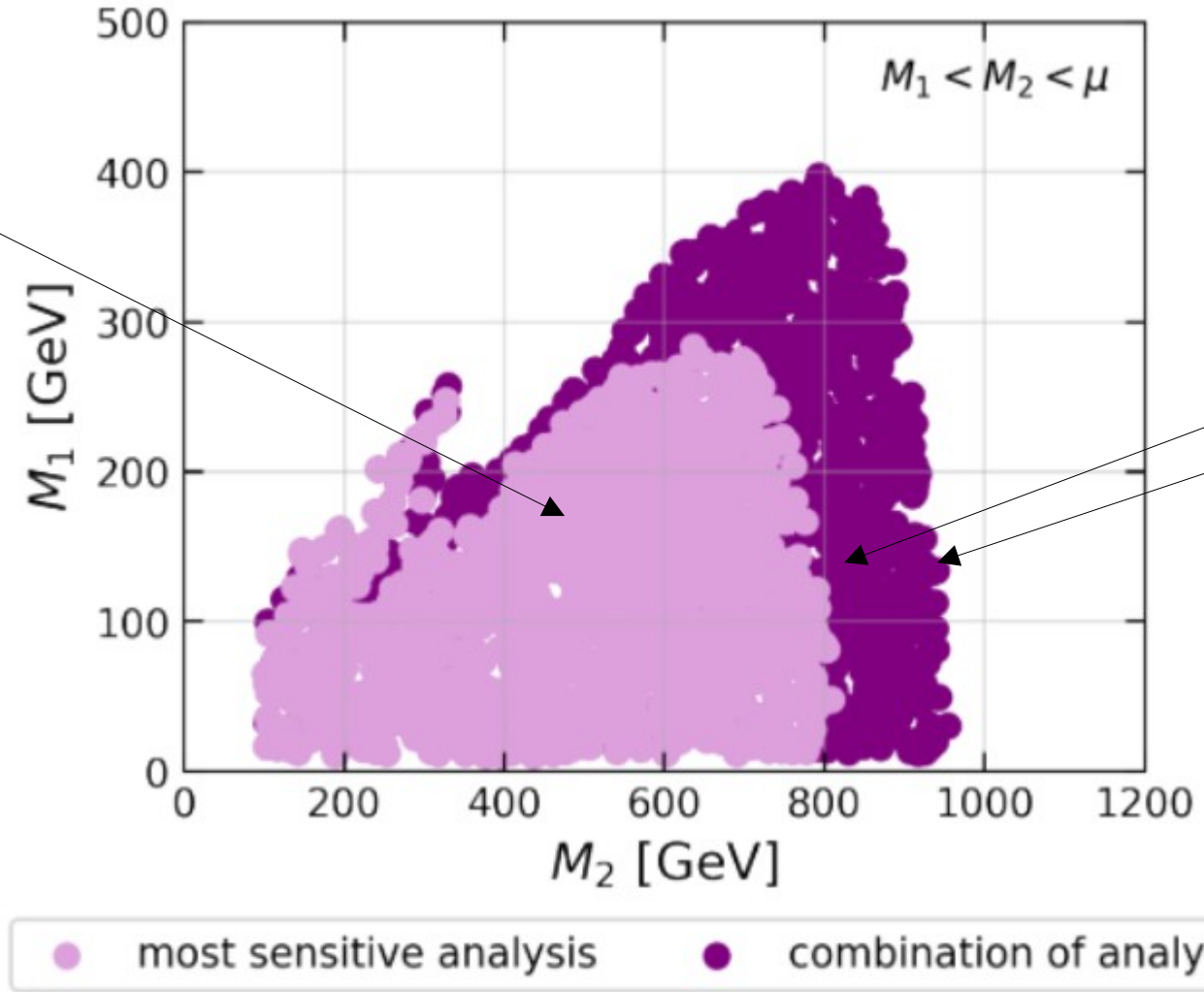
From the descriptions you try and construct precursor theories to the NSM that describe everything you really know about TeV-scale (and below) physics

Start here: You describe your experimental findings in a language amenable to theoretical physics, e.g. simplified models for on-shell effects (“searches”), effective field theories for off-shell effects (“measurements”).

Top-down: example

"Global LHC constraints on electroweakinos with SModelS v2.3"
Model: MSSM, but looking at its charginos and neutralinos only

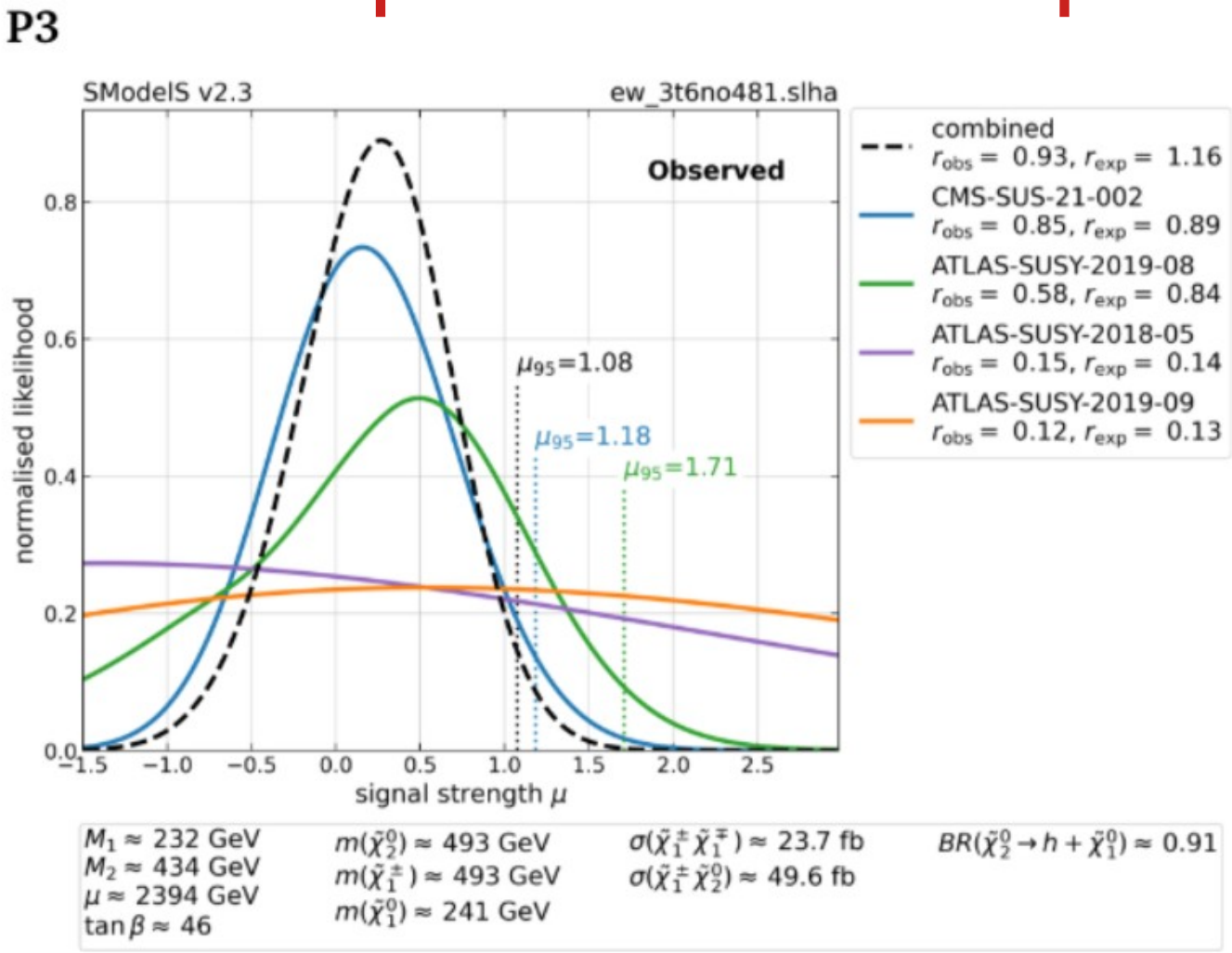
Excluded
Model points



gain in exclusion reach by combining (CMS and ATLAS) results

https://arxiv.org/abs/2312.16635

Top-down: example



Combination at work, for one model point. Likelihoods here are functions of a single parameter (the overall “signal strength”). Combining several likelihoods (colored lines) results in a narrower combined likelihood (dashed line).

https://arxiv.org/abs/2312.16635

Statistical Models in the SModelS Database

ATLAS, Run-2, Full Lumi only:

ID	Short Description	$\mathcal{L}$ [fb^{-1}]	UL_{obs}	UL_{exp}	EM	comb.
ATLAS-EXOT-2018-48	di-top resonance	139.0	✓	✓		
ATLAS-EXOT-2019-03	dijet resonance	139.0	✓	✓		
ATLAS-SUSY-2018-04	2 hadronic τ	139.0	✓		✓	PYHF
ATLAS-SUSY-2018-05	2ℓ + jets, EWK	139.0	✓		✓	PYHF
ATLAS-SUSY-2018-06	3ℓ, EWK	139.0	✓	✓	✓	
ATLAS-SUSY-2018-08	2 OS ℓ	139.0	✓		✓	
ATLAS-SUSY-2018-09	2 SS ℓ	139.0	✓			
ATLAS-SUSY-2018-10	1ℓ + jets	139.0	✓		✓	
ATLAS-SUSY-2018-12	0ℓ + jets	139.0	✓	✓	✓	
ATLAS-SUSY-2018-13	displaced jets	139.0			✓	SLv1
ATLAS-SUSY-2018-14	displaced vertices	139.0			✓	PYHF
ATLAS-SUSY-2018-16	2 soft ℓ + jets, EWK	139.0	✓	✓	✓	PYHF
ATLAS-SUSY-2018-22	multi-jets	139.0	✓		✓	
ATLAS-SUSY-2018-23	$Wh(\gamma\gamma)$, EWK	139.0	✓	✓		
ATLAS-SUSY-2018-31	2 b-jets + 2 h	139.0	✓		✓	PYHF
ATLAS-SUSY-2018-32	2 OS ℓ	139.0	✓		✓	PYHF
ATLAS-SUSY-2018-40	2 b-jets + 2 h	139.0	✓	✓	✓	
ATLAS-SUSY-2018-41	hadr. EWK	139.0	✓	✓	✓	SLv1
ATLAS-SUSY-2018-42	charged LLPs, dE/dx	139.0	✓	✓	✓	
ATLAS-SUSY-2019-02	2 soft ℓ, EWK	139.0	✓		✓	SLv1
ATLAS-SUSY-2019-08	$1\ell + h(bb)$, EWK	139.0	✓		✓	PYHF
ATLAS-SUSY-2019-09	3ℓ, EWK	139.0	✓	✓	✓	PYHF

Type of statistical model

Statistical Models

- **Only exclusions for particular models**
  If only “exclusion lines” are given, without upper limits, we can do nothing

- **Observed 95% CL upper limits on production cross sections only:**
  cannot construct likelihood, binary decision “excluded” / “not-excluded” only (“critic”)

- **Expected and observed 95% CL upper limits**
  can construct an approximate likelihood with truncated Gaussian, cannot combine topologies, very crude approximation

- **Signal Efficiency “maps” for “simplified models”**
  can construct a likelihood as Gaussian (for the nuisances) * Poissonian (for yields), can work per SR, and combine topologies in each SR [*]

- **Efficiency maps + correlation matrices**
  can combine signal regions via multivariate Gaussian * Poissonians

- **Efficiency maps + full likelihoods**
  full realism, correct statistical model

- **Efficiency maps + full likelihoods + naming convention + knowledge about “data overlaps”**
  can combine publications at full realism



Combos

Likelihoods

BETTER

[*] if efficiency maps are not supplied, we can try to produce them with recasting frameworks

Serialized Statistical Models

pyhf serializes the models into json files:

```
{
    "channel": {
        "type": "object",
        "properties": {
            "name": { "type": "string" },
            "samples": { "type": "array", "items": {"$ref": "#/definitions/sample"}, "minItems": 1 }
        },
        "required": ["name", "samples"],
        "additionalProperties": false
    },
}
```

Posterior distributions on nuisance parameters (such as jet energy scales, pdf uncertainties, reconstruction ineffciencies, etc etc) are described via “auxiliary measurements” –> keeping things frequentist (if we wish)!

Statistics Standardization comittee formed: **HS3**

https://github.com/hep-statistics-serialization-standard/hep-statistics-serialization-standard

These “closed world” approaches are general enough for users!

Simplifications and Interoperability

An ecosystem of tools around serialized statistical models will be crucial:

Currently we have:

Simplify: simplifies pyhf likelihoods, collapses nuisances.

https://pypi.org/project/simplify/

Seems to be working with only minor quirks, we are making use of it.

Combine2pyhf: converts combine datacards to pyhf and back

https://github.com/kskovpen/combine2pyhf

Orphaned, does not handle all cases.

Pyhf-to-combine converter:

https://github.com/peterridolfi/Pyhf-to-Combine-converter

Student project. Promising but incomplete.

In “User Space”: Machine Learning Surrogate Models

We are in the process of using surrogate models in SModelS (scheduled for SModelS v3.2.0)

If only likelihood is needed: simple, fully connected neural networks (Bayesian neural networks for adding errors on the estimates), e.g.:

CERN-TH-2019-187

The DNNLikelihood: enhancing likelihood distribution with Deep Learning

Andrea Coccaro[a], Maurizio Pierini[b], Luca Silvestrini[b,c], and Riccardo Torre[a,b]

[a] *INFN, Sezione di Genova, Via Dodecaneso 33, I-16146 Genova, Italy*
[b] *CERN, 1211 Geneva 23, Switzerland*
[c] *INFN, Sezione di Roma, P.le A. Moro, 2, I-00185 Roma, Italy*

If full statistical model including toys is needed as well: generative network such as normalizing flows. e.g:

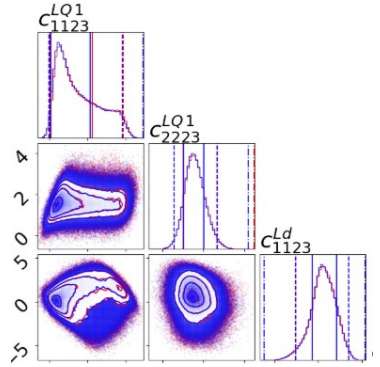
SciPost Physics Submission

The NFLikelihood: an unsupervised DNNLikelihood from Normalizing Flows

Humberto Reyes-González[1,2,⋆] and Riccardo Torre[2,†]

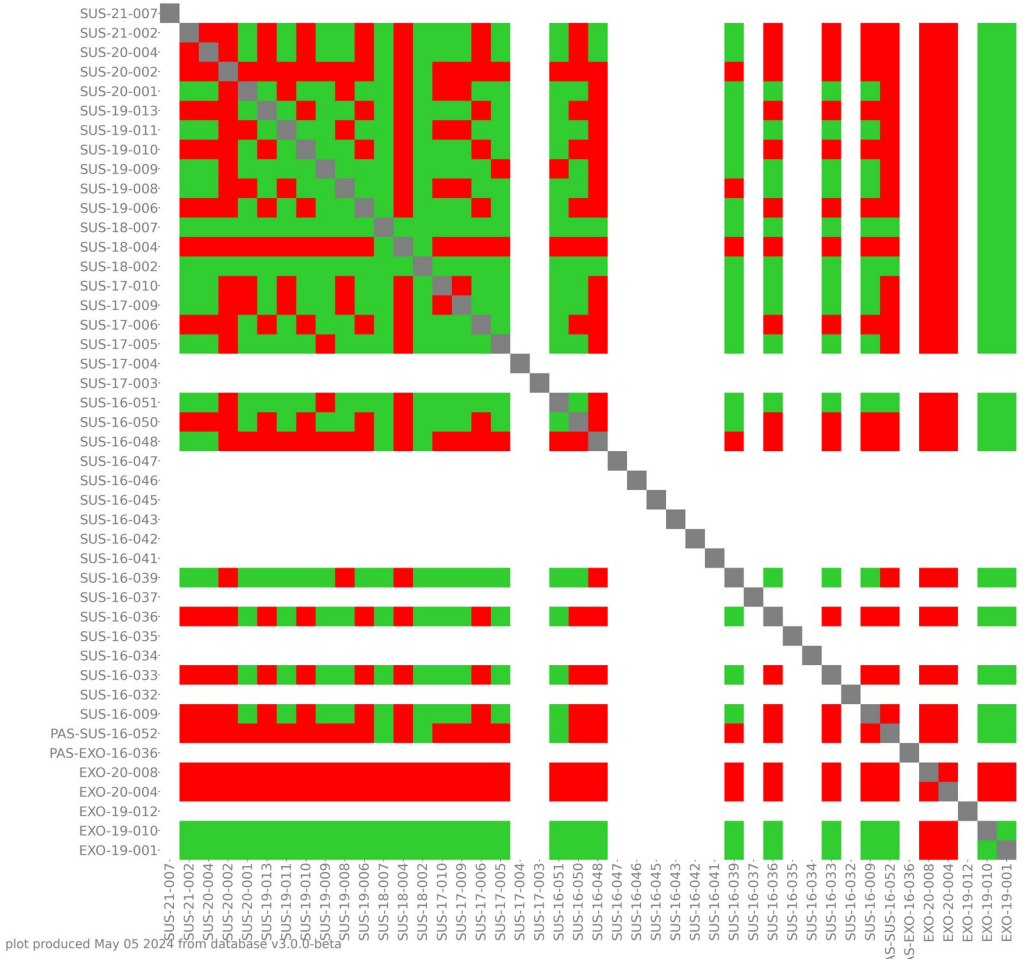
1 Department of Physics, University of Genova, Via Dodecaneso 33, 16146 Genova, Italy
2 INFN, Sezione di Genova, Via Dodecaneso 33, I-16146 Genova, Italy

⋆ humberto.reyes@rwth-aachen.de
† riccardo.torre@ge.infn.it



Combinability Matrix, and “recasts”

CMS, 13 TeV



- We keep track of which pairs of analyses have no overlaps in what LHC collision events enter the signal regions. In the “pheno community”, we combine only if no overlap.
- Currently most reliable source for the combinations matrix: analysis recast frameworks: generating events, track which signal regions are populated by which events
- Requires detailed documentation about the analysis
- Experimental collaborations could in principle produce this information as well (and better)

Long term vision: a naming convention is established, making it possible to correctly combine stats models!

Strength in numbers: Optimal and scalable combination of LHC new-physics searches

Jack Y. Araz, Andy Buckley, Benjamin Fuks, Humberto Reyes-Gonzalez, Wolfgang Waltenberger, Sophie L. Williamson, Jamie Yellen

SciPost Phys. 14, 077 (2023) · published 20 April 2023

doi: 10.21468/SciPostPhys.14.4.077 pdf BiBTeX RIS Submissions/Reports Check for updates

Reproducing an Analysis

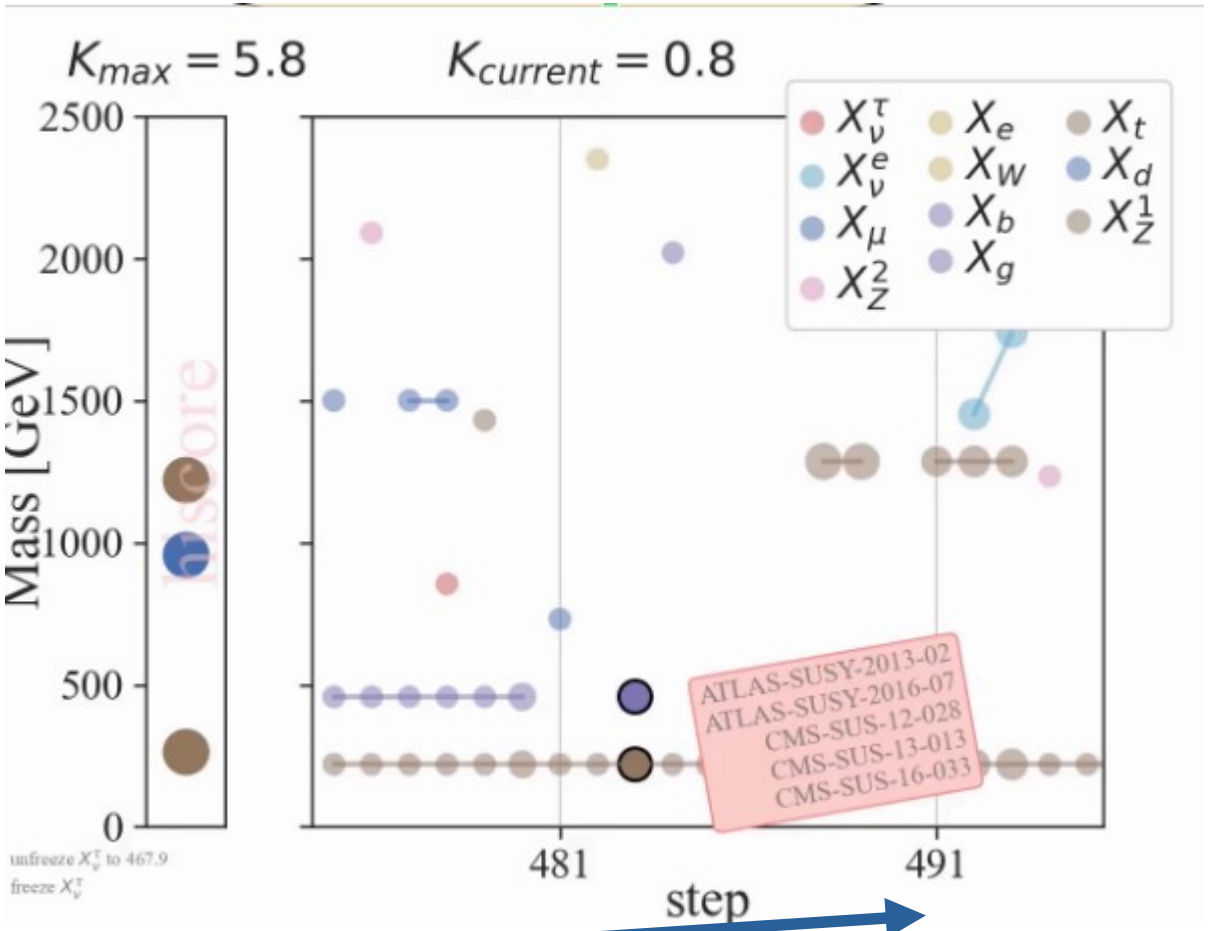
In order to facilitate reproduction of an entire analysis – to e.g. apply it to a new model, or to study analysis overap (previous slide), the analysis needs to be documented in great detail:

- Cutflow tables of the analysis cuts, reconstruction object efficiencies, machine-learning models (e.g. as onnx files) need to be given in digitized form, for e.g. recasts, studies of analysis overlaps.
- Implementation in a simple framework / language: e.g. SimpleAnalysis or Analysis Description Language (ADL)
- For Monte Carlo samples used in publication: generator datacards, etc
- Upper limit plots, signal efficiency maps (of simplified models) in digitized form
- **Statistical models in reusable digitized form**

Experimental Results → Protomodels

the test statistic

Particle spectra



A hiscore protomodel

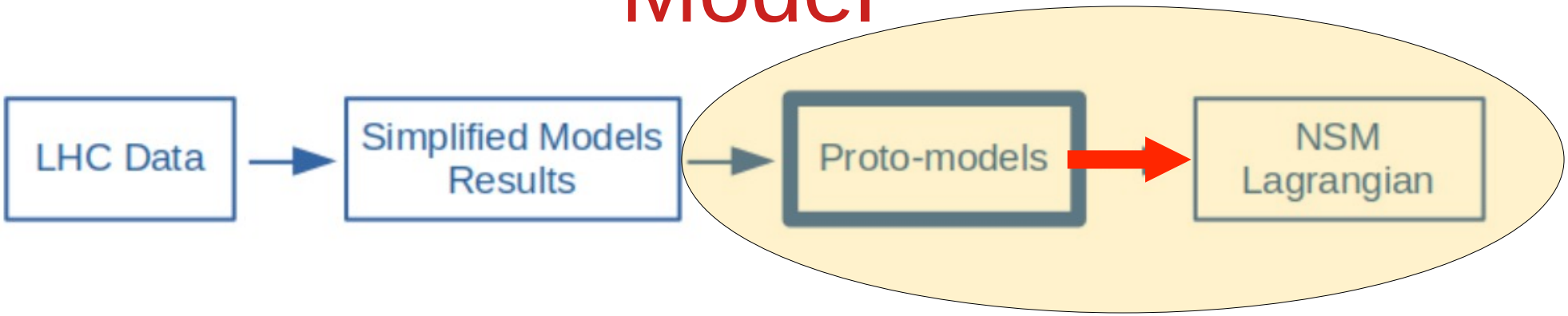
potential dispersed signals

Random modifications

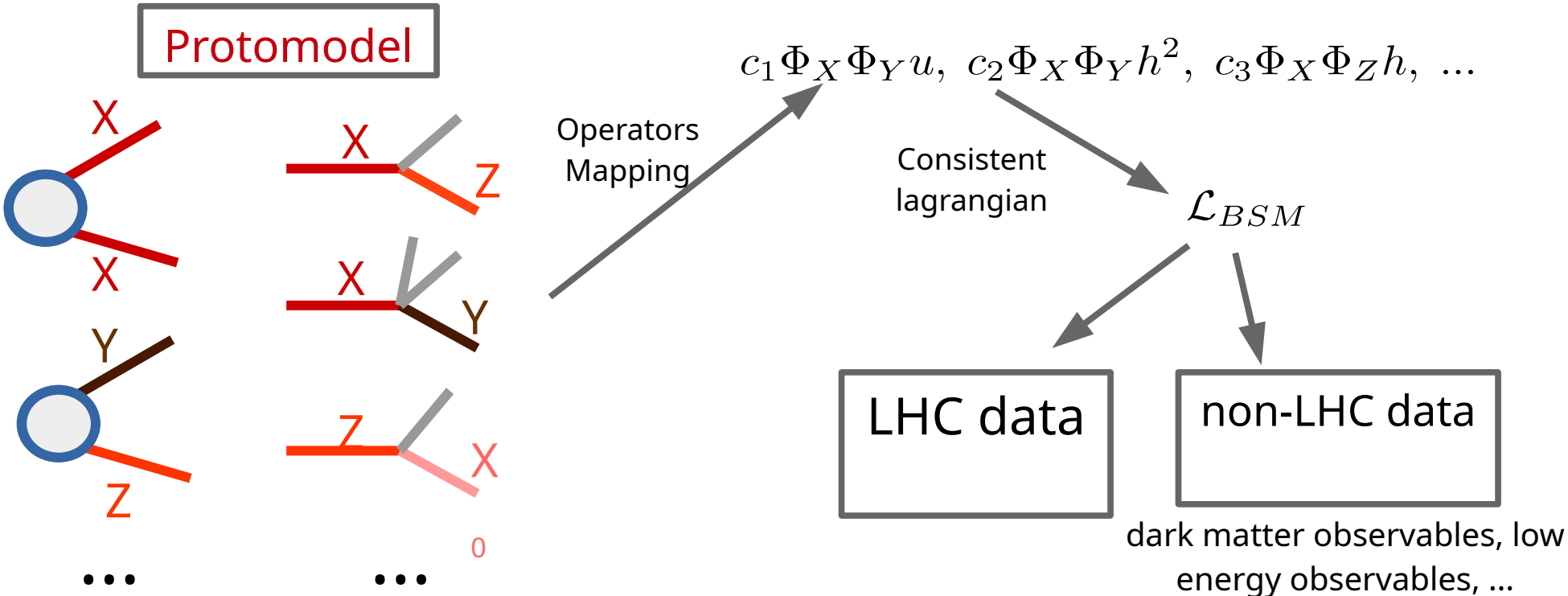
an MCMC-like walk

https://smodels.github.io/protomodels/videos

ProtoModels → Next Standard Model



Work from protomodels to UV complete theories has begun
(John Gargalionis, PostDoc in Valencia) – lot's of combinatorics!

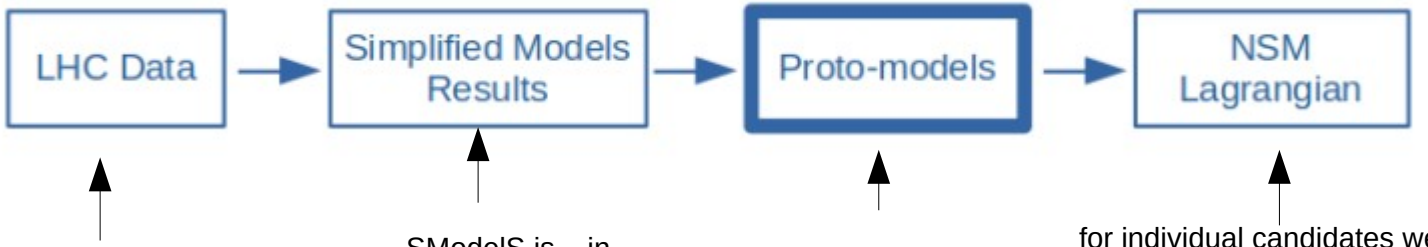


Summary

- Ideally, all important aspects of a physics result get published: cutflow tables, reconstruction object efficiencies, machine learned models, generator datacards, for SModelS: efficiency maps and: **serialized statistical models**!
- HEPdata is the collider physics results platform of choice
- DOIs for all digital objects
- Global fits are still an indispensible tool to obtain a “big picture” (see also gambit’s talk)
- The SModelS collaboration recently started to automate model building based on the statistical models

Differentiably if possible

If we had gradients we could perform gradient descent to find the best model, and we could use e.g. the Fisher information to infer the error on its parameters (or, alternatively we can then MCMC-sample).



described as likelihoods L that are differentiable with respect to the yields y_i

SModelS is – in principle -- differentiable

that's just a sum of simplified models → differentiable!

for individual candidates we can make this differentiable w.r.t fundamental parameters Θ_l, via neural networks

$$\frac{\partial L}{\partial \theta_l} = \frac{\partial L}{\partial y_i} \cdot \frac{\partial y_i}{\partial p_j} \cdot \frac{\partial p_j}{\partial (m_k, \Gamma_k, \sigma_k)} \cdot \frac{\partial (m_k, \Gamma_k, \sigma_k)}{\partial \theta_l}$$

Not yet required (theory space as well as space of measurements are still low-dimensional enough). Action item for now: keep choosing, supporting, developing technologies that support autograd